



BHARATA MATA COLLEGE
AUTONOMOUS | THRIKKAKARA, KOCHI-21

BOOK OF ABSTRACTS



CMMAI 2026

**INTERNATIONAL CONFERENCE ON COMPUTATIONAL
MATHEMATICS, MODELING & ARTIFICIAL INTELLIGENCE
(CMMAI-2026)**

**ORGANIZED BY
POSTGRADUATE AND RESEARCH DEPARTMENT OF MATHEMATICS**

This is the short version of the booklet for print use.
Rest all the details can be found at:
<https://cmmmai2026.bharatamatacollege.in/>

Scientific Advisory Committee

- **Prof. Wil Schilders**
Centre for Analysis, Scientific computing and Applications (CASA), Netherlands
- **Prof. Padmanabhan Seshaiyer**
George Mason University, USA
- **Prof. Dominic Clemence**
North Carolina A&T State University
- **Prof. M. K Kadalbajoo**
Indian Institute of Technology, Kanpur & LNMIIT Jaipur
- **Prof. S. Sundar**
Indian Institute of Technology, Madras & Director, National Institute of Technology, Mizoram
- **Prof. A. K. Nandakumaran**
Indian Institute of Science, Bangalore
- **Prof. G. D. Veerappa Gowda**
Tata Institute of Fundamental Research, Bangalore & Mahindra University
- **Prof. Peeyush Chandra**
Indian Institute of Technology Kanpur
- **Prof. Adi Adimurthi**
Centre for Applicable Mathematics, Tata Institute of Fundamental Research, Bangalore
- **Prof. Jichun Li**
School of Computing, Newcastle University, England
- **Prof. Olga Mula Hernandez**
TU Eindhoven, Netherlands
- **Prof. Ashish Awasthi**
National Institute of Technology, Calicut

Organising Committee

- **Dr. Ashish Awasthi**
Associate Professor NIT, Calicut
- **Rev. Fr. Dr. Abraham Oliyapurath**
Manager Bharata Mata College
- **Rev. Fr. Sen Kallungal**
Assistant Manager
- **Dr. SoumyaBharata Mata College Thomas**
Principal Bharata Mata College
- **Prof. Dr. Johnson K M**
Academic Director Bharata Mata College
- **Dr. Navin Tomy**
Head of department
Department of Mathematics Bharata Mata College
- **Dr. Lakshmi C**
Convenor
- **Dr. Ankit Ruhi**
Co-convenor
- **Dr. Seethu Varghese**
- **Dr. Joby Mackolil**
- **Dr. Toby B Antony**
- **Ms.Sruthi Raghothaman**
- **Ms.Riya Wilson**
- **Ms.Hanan P.T**
- **Ms.Amrutha P**

Contents

Scientific Advisory Committee	3
Organising Committee	4
Index of Contributory Talks	6
About	11
Preface	11
About CMMAI 2026	12
Areas of Interest	12
About Bharata Mata College (Autonomous)	13
About Department of Mathematics	13
Schedule	14
Thursday, 17 September 2026	14
Contributory Sessions — Day 1 (17 September)	15
Friday, 18 September 2026	18
Contributory Sessions — Day 2 (18 September)	19
Saturday, 19 September 2026	22
Contributory Sessions — Day 3 (19 September)	23
List of Abstracts	25
Invited Talks	25
Contributory Talks	43

Index of Contributory Talks

Alphabetical by presenter given name (ignoring Ms/Mr/Dr/Prof). Session numbers follow the contributory-talk schedule; page numbers point to the abstract in this booklet.

Presenter	Session	Title	Page
Aarav Singhania	10.06	ReturnReady: A Sensorized ACL Scaffold Digital Twin for Quantitative Rehabilitation Monitoring	44
Abishek P S	1.08	Shock Dynamics in Saturn's Multi-Ion Plasma with Suprathermal Electrons: Impact of Water ions	45
Achyutan Veeraraghavan	4.05	Psyber Mentalytics: A Novel Deep Learning Algorithm to Predict Mental Health Risks Due to Social Media, and Mathematical Modeling of Variability Over Time	46
Aditya Mohan	4.04	A Computational Framework for Wildfire Prediction and Emergency Response Using Machine Learning and Differential Equations	47
Amrutha Somireddy	7.03	Optimal Intervention Methodology to Advancing Quality Education Through Influencing Local Belief Dynamics	48
Anil Kumar B.T.	5.01	Statistics of Sunspot Cycles based on Multivariate Bootstrapping	49
Ann Maria George	1.07	Head-on Collision of Ion-Acoustic Solitons in Laser Produced Plasma with Kappa Distributed Hot Electrons and Boltzmann Cold Electrons	52
Ansamol M A	1.01	A Parameter-Uniform Time-Lag Linearization Method on a Shishkin Mesh for the Two-Dimensional Singularly Perturbed Burgers' Equation	53
Anupama K	4.07	Clifford Algebra Formulation of Quantum Iterated Function Systems	54
Arnav Krishnan	7.05	Vaccinating the Cleanup Crew: Mathematical Modeling and Optimal Control of Bioaugmented Remediation	55
Arya M V	5.04	Persistent Entropy of One-Dimensional Maps	56
Ashwinth Jeffrey A S	1.06	Numerical Simulation of Electroosmotic Maxwell Hybrid Nanofluid Flow over a Wedge using Triangular-Trapezoidal Fuzzy numbers	43
Aswin V S	6.04	A Graph Neural Network-Augmented Reinforcement Learning Framework for Adaptive Mesh Renement in Black-Scholes Option Pricing PDEs	57
Atharva Kalluraya	7.07	Optimizing Educational Attainment: An Econometric Optimization Approach to Infrastructure Aid Allocation	58

Athena Fleurquin	7.06	Beyond Contagion: A Policy-Coupled Compartmental Model for Forecasting Heatwave Health Escalation	59
Athira Vinay	3.08	A Multi-Metric Wavelet Basis Selection Framework and its Application in Image Segmentation	60
Avni Sriram	4.03	Computational Neuroscience and Machine Learning Analysis for Understanding Sex-Specific Differences in Microglial Morphology and Cognitive Function in Alzheimer's Disease	62
Bishwa Raj Adhikari	8.03	Multicriteria Approaches to Maximizing Static Multicommodity Network Flows	63
Charlotte Cheng	7.02	An Optimal Control Framework for Adaptive Virtual Reality Interventions in Drug Rehabilitation	64
Deepa V	3.07	DHFLatTop For Stock Market Analysis	65
Dipak Babu Amgain	2.01	Maximum Fuzzy Flows in Lossy Network	67
Durga Prasad Khanal	8.05	Budget constrained maximum flow with speed variation and intermediate storage	68
Giri Raj Paneru	8.04	Network Capacity Expansion Minimizing Total Cost	69
Grace Gao	10.02	A Coupled Dynamical Systems Model of Climate Stress, Technology Adaptation, and Corn Yield	70
Hari Rajagopal	6.01	Noise-Dependent Oversampling Trade-Offs in ESPRIT-Based Prony Recovery of Exponential Sums	71
Hrishikesh Shanmuganathan	4.01	Dynamical Systems Analysis of Diffusively Coupled FitzHugh–Nagumo Neural Networks: Synchronization, Stability, and Bifurcation on Weighted Graphs	72
Jicy N	8.07	Wiener index and Matrix Methods in Optimal network design	73
Jincy P Mathai	9.05	On Topological Indices of Ideal Sumset Labelled Signed Graphs	74
Kavitha Mathew	9.01	Non Split Tree Domination in the Total Graphs of some Simple Graphs	76
Krithika S	6.08	Automated DPDP Compliance Assessment Using Multi- Agent AI	77
Kumbinarasaiah S	3.05	A numerical investigation of a novel mathematical shigellosis model using the wavelet method	78
M. R. Lekshmi Priya	6.03	Malayalam Handwritten Character Recognition Model using Kolmogorov Arnold Networks	79
M. Shanmugapriya	3.10	Artificial Intelligence Integrated Numerical Study of Magneto-Casson Penta-Hybrid Nanofluid Flow with Activation Energy and Surface Catalytic Reactions	80

Malavika M	6.07	Perturbation Analysis and Numerical Stability of Quantized Rotary Positional Embeddings in Transformer Architectures	81
Maneesha K K	5.05	Vectorized Block Representation of Persistent Laplacian Diagram	83
Manoj Kumar Singh	3.01	A critical memory order in a delayed fractional rumour model	84
Manoj Kumar Singh	6.02	Degree-dependent conditioning of Chebyshev Kolmogorov–Arnold networks	86
Ms. Mary Pauly K	2.06	A Spatially Coupled q-Rung Orthopair Fuzzy TOPSIS Framework for Flood- Risk Prioritization in Idukki District	122
Miji Philip	9.06	A Study of Morphological Filtering Using Single-Valued Neutrosophic Hypergraphs	88
Mitakshara Rawat	3.04	Dynamics of a Fractional-Order Climate-Driven Vector-Borne Zoonotic Disease Model with Environmental Contamination	89
Mohan Parthasarathy	4.02	PINNLab: An Interactive Framework for Physics-Informed Parameter Estimation in Differential Equations	90
Muhammed Murshid A K	3.06	Analytical Investigation of Fracture in Porous Media Subjected to Moving Load	91
Najma Farsana A	9.03	A Preliminary Study of Reinforced Domination in Graphs	93
Naseema V A	9.07	Equitable Coloring of the Broom Corona Product $K_n \odot K_r$: Cyclic Latin-Rectangle Construction, Algorithmic Analysis, and Applications to Balanced Task Assignment	95
Neha	2.02	A Graph-Restricted Ćirić-Type Fixed Point Theorem (GRFTFT) in Complete Fuzzy b-Metric Spaces	96
Neha Jose	3.11	Depth-Adaptive LoRA for Transformer Architectures: A Comparative Study Against Uniform-Rank LoRA and PiSSA	97
Nehal Joshi	7.01	Is it Real? Optimal Control of an SIR Model of Misinformation through Educational Campaigns	99
Nidhi Thakur	1.04	Higher-Order Upwind Flux Decomposition MGD Equations Based on Kinetic Theory	92
Nihar Venkat Ramtej Chandu	8.02	Optimization of School Bus Transport using Application of Graph Theory and Python	100
Penelope Grewell	10.03	Targeting Operational Efficiency: Mathematical Modeling of Heavy Metals in Water – Treatment Optimization Through Difference Equations	101
Preema T S	2.04	Radical Construction and Characterization of r-Neutrosophic Primary Submodules	102
Radha Srinivasula	10.04	A Difference-Equation Framework for Modeling Organ Transplantation Dynamics	103

Ramsujithra R S	4.08	MDS Spectral Codes Beyond Generalized Reed-Solomon Codes	104
Resmi G Nair	2.03	K-Preorder, C-Preorder and Intuitionistic Fuzzy Filter	105
Rintu Jose	5.07	Convex Skeletons and Reliable Centrality: A Case Study on the 2025 Papal Conclave Network	106
Rithvik Sharma	4.06	Solving for Optimal Staffing: An Inverse Queuing Approach to Equity-Weighted Physician Allocation Across New Jersey Eds	108
Roselin Priscilla S	1.03	An ε -Uniform L1-Finite Element Method for Singularly Perturbed Time-Fractional Delay Parabolic Problems	109
Rovan Jonathan Vaz	3.09	Study of a Depleting Resource and Consumers Model Using Physics-Informed Neural Networks with Parameter Estimation	110
Sachin Wagle	8.01	Bi-Criteria Minimum-Charge FlowLoc Model with Excess Storage Using the ε -Constraint Method	111
Sahadeb Kuila	6.05	Comparative Study of Arterial and Venous Characteristics in the Riemann Problem for Blood Flow with Transport Sahadeb Kuila	112
Sahasra Rao	10.05	From Susceptibility to Survival: A Compartmental Approach to Maternal Health Under Climate Stress	113
Sandra K S	6.06	Uncertainty-Aware Diagnosis Support using Monte Carlo Dropout for Confidence-Calibrated Risk Prediction	114
Sangeetha C	3.12	A Trigonometric Quintic B-Spline-Based Approach for Two Parameter Singularly Perturbed Parabolic Problems with Non-Smooth Convection	—
Seethu Varghese	9.04	An Algorithm for γ_P -sets in Mycielskians of n-spiders	120
Shereeta D'cotha	2.05	A Mathematical Framework for Weighted Neutrosophic Principal Components in Machine Learning	115
Sherin K P	5.06	Targeted Containment vs. Structural Breakdown: Assessing Centrality Metrics for Epidemic Control in Real-World Networks	75
Siddharth Vishwanath	7.04	Modeling Tuberculosis Transmission, Policy, and Optimal Control in Resource-Limited Communities	117
Sona Santhosh	1.02	A Parameter Robust Nonstandard Finite Difference Scheme for the One-dimensional Burgers' Equation	118
Sourudra Nag	8.06	Profile-Conditioned Adaptive-Risk TD3 for Personalized Multi-Asset Portfolio Management	119

Suresh R	5.03	Limiting Behaviour of the Sample Autocorrelation and Partial Autocorrelation Functions in the Presence of a Doublet Outlier and Its Impact on Parameter Estimation	121
Unni M S	9.02	Some Upper Bound of Level Energy of a Rooted Tree	123
Vaisakh Venu	3.02	A Regularized Fractional Dynamical Embedding of the Midilli-Küçük Drying Model ($n = 1$): Well-Posedness and Global Existence	124
Vanisha Waikul	10.01	Modeling the Dynamics of Behavioral Episodes and Applied Behavioral Analysis in Autism Spectrum Disorder	125
Varun Soman	5.02	Precursor-Informed DTW Analog Ensembles for Early Forecasting of Quasi-Periodic Time Series	126
Vashavi A K	1.05	Interplay of Gyrotactic Motility and Solid Particles in Bio-Thermal Convective Instability	128
Zainab Kammappa	3.03	A Cubic Trigonometric B Spline Collocation Method for Time-Fractional Convection-Diffusion Equations with Variable Coefficients	129

Preface

It gives me great pleasure to present the Book of Abstracts of the International Conference on Computational Mathematics, Modelling and Artificial Intelligence (CMMAI 2026), organised by the Postgraduate and Research Department of Mathematics, Bharata Mata College (Autonomous), Thrikkakara, Kerala, from 17 to 19 September 2026. CMMAI 2026 brings together researchers, academicians, scientists, industry professionals, research scholars and students to share recent developments and ideas at the interface of computational mathematics, mathematical modelling and artificial intelligence.

The conference provides a platform for exploring emerging methodologies, interdisciplinary applications and new directions in these rapidly evolving areas. This volume presents the abstracts of the invited and contributed presentations featured in the conference. The contributions reflect the breadth of research in areas including numerical analysis, scientific computing, mathematical modelling, optimisation, dynamical systems, data-driven modelling, scientific machine learning and the mathematical foundations of AI and data science. We hope that the discussions and interactions during CMMAI 2026 will foster meaningful academic exchange, encourage new collaborations and inspire further research.

I express my sincere gratitude to all the invited speakers, participants, session chairs, researchers and students for their valuable contributions. I also extend my heartfelt appreciation to the Management and Administration of Bharata Mata College, faculty members, research scholars, student volunteers and members of the organising committees for their wholehearted support and dedication in making this conference possible.

It is our pleasure to welcome all participants to CMMAI 2026 and to Bharata Mata College (Autonomous), Thrikkakara. We hope that the conference will be an enriching and memorable academic experience for everyone.

Dr. Lakshmi C

Convenor, CMMAI 2026

Postgraduate and Research Department of Mathematics

Bharata Mata College (Autonomous), Thrikkakara,

Kerala, India

About CMMAI 2026

The International Conference on Computational Mathematics, Modelling and Artificial Intelligence (CMMAI 2026) provides a forum for researchers, academicians, and practitioners to exchange ideas at the intersection of mathematical theory, computational methods, modelling, and artificial intelligence. The conference brings together diverse perspectives to explore how mathematical and computational approaches are contributing to the understanding and solution of contemporary scientific and technological problems.

Computational mathematics has become an integral part of modern scientific research, providing tools for the analysis, simulation, prediction, and optimisation of complex systems. Advances in numerical analysis, scientific computing, mathematical modelling, stochastic methods, optimisation, and data-driven approaches have expanded the role of mathematics across the physical, biological, environmental, economic, and data sciences.

CMMAI 2026 seeks to promote interaction across these areas by bringing together researchers from universities, research institutions, and industry. Particular emphasis is placed on fostering dialogue between established mathematical frameworks and emerging computational paradigms, encouraging interdisciplinary collaboration and the exchange of ideas across traditional disciplinary boundaries.

The conference also provides an opportunity to explore emerging directions in computational mathematics and artificial intelligence, including quantum computing, quantum-inspired algorithms, scientific machine learning, and their underlying mathematical foundations. Through invited talks, contributed presentations, and academic discussions, CMMAI 2026 aims to foster a collaborative research environment driven by mathematical rigour, innovation, and curiosity.

Areas of Interest

The conference welcomes contributions spanning a broad spectrum of computational mathematics, mathematical modelling, and artificial intelligence, including but not limited to:

- Numerical Analysis and Scientific Computing
- Mathematical Modelling and Simulation of Physical Systems
- Optimisation, Variational Methods, and Control Theory
- Stochastic Modelling, Uncertainty Quantification, and Dynamical Systems

- Mathematical Modelling in Life and Environmental Sciences
- Mathematical Finance, Economics, and Decision Sciences
- Data-Driven Mathematical Modelling and Scientific Machine Learning
- Mathematical Foundations of Artificial Intelligence and Data Science
- Computational Mathematics in Imaging, Neuroscience, and Medicine
- Emerging Topics in Computational Mathematics and Artificial Intelligence

About Bharata Mata College (Autonomous)

Bharata Mata College (Autonomous), established in 1965 by His Eminence the late Joseph Cardinal Parecattil, is a nationally accredited first-grade arts and science institution in the aided sector, affiliated to Mahatma Gandhi University, Kottayam. Managed by the Archdiocese of Ernakulam-Angamaly under the Bharata Mata Educational Trust, the college earned an *A+* grade from NAAC in 2019 and is ranked by NIRF in the 100–150 band. Guided by its motto “For God and Country”, the college upholds the ideals of love, service and dedication to the nation, while fostering holistic education, responsible citizenship, and inclusivity across gender, caste, and socio-economic backgrounds.

About Department of Mathematics

The Postgraduate and Research Department of Mathematics at Bharata Mata College, Thrikkakara, established in 1965, is distinguished for academic excellence and vibrant research culture. Offering undergraduate, postgraduate, and doctoral programmes, the department has produced ten Ph.D. graduates and over 100 research publications in reputed journals. It regularly organises national and international conferences, seminars, and workshops, creating a dynamic platform for scholarly exchange and collaboration. Research thrust areas include Fluid Dynamics, Numerical Analysis, Applied Mathematics, Mathematical Modeling, Graph Theory, Topology, Fuzzy Algebra and Fuzzy Topology.

Schedule

KT: Keynote talk, PL: Plenary talk, IT: Invited talk, CT: Contributory talk.

Thursday, 17 September 2026

8:30–9:25			Registration
9:30–10:15			Inauguration
10:15–10:30			Tea Break
10:30–11:10	KL	Prof. S. Sundar IIT Madras, India	Pedestrian Crowd Dynamics: Models, Analysis, Computation – A Review
			<i>Session Chair: Prof. A. K. Nandakumaran, IISc Bengaluru</i>
11:15–11:55	PL	Prof. T. Suman Kumar University of Hyderabad, India	A higher-order numerical scheme to a nonlinear McKendrick–Von Foerster equation with singular mortality
			<i>Session Chair: Dr. Ashish Awasthi, NIT Calicut</i>
12:00–12:40	PL	Dr. Ramesh Kolluru BosonQ Psi, Bengaluru, India	Physics-informed and Quantum-Inspired Computational Frameworks for High-Fidelity Fluid Flow Simulation
			<i>Session Chair: Dr. V. B. Kiran Kumar, CUSAT</i>
12:40–13:45			Lunch Break
13:45–14:25	KL	Prof. Padmanabhan Seshaiyer (ONLINE) George Mason University, USA	Computational Modeling of Biological, Bio-Inspired and Engineering Systems using Data-driven approaches in the age of AI
			<i>Session Chair: Prof. Ambat Vijayakumar, CUSAT</i>
14:30–15:10	PL	Prof. Shruti Dubey (ONLINE) IIT Madras, India	Orthogonal Neural Network solution for Implicit Boundary Value Problems

			Session Chair: Dr. Ankit Ruhi, DHSGSU, MP	
15:10–15:25	Tea Break			
15:25–15:55	IT	Dr. Ankit Ruhi – DHSGVV, MP – Radial Basis Functions and Its Achievements, Challenges, and a Path to Structure-Preserving Meshless Hyperbolic Schemes	Dr. Geethi Krishnan (ONLINE) – CATERPILLAR, Bengaluru – Mathematical methods for electrochemical energy storage modeling	
		Dr. Lakshmi C, BMC	Dr. Joby Mackolil, BMC	
16:00–18:30	CT	Contributory Session 1	Contributory Sessions 2 & 3	Contributory Session 4
		Chair: Dr. Ashish Awasthi	Chair: Prof. T. Suman Kumar	Chair: Dr. Ashwin V. S.
	Each contributory talk: 10 minutes presentation + 2 minutes Q&A			

Contributory Sessions — Day 1 (17 September)

Session 1 (Venue 1). Chair: Dr. Ashish Awasthi, NIT Calicut.

16:00–16:10	1.01	Ms. Ansamol M A	A Parameter-Uniform Time-Lag Linearization Method on a Shishkin Mesh for the Two-Dimensional Singularly Perturbed Burgers' Equation
16:12–16:22	1.02	Ms. Sona Santhosh	A Parameter Robust Nonstandard Finite Difference Scheme for the One-dimensional Burgers' Equation
16:24–16:34	1.03	Ms. Roselin Priscilla S	An ϵ -Uniform L1-Finite Element Method for Singularly Perturbed Time-Fractional Delay Parabolic Problems
16:36–16:46	1.04	Ms. Nidhi Thakur	Higher-Order Upwind Flux Decomposition MGD Equations Based on Kinetic Theory
16:48–16:58	1.05	Ms. Vashavi A K	Interplay of Gyrotactic Motility and Solid Particles in Bio-Thermal Convective Instability
17:00–17:10	1.06	Mr. Ashwinth Jeffrey A S	Numerical Simulation of Electroosmotic Maxwell Hybrid Nanofluid Flow over a Wedge using Triangular-Trapezoidal Fuzzy numbers

17:12–17:22	1.07	Ms. Ann Maria George	Head-on Collision of Ion-Acoustic Solitons in Laser Produced Plasma with Kappa-Distributed Hot Electrons and Boltzmann Cold Electrons
17:24–17:34	1.08	Mr. Abishek P S	Shock Dynamics in Saturn's Multi-Ion Plasma with Suprathermal Electrons: Impact of Water ions

Session 2 (Venue 2). Chair: Prof. T. Suman Kumar, University of Hyderabad.

17:36–17:46	2.01	Mr. Dipak Babu Amgain	Maximum Fuzzy Flows in Lossy Network
17:48–17:58	2.02	Dr. Neha	A Graph-Restricted Ćirić-Type Fixed Point Theorem (GRFTFT) in Complete Fuzzy b-Metric Spaces
18:00–18:10	2.03	Ms. Resmi G Nair	K-preorder, C-preorder and intuitionistic fuzzy filter
18:12–18:22	2.04	Ms. Preema T S	Radical Construction and Characterization of r-Neutrosophic Primary Submodules
18:24–18:34	2.05	Ms. Shereeta D'cotha	A Mathematical Framework for Weighted Neutrosophic Principal Components in Machine Learning
18:34–18:46	2.06	Ms. Mary Pauly K	A Spatially Coupled q-Rung Orthopair Fuzzy TOPSIS Framework for Flood-Risk Prioritization in Idukki District

Session 3 (Venue 2).

16:00–16:10	3.01	Dr. Manoj Kumar Singh	A critical memory order in a delayed fractional rumour model
16:12–16:22	3.02	Ms. Vaisakh Venu	A Regularized Fractional Dynamical Embedding of the Midilli-Kücük Drying Model ($n = 1$): Well-Posedness and Global Existence
16:24–16:34	3.03	Ms. Zainab Kammappa	A Cubic Trigonometric B-Spline Collocation Method for Time-Fractional Convection-Diffusion Equations with Variable Coefficients
16:36–16:46	3.04	Ms. Mitakshara Rawat	Dynamics of a Fractional-Order Climate-Driven Vector-Borne Zoonotic Disease Model with Environmental Contamination
16:48–16:58	3.05	Dr. Kumbinarasaiah S	A numerical investigation of a novel mathematical shigellosis model using the wavelet method
17:00–17:10	3.06	Mr. Muhammed Murshid A K	Analytical Investigation of Fracture in Porous Media Subjected to Moving Load
17:12–17:22	3.07	Ms. Deepa V	DHFLatTop for stock market analysis

17:24–17:34	3.08	Ms. Athira Vinay	A Multi-Metric Wavelet Basis Selection Framework and its Application in Image Segmentation
17:36–17:46	3.09	Mr. Rován Jonathan Vaz	Study of a Depleting Resource and Consumers Model Using Physics-Informed Neural Networks with Parameter Estimation
17:48–17:58	3.10	Dr. M. Shanmugapriya	Artificial Intelligence Integrated Numerical Study of Magneto-Casson Penta-Hybrid Nanofluid Flow with Activation Energy and Surface Catalytic Reactions
18:00–18:10	3.11	Ms. Neha Jose	Depth-Adaptive LoRA for Transformer Architectures: A Comparative Study Against Uniform-Rank LoRA and PiSSA
18:12–18:22	3.12	Dr. Sangeetha C	A Trigonometric Quintic B-Spline-Based Approach for Two Parameter Singularly Perturbed Parabolic Problems with Non-Smooth Convection

Session 4 (Venue 3). Chair: Dr. Ashwin V. S., Digital University of Kerala.

16:00–16:10	4.01	Hrishikesh Shanmuganathan	Dynamical Systems Analysis of Diffusively Coupled FitzHugh–Nagumo Neural Networks: Synchronization, Stability, and Bifurcation on Weighted Graphs
16:15–16:25	4.02	Mohan Parthasarathy	PINNLab: An Interactive Framework for Physics-Informed Parameter Estimation in Differential Equations
16:30–16:40	4.03	Avni Sriram	Computational Neuroscience and Machine Learning Analysis for Understanding Sex-Specific Differences in Microglial Morphology and Cognitive Function in Alzheimer’s Disease
16:45–16:55	4.04	Aditya Mohan	A Computational Framework for Wildfire Prediction and Emergency Response Using Machine Learning and Differential Equations
17:00–17:10	4.05	Achyutan Veeraraghavan	Psyber Mentalytics: A Novel Deep Learning Algorithm to Predict Mental Health Risks Due to Social Media, and Mathematical Modeling of Variability Over Time
17:15–17:25	4.06	Rithvik Sharma	Solving for Optimal Staffing: An Inverse Queuing Approach to Equity-Weighted Physician Allocation Across New Jersey EDs
17:30–17:40	4.07	Ms. Anupama K	Clifford Algebra Formulation of Quantum Iterated Function Systems
17:42–17:52	4.08	Ms. Ramsujithra R S	MDS Spectral Codes Beyond Generalized Reed-Solomon Codes

Friday, 18 September 2026

9:00–9:40	KL	Prof. A. K. Nandakumaran IISc Bengaluru, India	Ray-Radon Transform and Inverse Problems in Tomography
			<i>Session Chair: Dr. Ashish Awasthi, NIT Calicut</i>
9:45–10:25	PL	Dr. Mayur Prakash Bonkile IIT Kharagpur, India	Why Should Mathematicians Look into Battery Modelling?
			<i>Session Chair: Dr. Maruthi N. H., Siemens Digital Industries Software, Bengaluru</i>
10:25–10:40		Tea Break	
10:40–11:20	KL	Prof. Amiya K. Pani (ONLINE) BITS Pilani, K. K. Birla Goa Campus, India	On Backward Time-Fractional Diffusion Problems: A Unified Approach
			<i>Session Chair: Prof. A. K. Nandakumaran, IISc Bengaluru</i>
11:25–12:05	PL	Prof. Ritesh Kumar Dubey (ONLINE) SRM Institute of Science and Technology, Chennai, India	Outliers Are Not Liars: Data-Driven WENO Weights
			<i>Session Chair: Prof. T. Suman Kumar, University of Hyderabad</i>
12:10–12:50	PL	Dr. Ashish Awasthi NIT Calicut, India	Nonstandard Finite Difference Methods for Singularly Perturbed Problems: Fundamentals and Applications to One- and Two-parameter Models
			<i>Session Chair: Dr. Vijitha Mukundan, Sacred Heart College Chalakudy</i>
12:50–13:50		Lunch Break	

13:50–14:20	IT	Dr. Maruthi N. H. – Siemens Digital Industries Software, Bengaluru – ML Frameworks as Scientific Computing Infrastructure for CFD	Dr. Arun C Nair (ONLINE) – Mercedes-Benz R&D India, Bengaluru – From Equations to Energy: Mathematics Behind Power Electronics	
		<i>Dr. Mayur Prakash Bonkile, IIT Kharagpur</i>	<i>Dr. Joby Mackolil, BMC</i>	
14:25–16:00	CT	Contributory Session 5	Contributory Session 6	Contributory Session 7
		<i>Chair: Dr. Ankit Ruhi</i>	<i>Chair: Dr. Noufal Ashraf, CUSAT</i>	<i>Chair: Dr. Ashwin V. S.</i>
16:00–16:15	Tea Break			
16:15–16:55	KL	Prof. Peeyush Chandra (ONLINE) IIT Kanpur, India	Modeling Infectious Diseases with Mathematics – Why and How!	
			<i>Session Chair: Dr. Mayur Prakash Bonkile, IIT Kharagpur</i>	
17:00–17:40	KL	Prof. Sivaguru S. Sritharan (ONLINE) Air Force Institute of Technology, USA	Open Quantum Systems and Quantum Decoherence	
			<i>Session Chair: Dr. Ankit Ruhi, DHSGSU, MP</i>	
18:30–19:30	Cultural Evening			
19:30–	Conference Dinner			

Contributory Sessions — Day 2 (18 September)

Session 5 (Venue 1). Chair: Dr. Ankit Ruhi, DHSGSU, MP.

14:25–14:35	5.01	Mr. Anil Kumar B.T.	Statistics of Sunspot Cycles based on Multi-variate Bootstrapping
14:37–14:47	5.02	Mr. Varun Soman	Precursor-Informed DTW Analog Ensembles for Early Forecasting of Quasi-Periodic Time Series

14:49–14:59	5.03	Dr. Suresh R	Limiting Behaviour of the Sample Autocorrelation and Partial Autocorrelation Functions in the Presence of a Doublet Outlier and Its Impact on Parameter Estimation
15:01–15:11	5.04	Ms. Arya M V	Persistent Entropy of One-Dimensional Maps
15:13–15:23	5.05	Ms. Maneesha K K	Vectorized Block Representation of Persistent Laplacian Diagram
15:25–15:35	5.06	Ms. Sherin K P	Targeted Containment vs. Structural Breakdown: Assessing Centrality Metrics for Epidemic Control in Real-World Networks
15:37–15:47	5.07	Ms. Rintu Jose	Convex Skeletons and Reliable Centrality: A Case Study on the 2025 Papal Conclave Network

Session 6 (Venue 2). Chair: Dr. Noufal Ashraf, CUSAT.

14:25–14:35	6.01	Mr. Hari Rajagopal	Noise-Dependent Oversampling Trade-Offs in ESPRIT-Based Prony Recovery of Exponential Sums
14:37–14:47	6.02	Dr. Manoj Kumar Singh	Degree-dependent conditioning of Chebyshev Kolmogorov-Arnold networks
14:49–14:59	6.03	Ms. M. R. Lekshmi Priya	Malayalam Handwritten Character Recognition Model using Kolmogorov Arnold Networks
15:01–15:11	6.04	Dr. Aswin V S	A Graph Neural Network-Augmented Reinforcement Learning Framework for Adaptive Mesh Renement in Black-Scholes Option Pricing PDEs
15:13–15:23	6.05	Dr. Sahadeb Kuila	Comparative Study of Arterial and Venous Characteristics in the Riemann Problem for Blood Flow with Transport Sahadeb Kuila
15:25–15:35	6.06	Ms. Sandra K S	Uncertainty-Aware Diagnosis Support using Monte Carlo Dropout for Confidence-Calibrated Risk Prediction
15:37–15:47	6.07	Ms. Malavika M	Perturbation Analysis and Numerical Stability of Quantized Rotary Positional Embeddings in Transformer Architectures
15:49–15:59	6.08	Ms. Krithika S	Automated DPDP compliance assessment using Multi agent AI

Session 7 (Venue 3). Chair: Dr. Ashwin V. S., Digital University of Kerala.

14:25–14:35	7.01	Nehal Joshi	Is it Real?: Optimal Control of an SIR Model of Misinformation through Educational Campaigns
14:40–14:50	7.02	Charlotte Cheng	An Optimal Control Framework for Adaptive Virtual Reality Interventions in Drug Rehabilitation
14:55–15:05	7.03	Amrutha Somireddy	Optimal Intervention Methodology to Advancing Quality Education Through Influencing Local Belief Dynamics
15:10–15:20	7.04	Siddharth Vishwanath	Modeling Tuberculosis Transmission, Policy, and Optimal Control in Resource-Limited Communities
15:25–15:35	7.05	Arnav Krishnan	Vaccinating the Cleanup Crew: Mathematical Modeling and Optimal Control of Bioaugmented Remediation
15:40–15:50	7.06	Athena Fleurquin	Beyond Contagion: A Policy-Coupled Compartmental Model for Forecasting Heatwave Health Escalation
15:55–16:05	7.07	Atharva Kalluraya	Optimizing Educational Attainment: An Econometric Optimization Approach to Infrastructure Aid Allocation

Saturday, 19 September 2026

9:00–9:40	KL	Prof. María Angélica Salazar Aguilar (ONLINE) Universidad Autónoma de Nuevo León, Mexico	Mathematical Modeling and Computational Intelligence in Logistics: Bridging On-Demand Services and Optimization
			<i>Session Chair: Prof. Sarvesh Kumar, IIST Thiruvananthapuram</i>
9:45–10:25	PL	Prof. Chandhini G NITK Surathkal, India	Physics-Informed Machine Learning: PINNs, Their Failure Modes, and the Rise of Learned Operators
			<i>Session Chair: Dr. Rakesh Kumar, Mahindra University, Hyderabad</i>
10:25–10:40	Tea Break		
10:40–11:20	PL	Prof. Kaushik Mukherjee IIST Thiruvananthapuram, India	Higher-Order Methods for Singularly Perturbed Semilinear Parabolic PDEs
			<i>Session Chair: Dr. Ankit Ruhi, DHSGSU, MP</i>
11:25–12:05	PL	Dr. Asha K. Dond IISER Thiruvananthapuram, India	Adaptive FEM For Optimal Control Problems Involving a Point Source Governed by Biharmonic Equation
			<i>Session Chair: Prof. Chandhini G, NITK Surathkal</i>
12:10–12:50	PL	Dr. Rakesh Kumar Mahindra University, Hyderabad, India	Hybrid Numerical Schemes for Systems of Hyperbolic Conservation Laws
			<i>Session Chair: Prof. Kaushik Mukherjee, IIST</i>
12:50–13:50	Lunch Break		
13:50–14:30	PL	Dr. Sri Vallabha Deevi (ONLINE) Tiger Analytics, India	AIML for Scientific Computing
			<i>Session Chair: Dr. Ankit Ruhi, DHSGSU, MP</i>

14:35–15:15	PL	Prof. Sarvesh Kumar IIST Thiruvananthapuram, India	Virtual Element Methods: Recent Developments and Applications	
			<i>Session Chair: Dr. Asha K. Dond, IISER Thiruvananthapuram</i>	
15:20–16:45	CT	Contributory Session 8	Contributory Session 9	Contributory Session 10
		<i>Chair: Dr. Tanusree Pandit, CUSAT</i>	<i>Chair: Prof. Manju K. Menon, St. Paul's College</i>	<i>Chair: Dr. Ashwin V. S.</i>
16:45–17:00	Tea Break			
17:00–17:30	Valedictory Session			

Contributory Sessions — Day 3 (19 September)

Session 8 (Venue 1). Chair: Dr. Tanusree Pandit, CUSAT.

15:20–15:30	8.01	Mr. Sachin Wagle	Bi-Criteria Minimum-Charge FlowLoc Model with Excess Storage Using the ε -Constraint Method
15:32–15:42	8.02	Mr. Nihar Venkat Ramtej Chandu	Optimization of School Bus Transport Using Application of Graph Theory and Python
15:44–15:54	8.03	Mr. Bishwa Raj Adhikari	Multicriteria Approaches to Maximizing Static Multicommodity Network Flows
15:56–16:06	8.04	Mr. Giri Raj Paneru	Network Capacity Expansion Minimizing Total Cost
16:08–16:18	8.05	Dr. Durga Prasad Khanal	Budget constrained maximum flow with speed variation and intermediate storage
16:20–16:30	8.06	Mr. Sourudra Nag	Profile-Conditioned Adaptive-Risk TD3 for Personalized Multi-Asset Portfolio Management
16:32–16:42	8.07	Dr. Jicy N	Wiener index and Matrix methods in optimal network design

Session 9 (Venue 2). Chair: Prof. Manju K. Menon, St. Paul's College, Kalamassery.

15:20–15:30	9.01	Ms. Kavitha Mathew	Non Split Tree Domination in the Total Graphs of some Simple Graphs
15:32–15:42	9.02	Mr. Unni M S	Some Upper Bound of Level Energy of a Rooted Tree
15:44–15:54	9.03	Ms. Najma Farsana A	A Preliminary Study of Reinforced Domination in Graphs

15:56–16:06	9.04	Dr. Seethu Varghese	An Algorithm for γ_P -sets in Mycielskians of n-spiders
16:08–16:18	9.05	Ms. Jincy P Mathai	On Topological Indices of Ideal Sumset Labelled Signed Graphs
16:20–16:30	9.06	Ms. Miji Philip	A Study of Morphological Filtering Using Single-Valued Neutrosophic Hypergraphs
16:32–16:42	9.07	Ms. Naseema V A	Equitable Coloring of the Broom Corona Product $K_n \odot K_r$: Cyclic Latin-Rectangle Construction, Algorithmic Analysis, and Applications to Balanced Task Assignment

Session 10 (Venue 3). Chair: Dr. Ashwin V. S., Digital University of Kerala.

15:20–15:30	10.01	Vanisha Waikul	Modeling the Dynamics of Behavioral Episodes and Applied Behavioral Analysis in Autism Spectrum Disorder
15:35–15:45	10.02	Grace Gao	A Coupled Dynamical Systems Model of Climate Stress, Technology Adaptation, and Corn Yield
15:50–16:00	10.03	Penelope Grewell	Targeting Operational Efficiency: Mathematical Modeling of Heavy Metals in Water – Treatment Optimization Through Difference Equations
16:05–16:15	10.04	Radha Srinivasula	A Difference-Equation Framework for Modeling Organ Transplantation Dynamics
16:20–16:30	10.05	Sahasra Rao	From Susceptibility to Survival: A Compartmental Approach to Maternal Health Under Climate Stress
16:35–16:45	10.06	Aarav Singhania	ReturnReady: A Sensorized ACL Scaffold Digital Twin for Quantitative Rehabilitation Monitoring

List of Abstracts

Invited Talks

Pedestrian Crowd Dynamics: Models, Analysis, Computation - A Review

S. Sundar

KL

Department of Mathematics Indian Institute of Technology Madras Chennai 600036,
INDIA

In this talk I will be covering a tour on Pedestrian Crowd Dynamics Models of two different scales with a through analysis on the various models, emphasizing the importance of developing sophisticated numerical models in order to make the models relevant.

Computational Modeling of Biological, Bio-Inspired and Engineering Systems using Data-driven approaches in the age of AI

Padmanabhan Seshaiyer

KL

Professor and Director Presidential Advisor for Strategic Partnerships George Mason University

The rapid convergence of mathematical modeling, computation, data science, and artificial intelligence is transforming how we understand, predict, and design complex systems. This talk presents a unified computational framework for studying biological, bio-inspired, and engineering systems across multiple scales, combining mathematical principles with modern data-driven approaches to move beyond descriptive analysis toward prediction, inference, and discovery. Drawing on ideas developed in Quantitative Biology: Mathematical Modeling and Computation, we will explore a range of modeling paradigms, including compartmental, dynamical, network-based, and mechanistic approaches, and their applications to complex systems arising in biology, epidemiology, neural systems, bio-inspired interactions, and engineering. Particular emphasis will be placed on the interplay between models, computation, and data. Numerical simulation, parameter estimation, uncertainty quantification, optimization, and machine learning provide complementary tools for extracting insight from increasingly rich and heterogeneous data. We will examine how data-driven methods can be integrated with mechanistic and physics-informed models to improve prediction, uncover emergent behavior, and address nonlinear, multiscale systems for which purely analytical or purely data-driven approaches may be insufficient. Through examples spanning biological, bio-inspired, and engineering systems, the talk will illustrate how the integration of theory, computation, and data creates a common language for understanding complex phenomena and designing new solutions.

Ray-Radon Transform and Inverse Problems in Tomography

A. K. Nandakumaran

KL

Department of Mathematics, Indian Institute of Science, Bangalore- 560 012

Tomography is a cross-sectional imaging of an object or tissue from the mathematical modeling and measured experimental data. The study of tomography involves mathematical modeling, analysis, development of numerical algorithms, its implementation and experimental validation. These problems lead to a class of inverse problems, where the mathematical analysis and computations are rather complicated, but has tremendous applications in various branches of science including medical applications like CT scan and other areas. It is truly a multi-disciplinary area and requires help from experts on various subjects. In this talk, we begin by introducing Ray and Radon transforms in the context of Tomographic reconstruction in X-ray tomography (CT scan). We will see how Fourier transform is used in inverting the Radon Transform. This was initially introduced by Radon in 1917 and he presented a procedure to obtain a two-dimensional unknown function from one-dimensional cumulative projections known as Radon transform. There are higher dimensional versions as well. This is what essentially used in CT scan. In the second part of the talk depending on the time available, we will give a general introduction to various tomographies across disciplines, issues encountered, historical developments etc. Except X-ray tomography, where we invert an integral, the other tomographies lead to inversion of PDEs which are rather complicated which requires the analysis of modern PDEs. The second part will be addressed to cater the general audience both from engineering and science. Indeed, if time permits, we may discuss other tomographies which requires help from modern analysis of PDEs and try to present some results of solving a spectral problem.

On Backward Time-Fractional Diffusion Problems: A Unified Approach

Samir Karaa, Mariam Al-Maskari¹ and Amiya Kumar Pani

KL

¹Department of Mathematics, Sultan Qaboos University, Muscat Oman, Department of Mathematics,
BITS-Pilani, Goa Campus, Goa India

Inverse fractional diffusion problems, particularly, backward problems have received considerable attention in recent years, the resulting literature is diverse and sometimes fragmented due to the variety of models and numerical techniques employed. In this talk, it is intended to present a unified analysis of the quasi-boundary value method for solving backward problems associated with a broad class of time-fractional diffusion models. The regularized problem can be discretized using diverse spatial and temporal strategies; as a representative example, we analyze a piecewise linear finite element method in space and provide a hint for rigorous error analysis at least for the semidiscrete scheme and propose practical guidelines for selecting discretization and regularization parameters relative to the noise level. The unified character of our analysis ensures its applicability to various fractional models, spatial discretizations, etc.

Modeling Infectious Diseases with Mathematics – Why and How?

Peeyush Chandra

KL

Professor (Retired), Department of Mathematics and Statistics, Indian Institute of Technology Kanpur, Kanpur-208025, India

Mathematics has been playing important part in various fields in our life. One such area is Epidemiology - the study of infectious diseases. Mathematical Epidemiology refers to using mathematical tools to understand the underlying mechanism so as to curtail or eradicate an infectious disease. The recent outbreak of COVID-19 has re-emphasized the importance and usefulness of this area. This talk focuses on an evolution of epidemiology and an introduction to basic deterministic mathematical models for infectious diseases with two specific cases – modeling for Malaria and COVID 19.

Measurement And Control Of Nature's Dynamics: Commutative And Noncommutative

Sivaguru S. Sritharan

KL

Air Force Institute of Technology, USA

Filtering and control are two faces of the same problem: extracting the best estimate of a hidden state from noisy observations, and steering that state toward a desired outcome. This duality plays out in strikingly parallel ways across classical and quantum systems, distinguished mathematically by whether the underlying algebra of observables is commutative or noncommutative. On the commutative side, the speaker and collaborators developed nonlinear filtering for infinite-dimensional and jump-diffusion signals — deriving Zakai and Kushner–Stratonovich equations for systems as complex as stochastic Navier–Stokes flows and nonlinear wave equations — and paired this with a rigorous dynamic programming theory, establishing infinite-dimensional Hamilton–Jacobi–Bellman equations via viscosity solutions for optimal feedback control of these same systems. On the noncommutative side, V. P. Belavkin built the quantum analogue: filtering theory for continuously monitored open quantum systems, formulated via quantum stochastic calculus on operator algebras, yielding the quantum Zakai and Belavkin (quantum Kushner–Stratonovich) equations, together with a quantum control theory built directly atop the filtered state. This framework hinges on structural features with no classical counterpart — the non-demolition property, which alone makes quantum conditioning well-posed — and demands operator-algebraic techniques (martingale problems, Fagnola–Rebolledo conservativity conditions) to handle the unbounded generators ubiquitous in real physical models. This talk develops both theories in parallel, showing that they share a common architecture — linear unnormalized filter, nonlinear normalized filter via an innovations process, and dynamic programming over the filtered state — while diverging exactly at the point where probability itself becomes noncommutative. The aim is a unified view of measurement and control as they act on nature's dynamics, from turbulent fluids to open quantum systems.

Mathematical Modeling and Computational Intelligence in Logistics: Bridging On-Demand Services and Optimization

María Angélica Salazar Aguilar

KL

Graduate Program in Artificial Intelligence and Optimization, Facultad de Ingeniería Mecánica y Eléctrica, Universidad Autónoma de Nuevo León, San Nicolás de los Garza, Nuevo León, Mexico

Modern logistics and on-demand transportation systems are increasingly defined by highly dynamic demands, complex spatial-temporal constraints, and the critical need for operational efficiency. Bridging the gap between flexible, user-centric on-demand services and global fleet optimization requires a powerful combination of rigorous mathematical modeling and computational intelligence. This plenary talk explores this intersection by synthesizing two challenging, real-world optimization problems: the Dial-a-Tour Problem (DATP) and the Team Orienteering Problem with Nonidentical Agents and Balanced Score (TOP-NABS). First, we address the Dial-a-Tour Problem (DATP), a novel on-demand transportation problem arising from popular tourist sectors where multiple tourist groups must be routed to sequential attraction sites. We formulate this using Mixed-Integer Linear Programming (MILP) and Constraint Programming (CP) models, which optimize vehicle occupancy and routing under strict precedence constraints and Quality of Service (QoS) guarantees. To resolve real-size instances, we introduce a descent heuristic driven by a novel k -buffer insertion operator. Second, we extend our logistical framework to workforce scheduling through TOP-NABS, which schedules service orders among heterogeneous, nonidentical crews. This problem integrates task compatibility constraints (skills matching) while optimizing workload and wage equity among workers using max-min and min-max balance metrics. The problem is modeled mathematically and solved using an adaptive multi-start heuristic (AMH) that incorporates a self-tuning reactive GRASP learning mechanism. Finally, we highlight the computational results from both studies, demonstrating that while exact formulations (MILP and CP) provide foundational mathematical rigor, computational intelligence-driven metaheuristics can achieve comparable or superior solutions with dramatic computational savings compared to exact solvers. This synthesis provides a practical and mathematical roadmap for decision-makers looking to balance customer satisfaction, operational costs, and workforce equity in modern logistical networks.

A higher-order numerical scheme to a nonlinear McKendrick-Von Foerster equation with singular mortality

T. Suman Kumar

PL

University of Hyderabad

In this talk, we present higher-order finite-difference numerical schemes to a nonlinear McKendrick-Von Foerster equation when the death rate has a singularity at the maximum age. The third- and fourth-order schemes that are proposed here are based on the characteristics, which are non-intersecting lines in this case, and are multi-step methods with appropriate corrections at each step. An outline of the proof of the error estimates will be presented.

Physics-informed and Quantum-Inspired Computational Frameworks for High-Fidelity Fluid Flow Simulation

Ramesh Kolluru

PL

BosonQ Psi, Bengaluru, India

High-fidelity computational fluid dynamics remains essential for predicting complex flow phenomena involving shocks, boundary layers, turbulence, vortex dynamics, multiphase interfaces, and coupled transport processes. However, the computational demands of high-resolution three-dimensional simulations, repeated design evaluations, uncertainty quantification, data assimilation, and inverse problems continue to motivate new computational paradigms. This invited talk examines two complementary directions: physics-informed machine learning and quantum-inspired tensor-network computing. Physics-informed methods incorporate governing equations, conservation laws, constitutive relations, and boundary or initial conditions into data-driven models. They provide promising tools for sparse-data reconstruction, parameter estimation, inverse modelling, differentiable simulation, and targeted closure correction. Their use in demanding CFD settings nevertheless requires care: low training residuals do not automatically guarantee accurate shocks, forces, long-time dynamics, or turbulent spectra. The talk discusses why robust hybrid formulations—where learning corrects or augments a trusted numerical component—are often more credible than unconstrained replacement of conventional solvers. The second direction uses quantum-inspired tensor representations, including Tensor Trains, Matrix Product States, and Quantics Tensor Trains, to compress high-dimensional flow fields and operators on classical computing platforms. These methods seek to exploit multiscale correlations and low-rank structure to reduce memory requirements and accelerate repeated PDE operations. Their effectiveness depends on controlled tensor-rank growth, adaptive truncation, conservative formulations, and careful treatment of shocks, moving interfaces, complex geometry, and turbulence. A unified framework is proposed in which conservative CFD discretisation provides the physical backbone, physics-informed learning addresses uncertain or data-dependent components, and tensor-network representations compress states and operators. The presentation emphasises verification through conservation, positivity, entropy consistency, grid and rank convergence, engineering quantities of interest, robustness beyond training conditions, and total computational cost. The central message is that future CFD capability will likely emerge from hybrid, structure-aware methods: learning should infer what is unknown, tensor methods should compress what is structured, and numerical CFD should enforce what must remain physically correct.

Orthogonal Neural Network solution for Implicit Boundary Value Problems

Shruti Dubey

PL

IIT Madras

This study presents an orthogonal polynomial-based neural network with an Extreme Learning Machine (ELM) algorithm for obtaining approximate solutions of differential equations with implicit boundary conditions. The proposed framework employs Legendre, Laguerre, Chebyshev, and Hermite polynomials within a functional expansion block to construct the neural network. The approach is applied to higher-order differential equations, systems of differential equations, and a fluid mechanics problem. The obtained solutions are compared with exact solutions and conventional numerical methods to assess accuracy and effectiveness. Numerical results demonstrate that the proposed method provides accurate solutions with a simple and robust computational framework, highlighting its potential for solving differential equations with complex boundary conditions.

Why Should Mathematicians Care About Battery Modelling?

Mayur Prakash Bonkile

PL

School of Energy Science and Engineering, IIT Kharagpur, Kharagpur West Bengal, India

Batteries are complex, nonlinear, and multiscale electrochemical energy storage systems. Their behaviour is governed by coupled electrochemical, thermal, mechanical, and transport phenomena. These characteristics make battery modelling a challenging problem for mathematical analysis and scientific computing. This talk introduces the mathematical foundations of physics-based battery models. It covers conservation laws, diffusion equations, nonlinear electrochemical kinetics, coupled partial differential equations, and multiphysics interactions. The discussion will then move to battery degradation, including solid-electrolyte interphase growth, loss of active material, mechanical stress, and electrode expansion. Particular attention will be given to how seemingly simple questions: How much energy can a battery deliver? Why does it degrade? Why does temperature matter? Why does an electrode particle expand and crack?, lead to complex mathematical models involving PDEs, nonlinear algebraic equations, moving interfaces, parameter estimation, model reduction and numerical simulation. Drawing on examples from physics-based lithium-ion battery modelling, the talk will demonstrate how mathematics can provide not only computational tools but also physical insight into battery behaviour. The talk will further discuss the emerging interface between battery modelling, machine learning and digital twins, where mathematical models can be combined with data-driven approaches to develop faster battery models.

Outliers Are Not Liars: Data-Driven WENO Weights

Ritesh Kumar Dubey

PL

Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, Chengalpattu, Tamilnadu, India-603203

High-order Weighted Essentially Non-Oscillatory (WENO) schemes are widely used for approximating solution of hyperbolic conservation laws (HCL) which inevitably contains discontinuities. These discontinuities in the solution of HCL are finitely countable and occupy only a very small region of the entire solution therefore, they can be considered as outliers in the solution region. Consequently, the WENO weights associated with these discontinuities are regarded as outlier weights in the distribution of WENO weights. A data-driven framework is proposed to detect such outlier WENO weights, which deviate significantly from the optimal linear weights. To statistically quantify these deviated weights, extensive WENO weights datasets are generated and threshold cutoff values corresponding to discontinuous weights are estimated using quantile-based analysis and kernel density estimation. Based on these statistically computed thresholds, a data-driven correction is introduced to modify the classical non-linear weights which achieve formal order of accuracy with high resolution without compromising inherited non-oscillatory ENO stability property of baseline WENO weights. Comprehensive numerical experiments demonstrate that such data driven modified WENO schemes consistently outperform classical WENO schemes, achieving improved resolution of discontinuities and enhanced overall accuracy.

Nonstandard Finite Difference Methods for Singularly Perturbed Problems: Fundamentals and Applications to One- and Two-Parameter Models

Ashish Awasthi

PL

Department of Mathematics, National Institute of Technology Calicut, Kerala, India

Singularly perturbed problems (SPPs) arise naturally in mathematical models involving convection–diffusion processes, chemical flow reactors, fluid dynamics, heat and mass transfer, and boundary-layer phenomena. The presence of one or more small perturbation parameters often produces sharp boundary or interior layers, making conventional numerical methods unreliable unless prohibitively fine meshes are employed. Nonstandard finite difference (NSFD) methods provide an effective alternative by modifying conventional difference approximations in a manner that better reflects the qualitative and asymptotic behaviour of the underlying differential equation. In particular, appropriately constructed denominator functions can incorporate perturbation parameters and characteristic layer behaviour directly into the discrete formulation, leading to stable and parameter-uniform approximations. This work presents an introduction to the fundamental principles and construction of NSFD schemes and discusses their application to one-parameter and two-parameter singularly perturbed problems. For one-parameter convection–diffusion and reaction–diffusion problems, NSFD discretizations are developed to accurately resolve steep boundary layers without requiring excessive mesh refinement. The framework is subsequently extended to two-parameter singularly perturbed problems, where the interaction between the perturbation parameters may generate different layer structures and substantially increase the numerical complexity. Both steady and time-dependent formulations are considered, with suitable time discretizations such as the Crank–Nicolson method combined with NSFD approximations in space. Particular emphasis is placed on the construction of parameter-dependent denominator functions, stability, consistency, discrete maximum principles, and parameter-uniform convergence. Representative numerical experiments illustrate the performance of NSFD methods for different perturbation regimes and demonstrate their ability to capture layer phenomena accurately. The study highlights NSFD as a simple, robust, and computationally attractive framework for the numerical treatment of a broad class of singularly perturbed one- and two-parameter differential equations.

Physics-Informed Machine Learning: PINNs, Their Failure Modes, and the Rise of Learned Operators

Chandhini G

PL

Dr. Chandhini. G, Department of Mathematical and Computational Sciences, NIT Karnataka, Surathkal, Karnataka - 575025

Physics-Informed Neural Networks (PINNs) promised a mesh-free route to solving partial differential equations by embedding governing physics directly into a network's loss function. Yet a substantial body of work has shown that PINNs can fail distinctively: the PDE residual loss converges to near-zero while the learned solution remains far from the true one, particularly for stiff, advection-dominated, or multi-scale systems. These failures are generally attributed to optimizing challenges, overfitting, and architectural limitations, and recent controlled comparisons show that no single explanation, nor its proposed remedy, fully resolves the phenomenon. Alongside the discussion on PINN, we also brief on another development in the direction of learned operators (DeepONet, for eg), where solution maps across families of PDEs are learned from data rather than minimizing a residual for a single instance. However, we argue that if physics constraints are reintroduced in the operator learning frameworks (Physics-Informed Neural Operators), similar residual-optimization difficulties can resurface.

Higher-Order Methods for Singularly Perturbed Semilinear Parabolic PDEs.

Kaushik Mukherjee

PL

Department of Mathematics, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram-695547, Kerala

In this talk, we focus on convergence analysis towards higher-order numerical approximations for a class of singularly perturbed semilinear parabolic PDEs exhibiting a regular boundary layer. Firstly, we discuss about two novel fitted mesh methods (FMMs) (the fully-implicit FMM and the implicit explicit FMM) followed by a post-processing technique for one-dimensional semilinear parabolic convection-diffusion problem. Next, we address two novel operator-splitting FMMs followed by a post-processing technique for cost-effective higher-order approximations of two-dimensional semilinear parabolic convection-diffusion problem with time-dependent boundary data.

Adaptive Fem For Optimal Control Problems Involving A Point Source Governed By Biharmonic Equation

Asha K. Dond*, Subham Nayak, Neela Nataraj

PL

*IISER Thiruvananthapuram, India

The adaptive finite element method (AFEM) is an elegant and powerful technique used to compute the numerical solution to differential equations with a minimal computational cost. Optimality analysis is essential to determine convergence rates and ensure the desired accuracy is achieved efficiently. In the talk, the quasi-optimality of adaptive nonconforming finite element methods will be discussed for optimal control problems involving Dirac measures governed by the biharmonic equation. The nonconforming Morley finite elements are employed for discretising both the state and adjoint variables. A modified right-hand side through a companion operator that maps Morley finite elements to a conforming space helps to overcome the challenge in handling point sources on the right-hand side. Optimal convergence rates for AFEM are established using an axiomatic framework: by proving key properties such as stability, reduction, discrete reliability, and quasi orthogonality. Numerical experiments will be discussed to validate the theoretical results.

Hybrid Numerical Schemes for Systems of Hyperbolic Conservation Laws

Rakesh Kumar

PL

Department of Mathematics, Ecole Centrale School of Engineering Mahindra University, Hyderabad-500043, India.

Systems of hyperbolic conservation laws arise in many scientific and engineering applications, where solutions may develop discontinuities in finite time. Classical high-order schemes often generate spurious oscillations near discontinuities, while non-oscillatory schemes such as WENO methods are computationally expensive. In this work, we develop an adaptive UFDM-WENO scheme for systems of hyperbolic conservation laws. The proposed method combines the high-order accuracy and non-oscillatory properties of WENO schemes with the efficiency of upwind finite difference methods. A troubled-cell indicator is used to detect discontinuities, allowing the scheme to adaptively apply WENO reconstruction near nonsmooth regions and use an efficient upwind discretization in smooth regions. Several one- and two-dimensional benchmark problems are presented to demonstrate the accuracy, stability, and computational efficiency of the proposed method.

AIML for Scientific Computing

Sri Vallabha Deevi, Director

PL

Data Science, Tiger Analytics

Artificial Intelligence Machine Learning have had a significant impact on various business areas involving human preferences. Their mathematical rigour coupled with flexible modeling approaches have enabled good prediction of purchases, music choices, reading habits and dietary preferences. Given their performance in predicting complex phenomena like human choices, practitioners have started exploring their utility in scientific computing. AI/ML methods are being applied to many areas of scientific computing where performance of traditional algorithms have reached a saturation. AIML methods bring in fast approximations that reduce the time to reach the same solution or enable finding better optima than the ones identified by traditional methods. Physics informed Neural Networks, Reinforcement Learning based Optimization etc. are examples of such methods. This talk will focus on application of AIML methods for Scientific Computing Applications and how AIML is leading to faster iterations and better designs.

Virtual Element Methods: Recent Developments and Applications

Sarvesh Kumar

PL

IIST Thiruvananthapuram, India

The Virtual Element Method (VEM) is a relatively recent numerical technique for solving partial differential equations on general polygonal and polyhedral meshes, without requiring the explicit construction of basis functions inside the elements. This talk presents the basic formulation and key features of VEM, with particular emphasis on consistency and stability. Applications to the Stokes and Navier–Stokes equations will be discussed, focusing on the stability and convergence of the proposed schemes. We will also consider poroelasticity problems, highlighting the suitability of VEM for coupled flow–deformation systems. If time permits, we will further discuss the discontinuous counterpart of VEM for solving hyperbolic conservation laws.

Radial Basis Functions and Its Achievements, Challenges, and a Path to Structure-Preserving Meshless Hyperbolic Schemes

Ankit Ruhi

IT

Department of mathematics and statistics, Dr Harisingh Gour Vishwavidyalaya

Mesh-based schemes remain the industrial standard for many PDEs, yet mesh generation and remeshing often dominate the cost—or become impractical—when the geometry is complex, evolving, or available only as a point cloud (free-surface and fluid–structure problems, fracture, imaging-based domains, and scattered observational data). Radial basis functions (RBFs) provide a natural, dimension-independent language for approximation on such scattered nodes and are already widely used in terrain modelling, geostatistics, image registration, surrogate modelling, and kernel methods. Among RBF routes to PDEs, radial basis function finite differences (RBF-FD) are particularly attractive for large-scale simulation: they retain the sparse, local character of classical finite differences while allowing irregular stencils, with polyharmonic splines augmented by polynomials offering high-order consistency without a shape parameter. Hyperbolic conservation laws, however, demand more than accurate differentiation. Generic RBF-FD operators need not be antisymmetric, so naive semi-discrete updates need not conserve mass or admit an energy estimate—deficiencies that become critical at shocks and contacts. This lecture introduces summation-by-parts (SBP) structure as a discrete analogue of integration by parts that can be imposed on the RBF-FD stencil graph, yielding pairwise-conservative flux formulations while preserving meshless sparsity. The resulting RBF-FD–SBP framework is presented as a structured option for meshless hyperbolic simulation: geometrically flexible, algebraically conservative, and compatible with standard Riemann fluxes and shock-capturing strategies. Representative tests (from smooth advection to Sod-type Riemann problems) and the remaining open issues—positivity, oscillation control, and point-cloud sensitivity—are discussed.

Mathematical Modelling and State Estimation of Battery Systems: From Nonlinear Dynamics to Physics-Informed AI

Geethi Krishnan

IT

Caterpillar, Bengaluru, India

Battery systems exhibit strongly nonlinear and time-varying behaviour influenced by state of charge, temperature, ageing, operating conditions, and electrochemical dynamics. Developing models that remain accurate across these conditions is therefore a challenging mathematical problem with direct relevance to the design and management of modern energy storage systems. This talk explores how nonlinear mathematical models can represent battery dynamics and enable estimation of internal states that cannot be directly measured. The role of observer-based methods for State of Charge (SOC) estimation and mathematical model-based approaches for State of Health (SOH) estimation will be discussed through practical battery applications. The talk will also examine the growing use of artificial intelligence in battery modelling and highlight the advantages of combining physical and mathematical models with data-driven methods, rather than relying solely on experimental data. Through these examples, the talk illustrates how nonlinear modelling, state estimation, numerical methods, and physics-informed AI provide a powerful mathematical framework for understanding, predicting, and managing complex battery systems.

ML Frameworks as Scientific Computing Infrastructure for CFD

Maruthi N. H.

IT

Siemens Digital Industries Software, Bengaluru, India

Traditional CFD codes are mostly written in Fortran, C, or C++. They usually need separate code and optimizations for CPUs, NVIDIA GPUs, and AMD GPUs. At the same time, machine-learning frameworks such as PyTorch and JAX have become powerful, device-agnostic platforms. This raises a natural question: can these frameworks also be used to build practical CFD software? This talk explores the idea with a simple 1D compressible Euler solver. We will implement the solver in pure Python using PyTorch and show that the same code runs on both CPUs and GPUs—without writing any CUDA or HIP kernels. We will also briefly compare the approach with JAX and highlight existing CFD and scientific-computing projects built on these frameworks. Because the numerical solver is expressed as differentiable tensor operations, the same code naturally supports automatic differentiation. This opens the door to gradient-based optimization, inverse problems, parameter estimation, uncertainty quantification, and physics-informed or data-driven modeling. As a simple demonstration, we will use the Euler solver itself to train a neural network that learns artificial viscosity—without first generating a separate training dataset. Finally, we will discuss how this programming model can scale beyond a 1D example, covering multi-dimensional solvers, distributed computing with MPI, and execution on accelerators. We will also examine the practical limitations of using ML frameworks for production CFD, including performance overheads, memory usage, challenges with distributed unstructured meshes, and the current maturity of the ecosystem.

From Equations to Energy: Mathematics Behind Power Electronics

Arun C

IT

Mercedes-Benz Research and Development India, Bangalore

Power electronics plays a central role in modern energy systems, enabling applications ranging from electric vehicles and renewable energy integration to battery storage and power grids. Behind the semiconductor switches and electrical circuits lies a rich mathematical foundation that allows these systems to be modeled, analyzed, controlled, and optimized. This talk explores selected mathematical concepts through practical examples from power electronics. Differential equations are used to describe the dynamic behavior of inductors, capacitors, and power converters, while the switching nature of semiconductor devices introduces piecewise and hybrid dynamical systems. Averaging techniques demonstrate how rapidly switching circuits can be transformed into simpler continuous models suitable for analysis and control. Fourier analysis provides a powerful framework for understanding pulse-width modulation, harmonic generation, and waveform quality. The talk will also illustrate the role of numerical methods in simulating power-electronic systems, where accuracy, computational effort, and widely different time scales present interesting challenges. Finally, optimization will be discussed in the context of converter design and operation, highlighting practical trade-offs between efficiency, size, losses, and performance. Through these examples, the talk aims to demonstrate how fundamental mathematical ideas move beyond theoretical formulations to become practical tools for controlling and optimizing the flow of electrical energy in modern engineering systems.

Contributory Talks

Numerical Simulation of Electroosmotic Maxwell Hybrid Nanofluid Flow over a Wedge using Triangular-Trapezoidal Fuzzy numbers

A.S. Ashwinth Jeffrey¹, M. Shanmugapriya^{2,*}, R. Sundareswaran¹

CT

¹Department of Mathematics, Sri Sivasubramaniya Nadar College of Engineering, Chennai - 603 110, India.

²School of Science and Humanities (Mathematics), Shiv Nadar University, Chennai - 603 110, India.

This research analyses how uncertainty in nanoparticle volume fractions affects the energy and mass transport in the visco-elastic non-Newtonian Maxwell fluid flow over a shifting wedge. The dual nanoparticle suspended mixtures are employed in automotive and energy systems, offering amplified support for heat conduction. The prominent nano-sized particles Fe_3O_4 & TiO_2 are incorporated in our numerical simulation, considering pivotal effects including electro- Magnetohydrodynamic & osmotic forces, thermophoresis, activation energy, thermal radiation and Brownian motion. With the above applications in mind, the uncertain volume fractions are studied as triangular and trapezoidal fuzzy numbers to analyse their upshots on drag force, heat & mass transport. In the numerical study the dimensionless variables are employed to simplify the governing differential equations which are then altered to fuzzy differential equations using the -cut method. The heat transfer rate of the present hybrid mixture is computed numerically and compared. The dual nanoparticle mixture exhibited improved fuzzy profiles for velocity, temperature and concentration. The electro- Magnetohydrodynamic & osmotic forces were found to significantly enhance the drag force, while the combined effects of Brownian motion and thermophoresis led to a notable increase in heat transfer. Upon evaluating the Triangular Fuzzy number and Trapezoidal Fuzzy number, it was observed that the Trapezoidal Fuzzy number effectively improved the accuracy and reliability of estimating the drag force, energy and mass transport. The dual nanoparticle mixture ($Fe_3O_4 + TiO_2/Blood$) emerged as the most thermally efficient, achieving an 8.15% higher heat transfer rate compared to ($Fe_3O_4/Blood$), a 5.82% improvement over ($TiO_2/Blood$) and a 6.91% increase compared to the blood.

References

1. Verma, L., & Meher, R. (2023). Numerical study on uncertain effects of shapes and heat transfer on MHD nanofluid flow with a fuzzy volume fraction. *Pramana*, 97(4), 155.
2. Alrihieli, H., Aldhabani, M. S., Alshaban, E., & Alatawi, A. (2025). Thermal- hydrodynamic analysis of a Maxwell fluid with controlled heat/mass transfer over a Riga plate: A numerical study with engineering applications. *Results in Engineering*, 104801.

ReturnReady: A Sensorized ACL Scaffold Digital Twin for Quantitative Rehabilitation Monitoring

Aarav Singhania and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Nearly one in four young athletes who return to sports after anterior cruciate ligament reconstruction sustain another ACL injury. This raises a critical question: how can an athlete know whether a recovering ligament or scaffold is mechanically prepared for the forces of running, cutting, and jumping before re injury occurs? Current rehabilitation assessments are expensive and primarily evaluate external performance at isolated clinical visits, providing limited information about how the recovering structure responds to repeated loading over time. In this work we develop a digital twin which simulates an ACL by employing a time-varying spring-mass-damper model for scaffold dynamics. In order to simulate healing the scaffold we utilize a ODE, $Q(t)$, which natural tissue regeneration. The stiffness and damping coefficient are both coupled to the healing equation which allows the model to dynamically change over time. In order to simulate our model we utilized a Finite Difference Method which predicted stiffness and damping and strain under a force profile, We generated synthetic data with sample forces and strain then utilizing a least squares recursive optimizer were able to update the parameters of stiffness and damping until it matched the synthetic data. A proof-of-concept benchtop acl tester was made which could pull on a acl scaffold unilaterally with different force profiles, it features a load force sensor and a strain gauge sensor which allowed us to validate our digital twin model on a physical prototype

Shock Dynamics in Saturn's Multi-Ion Plasma with Suprathermal Electrons: Impact of Water ions

Abijith Hariprasad¹, Kshama P Moosathu¹, Hethal Suresh¹, P S Abishek^{1,*}, Ann Maria George¹ and Manesh Michael¹ CT

¹Department of Physics, Bharata Mata College, Thrikkakara, Kochi, India: 682021

This study focuses on ion-acoustic shock waves in a suprathermal multi-ion plasma relevant to Saturn's magnetosphere. The plasma system considered here consists of hydrogen ions (H^+), water group ions (OH^+ and H_3O^+), along with Kappa (κ) distributed suprathermal electrons. Using the reductive perturbation method, the Korteweg–de Vries–Burgers (KdVB) equation is derived from the basic fluid equations together with Poisson's equation in order to study the nonlinear propagation of ion-acoustic shock structures. The obtained equation is solved analytically using the tanh method and the variation of the shock profile is analysed for different plasma parameters. It is observed that increasing OH^+ ion concentration and ion temperature enhance the shock amplitude, whereas higher H_3O^+ density, larger κ values and increased charge ratio reduces the strength of the shock structure. The viscosity parameter also plays an important role by modifying the dissipative nature and stability of the shock waves. Our work highlights the influence of suprathermal particles and water group ions on nonlinear wave dynamics in Saturn's magnetosphere, and this study may contribute to future investigations of planetary plasma environments and space exploration

References

- [1] P. S. Abishek and M. Michael, Ion-acoustic shock waves in a multi-ion plasma of Saturn's magnetosphere with suprathermal electrons and ions, *Physics of Plasmas*, 2025, Vol. 32, no. 4, p. 042301
- [2] M. F. Thomsen et al, Survey of ion plasma parameters in Saturn's magnetosphere, *Journal of Geophysical Research: Space Physics*, 2010, Vol. 115, A10220
- [3] S. Sultana, G. Sarri, and I. Kourakis, Electrostatic shock dynamics in suprathermal plas mas, *Physics of Plasmas*, 2012, Vol. 19, no. 1, p. 012310

Psyber Mentalytics: A Novel Deep Learning Algorithm to Predict Mental Health Risks Due to Social Media, and Mathematical Modeling of Variability Over Time

Achyutan Veeraraghavan and Padmanabhan Seshaiyer

CT

George Mason University

Approximately 95% of adolescents in the United States use smartphones and engage with at least one social media platform, with many reporting negative impacts on self-esteem, productivity, and overall mental well-being. As concerns about youth mental health continue to rise, there is an increasing need for predictive tools that identify individuals at risk while also modeling how mental health changes over time across a population. Existing approaches primarily assess current risk and often fail to capture the long-term effects of behavioral patterns and intervention strategies. This project develops an integrated framework that combines machine learning with mathematical modeling to predict and analyze mental health risks associated with social media usage. A Multi-Layer Perceptron (MLP) neural network was trained using an open-source Kaggle dataset containing social media usage behaviors and mental health indicators. The model predicts mental health risk using behavioral features including social comparison, restlessness, and sleep disruption. The trained model demonstrated strong predictive performance, achieving approximately 95% accuracy, precision, recall, and F1-score across multiple feature combinations and identifying a strong association between excessive social media use and elevated mental health risk. To extend these predictions beyond individual assessments, an SIR-inspired compartmental model of linear differential equations, called NLMH, is being developed to simulate transitions among no-risk, low-risk, medium-risk, and high-risk populations over time. In this formulation, escalation between risk compartments depends only on the source population size, modeling individual-level behavioral exposure without peer contagion effects. The model incorporates escalation rates, natural and assisted recovery, educational initiatives, and policy interventions as dynamic inputs. An accompanying quadratic optimal control framework, solved via the Forward-Backward Sweep algorithm, identifies intervention strategies that minimize societal and financial costs. By integrating artificial intelligence, mathematical modeling, and optimal control, this research provides a novel framework for both individualized risk prediction and population-level forecasting. The resulting decision-support tool has the potential to assist healthcare professionals, educators, and policymakers in designing evidence-based interventions that promote healthier social media engagement and improve long-term mental health outcomes. This project focuses on UN Sustainable Development Goal 3: Good Health and Well-Being.

A Computational Framework for Wildfire Prediction and Emergency Response Using Machine Learning and Differential Equations

Aditya Mohan and Padmanabhan Seshaiyer

CT

George Mason University

Wildfires are increasingly destructive natural hazards whose rapid and nonlinear progression presents significant challenges for early detection, emergency response, and resource allocation. While existing computational approaches can predict wildfire occurrence or simulate fire behavior, prediction alone does not necessarily determine how emergency resources should be deployed as conditions evolve. Furthermore, response effectiveness may change over time as resources become limited, firefighter fatigue accumulates, and evacuation demands increase. To address these challenges, this study develops an integrated computational framework combining data-driven wildfire prediction with a system of coupled ordinary differential equations to model the dynamics of wildfire progression and emergency response. A K-nearest neighbours (KNN) classification algorithm is first used to identify environmental conditions associated with wildfire occurrence using historical wildfire and environmental data. This predictive component is coupled conceptually with a compartmental mathematical model describing interactions among wildfire growth, suppression efforts, available emergency resources, firefighter fatigue, and evacuation dynamics. The resulting system of nonlinear ordinary differential equations provides a framework for examining how changes in response capacity and operational conditions influence wildfire outcomes over time. Mathematical analysis is performed to establish nonnegativity and boundedness of the model solutions, ensuring that the state variables remain physically meaningful throughout the simulations. Forward numerical simulations are then used to investigate the behavior of the system under different parameter values and response scenarios. A global sensitivity analysis based on Sobol indices is conducted to quantify the influence of model parameters and identify the factors that contribute most strongly to variations in wildfire and response outcomes. Results indicate that parameters governing wildfire growth and initial response effectiveness, including β and α , are among the most influential components of the system, while the model demonstrates comparatively lower sensitivity to variations in several fatigue-related parameters. Parameter estimation is additionally explored using data from multiple historical wildfire events to connect the mathematical model with observed wildfire behavior and assess its ability to represent different fire scenarios. By integrating machine learning, differential equations, sensitivity analysis, parameter estimation, and mathematical validation, this framework provides a quantitative approach for studying both wildfire risk and emergency response dynamics. The model may ultimately support the development of adaptive response strategies that improve resource allocation, anticipate operational limitations, and enable earlier intervention during rapidly evolving wildfire events.

Optimal Intervention Methodology to Advancing Quality Education Through Influencing Local Belief Dynamics

Amrutha Somireddy and Padmanabhan Seshaiyer

CT

George Mason University

Modern innovations and technological advancements are concealing how distant society is from its absolute potential. According to findings through Harvard University and Asian Development Bank, previously, 6.7% of adults globally held a bachelor's degree or higher leaving only this minimal population behind the existing rapid modernization. Education is correlated with not merely success as an individual but advancement of society as a whole; however, in the United States, a center of substantial industrialization, less than 40% of the population holds a college degree despite most citizens having access to attain one (Hanson, 2026). The data suggests that the issue is not access but a locality's belief in the value of education. The lack of educational belief across a locality's population and its government serves as a leading cause in the consistency of low educational attainment rates and the resulting loss of potential present in society. This study works to identify the appropriate optimal intervention approach needed through campaigning and awareness to influence societal belief and the resulting resource allocation to advance quality education while minimizing costs accordingly. Addressing United Nations Sustainable Development Goal 4, the model assesses the mutualistic interplay of standardized belief in the value of education, quality education itself, and resource distribution indices across multiple representative societies through analyzing well-posedness with Picard-Lindelöf Theorem, non-negativity, and equilibria with stability through the Routh-Hurwitz Criterion. Moreover, the model employs three-dimensional plot analyses to extend implications of the intervention methodology to multiple societies once more. Throughout, the study analyzes the pronounced efficacy of intervention in societies while determining the correct nonlinear intervention methodology needed by utilizing Pontryagin's Principle to both reduce total strain and optimize quality education and society's potential as a whole.

Statistics of Sunspot Cycles based on Multivariate Bootstrapping

Anilkumar B.T^{1,2,*} and Amaranathan Sabarinath³

CT

¹Department of Mathematics, College of Engineering Trivandrum, Thiruvananthapuram, Kerala, India

²Research Scholar, University of Kerala, Thiruvananthapuram, Kerala, India

³Vikram Sarabhai Space Centre, Thiruvananthapuram, Kerala, India

Sunspot cycle prediction is a fundamental activity for all spacefaring nations, as solar activity directly governs atmospheric density, drag forces, radiation levels, and communication reliability in near-Earth space. The consequences of inaccurate forecasts are tangible: underestimating solar activity shortens satellite lifetimes through enhanced atmospheric drag, increases radiation exposure for crewed missions, and complicates orbital debris management. Conversely, overly conservative predictions inflate mission costs and constrain design choices unnecessarily. Reliable, uncertainty-quantified sunspot cycle forecasts are therefore of direct operational value to mission planners, satellite operators, and space weather services. The current running cycle is Solar Cycle 25. While the 24 completed historical cycles share an approximate 11-year periodicity, each cycle is morphologically distinct in terms of amplitude, ascent duration, descent duration, asymmetry, and overall shape. This inherent variability makes it difficult to identify a single representative template for prediction, motivating a data-driven classification of historical cycles into homogeneous subgroups before any forecasting is attempted. **Fuzzy Clustering of Historical Cycles.** In our previous study [1], we applied the Fuzzy c-means (FCM) algorithm to classify all 24 historical sunspot cycles based on morphological pattern similarity. FCM differs from conventional hard-clustering methods in that it assigns each cycle a continuous membership value representing its degree of belongingness to each cluster, rather than forcing an exclusive assignment. This is particularly appropriate for sunspot cycle data, where cycles transition gradually between types and share overlapping characteristics. Our analysis identified two optimal groups: the Small Group, comprising cycles 1, 5, 6, 7, 12, 13, 14, 15, 16, and 24, and the Large Group, comprising cycles 2, 3, 4, 8, 9, 10, 11, 17, 18, 19, 20, 21, 22, and 23. The fuzzy cluster centers derived from this partitioning serve as representative morphological templates for each group and form the basis for predicting future cycles. Beyond classifying cycles by amplitude and shape, we examined the temporal ordering of group memberships across the historical record. This analysis revealed a structured periodicity in the recurrence of large and small cycle types—a pattern that does not appear to be random. Based on this periodicity, we classify Cycle 25 as a member of the Small Group, and project that Cycles 26 through 32 will also belong to the Small Group. If sustained, this projected sequence of smaller cycles would imply reduced overall solar activity levels for the coming decades, with practical implications for satellite design margins, orbital decay rates, and radiation shielding requirements. **Data and Preprocessing.** The analysis uses monthly averaged smoothed sunspot numbers obtained from the SILSO data repository at the Royal Observatory of Belgium. The smoothed values, produced by

applying a 13-month running mean to raw monthly counts, are the standard metric for characterizing cycle amplitude and evolution. Because historical cycles range in duration from approximately 9 to 14 years, each cycle was first normalized to a standard length of 132 months (11 years) via interpolation, ensuring direct comparability across cycles. To reduce computational complexity while preserving essential morphological structure, each normalized cycle was then represented by 15 strategically selected points capturing the cycle start, peak, end, and key inflection points along the ascending and descending phases. This compact representation retains the features most relevant to prediction—maximum amplitude, ascent length, descent length, and overall shape—while enabling efficient multivariate statistical analysis.

Bootstrap Uncertainty Quantification. To quantify the uncertainty in our predictions, we employed multivariate bootstrapping, a computer-intensive resampling method introduced by Efron [2]. Bootstrapping is well suited to the present problem: the historical record comprises only 24 complete cycles, making traditional parametric inference unreliable, and the underlying population distribution of sunspot cycles is unknown. The procedure generates a large number of pseudo-samples by drawing with replacement from the observed data, then computes the statistic of interest on each pseudo-sample. Aggregating across all replications yields an empirical sampling distribution from which standard errors, confidence intervals, and bias-corrected estimates are obtained without assuming a particular distributional form. Bootstrapping was performed separately on three datasets: the full set of 24 cycles, the Small Group (10 cycles), and the Large Group (14 cycles). For each dataset, we computed the mean cycle, median cycle, and mode cycle from every bootstrap replicate. The mean provides a measure of central tendency, while the median and mode offer greater robustness to outliers and skewed distributions. The resulting empirical sampling distributions yield confidence bands around each predicted cycle profile, providing a quantitative measure of forecast uncertainty that is directly usable in risk assessments for space mission planning.

Comparison with Monte Carlo Sampling. In parallel, we estimated prediction bands using a Monte Carlo approach, in which synthetic cycle profiles are generated by sampling from a parametric model fitted to the observed data. Comparing the two methods reveals a consistent finding: Monte Carlo sampling produces noticeably wider prediction bands than bootstrapping across all three datasets. This result indicates that the additional uncertainty introduced by parametric model assumptions—which may not accurately represent the true data-generating process—inflates the resulting bands. The bootstrap method, being non-parametric, adapts directly to the empirical data and produces tighter, more informative confidence estimates. For the purposes of operational forecasting, tighter bands are preferable provided they remain honest, and the bootstrap's ability to achieve this without parametric assumptions makes it particularly attractive for sunspot cycle applications where the sample size is small.

Summary and Outlook. This study demonstrates that combining FCM clustering with multivariate bootstrap resampling yields a robust and statistically rigorous framework for sunspot cycle forecasting. Clustering first identifies homogeneous subgroups among the historical cycles, ensuring that the reference template used for prediction is drawn from the most relevant subset of the data. Bootstrap resampling then provides honest, non-parametric uncertainty quantification

calibrated to the limited historical sample. The projected membership of Cycles 25–32 in the Small Group, if confirmed, suggests a period of moderate solar activity favorable to long-duration space missions, though operators should remain alert to unexpected deviations. Future work will focus on incorporating solar magnetic field observations and precursor- based indicators as additional inputs, and on extending the prediction framework to complementary solar activity indices—including solar radio flux (F10.7) and geomagnetic activity indices—that are of direct relevance to operational space weather forecasting.

References

- [1] Sabarinath, A., Anilkumar, A.K., A Stochastic Prediction Model for the Sunspot Cycles, *Solar Physics*, 255-265, 2011
- [2] Efron, B. Nonparametric estimates of standard error: The jackknife, the bootstrap and other methods. *Biometrika*, 68(3), 589-599, 1981
- [3] David H Hathaway, R.M.Wilson, E.J.Reichmann, The shape of the sunspot cycle, *Solar Physics*, 177-190, 1994
- [4] Du, Z. The shape of solar cycle described by a modified Gaussian function. *Solar Physics*, 273(1), 231-253, 2011
- [5] George J Klir, Bo Yuan, *Fuzzy Sets and Fuzzy Logic-Theory and Applications*, Prentice- Hall India, 2004
- [6] Zhanle Du, Predicting the shape of solar cycle 25 using a similar cycle method, *Solar Physics*, 295:134, 2020

Head-on Collision of Ion-Acoustic Solitons in Laser Produced Plasma with Kappa Distributed Hot Electrons and Boltzmann Cold Electrons

Ann Maria George^{1,}, S. Shilpa², P. S. Abishek¹ and Manesh Michael¹*

CT

¹Department of Physics, Bharata Mata College, Kochi, India.

²Department of Physics, Cochin University of Science and Technology, Kochi, India.

The head-on collision of two ion-acoustic solitary waves in a laser produced plasma is investigated using the extended Poincare–Lighthill–Kuo (PLK) method. The plasma model consists of degenerate hydrogen ions governed by quantum hydrodynamic (QHD) fluid equations, superthermal hot electrons obeying a kappa distribution, and cold electrons obeying a Boltzmann distribution. Employing the extended PLK method two-sided Korteweg Vries (KdV) equations governing the colliding solitons are derived, along with analytical expressions for the phase shifts arising from the collision. The results show that the amplitude of the solitons increases significantly during the interaction, producing a transient high-amplitude structure at the point of collision. Both the nonlinearity and dispersion coefficients are found to increase for lower hot-electron density and higher kappa index, indicating that a more superthermal hot electron population coupled with a smaller hot electron fraction enhances the nonlinear steepening and dispersive spreading of the solitons. Furthermore, the phase shift is found to increase for smaller values of the hot electron density and smaller hot electron temperature, highlighting the sensitivity of the collision dynamics to these parameters. These findings provide insight into the nonlinear wave dynamics of degenerate, quantum plasmas relevant to laser-plasma interaction experiments and inertial confinement fusion.

A Parameter-Uniform Time-Lag Linearization Method on a Shishkin Mesh for the Two-Dimensional Singularly Perturbed Burgers' Equation

Ansamol M A¹, Sona Santhosh.¹, and Ashish Awasthi¹

CT

¹Department of Mathematics, National Institute of Technology Calicut, Kerala, India

This paper develops an efficient parameter uniform numerical framework for the solution of the two-dimensional singularly perturbed Burgers' equation in the high-Reynolds-number regime. The nonlinear convective terms are treated using a time-lag linearization strategy, converting the original nonlinear problem into a sequence of linear problems that can be solved efficiently. An implicit Euler scheme is employed for temporal discretization, whereas the spatial discretization is performed using finite difference approximations on layer-adapted Shishkin mesh to accurately resolve the boundary layers. The proposed method is supported by a theoretical analysis establishing its consistency, stability, and convergence properties. Extensive numerical experiments are carried out to investigate the influence of the perturbation parameter and mesh refinement. The numerical results confirm that the proposed approach accurately captures steep solution gradients while maintaining robustness and computational efficiency, making it well suited for convection-dominated singularly perturbed Burgers' equations.

References

- [1] Miller J.J.H, O'Riordan, E., Shishkin, G. I. : Fitted numerical methods for singular perturbation problems. Revised. Error estimates in the maximum norm for linear problems in one and two dimensions, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ (2012)
- [2] M.K. Kadalbajoo, K.K. Sharma and A. Awasthi, A parameter-uniform implicit difference scheme for solving time-dependent Burgers' equations, *Applied Mathematics and Computation*, 170 (2005), 1365–1393.
- [3] A. Sreelakshmi, V.P. Shyaman and A. Awasthi, A robust numerical method for singularly perturbed Burgers-type problems, *Applied Numerical Mathematics*, 197 (2024), 71–96.
- [4] Richard E. Bellman and Robert E. Kalaba, *Quasilinearization and Nonlinear Boundary-Value Problems*, American Elsevier Publishing Company, INC. New York 1965

Clifford Algebra Formulation of Quantum Iterated Function Systems

Anupama K¹, Sasi Gopalan²

CT

¹Department of Mathematics, Cochin University of Science and Technology, Kochi, Kerala

²Department of Mathematics, Cochin University of Science and Technology, Kochi, Kerala
anupamakaiswarya1996@gmail.com, sgcusat@gmail.com

Quantum Iterated Function Systems (QIFS) extend the classical theory of Iterated Function Systems into quantum state spaces and provide a framework for studying complex quantum dynamics and self-similar structures. This work proposes a Clifford algebra formulation of Lorentz Quantum Iterated Function Systems for the generation and analysis of quantum fractals. The proposed framework is motivated by the correspondence between the Lorentz group and spin representations, where positive matrices in $SL(2, \mathbb{C})$ possess a dual interpretation as Lorentz boosts and quantum operators [2]. Through the isomorphism between $SL(2, \mathbb{C})$ and the spin group, Lorentz transformations induce Möbius transformations on the Bloch sphere (S^2), providing a natural mechanism for iterative evolution of quantum states. The study formulates Lorentz-induced QIFS using Clifford algebra and spinor representations, where quantum state evolution is described through probabilistic iterative mappings. Repeated application of these transformations generate self-similar quantum fractal structures on the Bloch sphere. The Clifford algebra formulation offers a geometric framework that can be used for studying nonlinear dynamical systems, conformal mappings, and fractal structures on spherical state spaces. Quantum fractal formation using Clifford algebra provides a geometric framework for extending Quantum Iterated Function Systems (QIFS) from the Bloch sphere to arbitrary n-dimensional spheres by employing the algebraic structure of Clifford algebras. Clifford algebra provides a unified algebraic framework in which geometric transformations, Möbius maps, Lorentz transformations and quantum-state operations can be represented consistently. Its ability to work naturally in arbitrary dimensions makes it particularly suitable for constructing QIFS on n-spheres and hyperbolic spaces.

References

- [1] Barnsley, M.F. Fractals Everywhere 2nd ed, Academic press professional, New York, 1993
- [2] A. Jadczyk, Quantum fractals From Heisenberg's Uncertainty to Barnsley's fractality, World Scientific Publishing Co. Pte. Ltd., 2014
- [3] A. Lozinski, K. Zyczkowski and W. Slomczynski, Quantum Iterated Function Systems Phys. Rev. E 68 (2003), 046110; <http://arXiv.org/abs/quant-ph/0210029>
- [4] A. Jadczyk, Quantum fractals on n-spheres, <http://arxiv.org/abs/quant-ph/0608117v2>
- [5] J. E. Gilbert and M. A. M. Murray, Clifford algebras and Dirac operators in harmonic analysis, (Cambridge University Press, Cambridge, 1991)

[6] Pierre Angles, Conformal Groups in Geometry and Spin Structures, Birkhauser, Progress in Mathematical Physics, Vol. 50 (2007) 2

Vaccinating the Cleanup Crew: Mathematical Modeling and Optimal Control of Bioaugmented Remediation

Arnav Krishnan, Mohan Parthasarathy and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Bioaugmented remediation uses pollutant-degrading bacteria to treat contaminated water, a low-cost alternative to chemical treatment. But many pollutants, like toluene, are toxic to the very bacteria meant to break them down. In species such as *Pseudomonas putida*, poisoned cells lyse, releasing both unmetabolized pollutant and a secondary toxin back into the system. We build a five-compartment ODE model coupling susceptible, intoxicated, and resistant degrader populations to pollutant and toxin levels, and prove that solutions stay nonnegative and bounded. Using the next-generation method, we derive a basic reproduction number that splits cleanly into two feedback pathways: pollutant recycling and toxin release. We show recycling alone is subcritical and can only initiate contamination, while the toxin pathway is what keeps it going; We prove the existence and uniqueness of an endemic equilibrium whenever the basic reproduction number is greater than one. Ablation experiments show that cutting the toxin loop clears the system, while cutting the pollutant loop only reduces total burden. The model exhibits persistent oscillations under baseline parameters and no dosing, meaning contamination keeps recurring without intervention. Applying Pontryagin's Maximum Principle with a forward-backward sweep algorithm, we identify an optimal early pulse dosing strategy for a protective agent, reducing total remediation cost compared with no intervention and continuous maximum dosing. These findings identify toxin-mediated feedback as the dominant driver of persistent contamination and provide quantitative guidance for designing efficient bioaugmentation strategies that support UN Sustainable Development Goal 6: Clean Water and Sanitation

Persistent Entropy of One-Dimensional Maps

Arya M V¹

CT

¹Department of Mathematics, APJ Abdul Kalam Technological University, Thiruvananthapuram, Kerala, India

Persistent entropy is a numerical invariant of a filtered topological space, defined from the lifetimes of persistent homology classes. It measures how evenly topological features are distributed across scales and has become a useful statistic in topological data analysis. This paper develops a rigorous framework for persistent entropy associated with one-dimensional dynamical systems. Given a continuous map $f : I \rightarrow I$ on a compact interval, we study filtrations naturally generated by orbit data, sublevel sets, and invariant sets, and we define the persistent entropy of such filtrations. We prove structural results showing stability under perturbations, monotonicity under refinement, invariance under topological conjugacy for dynamically natural filtrations, and asymptotic behavior for periodic, eventually periodic, and chaotic regimes. Special attention is given to piecewise monotone interval maps, where persistent entropy can be related to symbolic dynamics and growth of lap numbers. The results clarify how persistent entropy captures finite-scale dynamical complexity while remaining distinct from classical topological entropy.

A Graph Neural Network-Augmented Reinforcement Learning Framework for Adaptive Mesh Renement in Black–Scholes Option Pricing PDEs

Aswin V. S.¹

CT

¹School of Digital Sciences, Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram, Kerala, India

The Black Scholes partial differential equation (PDE) [1] and its extensions, including American early exercise problems, multi asset baskets, and stochastic volatility models, underpin derivative pricing and risk management. Closed form solutions exist only for the simplest European contracts. In practice, financial institutions solve the pricing PDE numerically under tight latency constraints for real time computation of prices and sensitivities (Greeks) across large option portfolios. The pricing PDE exhibits localised, evolving features that uniform meshes resolve poorly. These include a payoff kink at the strike price K , a sharp Gamma peak that migrates as time to maturity τ shrinks toward zero, and, for American options, a free early exercise boundary. Adaptive Mesh Refinement (AMR) is the standard remedy. However, existing AMR strategies rely on hand designed, reactive error indicators, such as the gradient recovery estimator of Zienkiewicz and Zhu [2], that refine only after error has already appeared. They also require manual recalibration for every new parameter combination (σ, r, K, τ) and, for multi asset baskets, they ignore correlations across the joint solution surface. This absence of a learned, transferable method indicates that no existing AMR framework has been formulated for financial PDEs. Prior graph neural or reinforcement learning AMR work, such as the swarm based approach of Freymuth et al. [3], targets only physics benchmarks such as diffusion, elasticity, and advection. We propose a Graph Neural Network (GNN) augmented Reinforcement Learning (RL) framework that recasts the AMR decision problem for Black Scholes type PDEs as a Markov Decision Process. The mesh at each time step is represented as a graph, with cells as nodes and shared boundaries as edges. We have adapted a message passing and graph attention policy so that it processes graphs of arbitrary and time varying size without any fixed dimensionality assumption. Node features combine the local option value, first and second order sensitivities (Delta, Gamma), a posteriori error indicator, and the contract parameters $(K, \sigma, r, \tau, \text{ and inter asset correlation})$, making the policy explicitly parameter conditioned. The agent selects refine, coarsen, or hold actions and is trained via Proximal Policy Optimisation with a reward balancing pricing error reduction against computational cost, across European, American, and multi asset basket benchmarks. This design enables anticipatory, budget aware refinement that is unavailable to reactive estimators. Building on strong precedent from GNN driven RL for AMR applied to scalar diffusion and elasticity PDEs [3], where learned policies have matched costly oracle refinement while using far fewer degrees of freedom than uniform meshes, we report preliminary architecture validation on the one dimensional European Black Scholes benchmark. The trained policy concentrates mesh nodes at the strike price kink and the peak Gamma region, consistent with the analytical solution, without any hand coded threshold. From this we observe

that the framework can identify the regions requiring refinement purely from learned representations. The full framework, trained across a distribution of contract parameters, is expected to generalise without retraining across option parameters, adaptively track the American free exercise boundary, and scale to multi asset baskets through attention that accounts for asset correlation. This would constitute the first learned and transferable AMR framework for computational finance.

References

[1] Black, F., Scholes, M., The Pricing of Options and Corporate Liabilities, Journal of Political Economy, 81(3) (1973), 637–654

[2] Zienkiewicz, O.C., Zhu, J.Z., A Simple Error Estimator and Adaptive Procedure for Practical Engineering Analysis, International Journal for Numerical Methods in Engineering, 24(2) (1987), 337–357

[3] Freymuth, N., Dahlinger, P., Wurth, T., et al., Swarm Reinforcement Learning for Adaptive Mesh Refinement, Advances in Neural Information Processing Systems (NeurIPS), 2023

Optimizing Educational Attainment: An Econometric Optimization Approach to Infrastructure Aid Allocation

Atharva Kalluraya and Padmanabhan Seshaiyer

CT

[George Mason University](#)

In developing countries and aid organizations, constrained infrastructure budgets require choices between investing in electrical grids or digital broadband. Motivated by UN SDGs 4 and 9, I modeled the relationship between utility access and educational attainment to support data-driven resource allocation. I merged datasets from Our World in Data tracking GDP per capita, electricity access, broadband penetration, and mean schooling years. Using KNIME, I built a multivariate linear regression model using File Reader, Row Splitter, Linear Regression Learner, and Numeric Scorer nodes to predict mean schooling years (Y) from electricity access (X_1) and broadband penetration (X_2), controlling for GDP per capita (X_3). The model ($Y = 5.12 + 0.042X_1 + 0.106X_2 + 0.000X_3$, $R = 0.63$) shows a one-percentage-point increase in broadband is associated with a larger change in predicted schooling years than electricity access. Although GDP displays a rounded coefficient of 0.000 due to its large raw scale relative to percentage points, retaining it as a control variable isolates baseline wealth effects. However, empirical data revealed a critical physical condition: broadband access never exceeded baseline electricity access ($X_2 \leq X_1$). Incorporating this "Foundation Rule," I built a Python solver in Google Colab that tests thousands of spending combinations to turn econometric estimates into practical aid strategies ensuring digital investments do not outpace power grids

Beyond Contagion: A Policy-Coupled Compartmental Model for Forecasting Heatwave Health Escalation

Athena Fleurquin and Padmanabhan Seshaiyer

CT

George Mason University

Heatwaves kill more people annually than floods, hurricanes, earthquakes, and wildfires combined, with heat deaths among adults over 65 up 85% since the 1990s, yet heat-health response still relies on retrospective mortality counts rather than forecasting models. Motivated by UN SDG 13.1 and 3.d, this work adapts infectious disease compartmental models, where escalation is governed by a threshold parameter R , to heatwave-driven public health impact. We propose a five-variable system (h, V, I, R, P) in which heat drives vulnerable individuals into an impacted state, impacted individuals leave through policy or recovery, and recovered individuals lose protection over time and re-enter the vulnerable pool, closing the system into a feedback loop, while policy itself grows with impact and relaxes as the crisis subsides. We establish well-posedness via ODE theory, derive the unique equilibrium, and rigorously re-derive the outbreak threshold via the next-generation matrix method. The equilibrium's Jacobian is non-hyperbolic; we resolve this by identifying an exact conservation identity and proving global, not merely local, asymptotic stability, confirmed numerically across diverse initial conditions. We formulate and solve, via Pontryagin's Maximum Principle and a forward-backward sweep, an optimal policy control problem, reducing peak impact 62.6% beyond reactive policy alone. A structural identifiability analysis reveals only 6 of 8 parameters are recoverable from impact data alone, precisely explaining a limitation encountered when fitting real mortality data. We benchmark a physics-informed neural network against classical estimation on real French mortality data, and independently validate a reconstructed excess-mortality series against Santé Publique France's official figure within 1.7%. Together these results demonstrate that epidemic threshold theory extends rigorously, and prescriptively, to climate-driven public health forecasting.

A Multi-Metric Wavelet Basis Selection Framework and its Application in Image Segmentation

Athira Vinay^{a,}, Sasi Gopalan^a, and Tessy Ninan^b*

CT

Department of Mathematics, Cochin University of Science and Technology, Kerala, India

^bDepartment of Electronics, Cochin University of Science and Technology

Wavelet transforms has become one of the important mathematical tools in image processing as it has the ability to simultaneously represent spatial and frequency information at multiple resolution [4, 9]. Wavelet transforms are employed in the JPEG 2000 image compression[11]. David and Donoho [?] proposed an effective denoising approach using wavelet transforms. Many deep learning models have also made use of wavelet transforms for different tasks. For example, incorporating wavelets into U-Net[10] for segmentation tasks can be beneficial as the inherent multi resolution and frequency localization properties of wavelet based models can improve the segmentation by preserving high frequency information and fine spatial details that are often lost during conventional pooling operations[5]. However, the effectiveness of wavelet-based models depends on the choice of wavelet basis. Different wavelets exhibit varying levels of compactness, regularity, vanishing moments, and spectral properties[4, 9], each of which can significantly influence the quality of extracted features. Coifman & Wickerhauser[3] introduced the entropy-based basis selection algorithm, selecting wavelet packet bases that minimize entropy, a measure of sparsity and information compactness, for optimal signal representation. While, Altay & Kalenderli[?] proposed a sub-band energy to entropy (SBETE) criterion, such that the preferred wavelet base maximizes the information-rich sub-band energy while minimizing entropy. Liu[7] proposed a wavelet packet basis selection approach that decomposes the signal into transient and residual subspaces, followed by individual basis selection for each, to enhance localized defect detection. Al-Kadi[1] proposed an adaptive best basis selection algorithm using the fractal dimension (FD) as the bases selection criterion. However in most existing studies, the choice of wavelet is fixed, or by domain expertise or by trial and error or chosen simply because it is computationally convenient[12, 8, 2, 6]. Hence the development of a multi-metric wavelet basis selection method is relatively underexplored. In this work, we address this gap by introducing a mathematically grounded multi-metric wavelet basis selection method. Our approach involves a comparative evaluation of candidate wavelet bases using a multi-metric scoring system that captures essential signal characteristics relevant to segmentation accuracy. But it's worth noting that in this work we are not actually seeking for a universally optimal wavelet, rather a representative basis for a collection of images. To evaluate the effectiveness of the proposed wavelet selection method, the selected wavelet basis is incorporated into a simple wavelet- enhanced U-Net architecture and validated on an image segmentation task.

References

[1] Al-Kadi O S, A fractal dimension based optimal wavelet packet analysis technique for classification of

meningioma brain tumours, in: 2009 16th IEEE International Conference on Image Processing (ICIP) (IEEE, 2009) pp. 4177–4180

[2] Chen D, Wu J, Zhang X Y and Wang D H, A frequency-spatial dual perception network for efficient and accurate medical image segmentation, *Sci. Rep.* 16 (2026) 7259

[3] Coifman R R and Wickerhauser M V, Entropy-based algorithms for best basis selection, *IEEE Trans. Inf. Theory* 38 (1992) 713–718

[4] Daubechies I, *Ten Lectures on Wavelets* (SIAM, Philadelphia, 1992)

[5] Li Y, Wang Y, Leng T and Zhijie W, Wavelet U-Net for medical image segmentation, in: *Artificial Neural Networks and Machine Learning – ICANN 2020* (eds I Farkas, P Masulli and S Wermter), *Lecture Notes in Computer Science*, Vol. 12396 (Springer, Cham, 2020) pp. 800–810

[6] Li Q and Shen L, WavesNet: Wavelet integrated deep networks for image segmentation, in: *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)* (Springer Nature Switzerland, Cham, 2022) pp. 325–337

[7] Liu B, Selection of wavelet packet basis for rotating machinery fault diagnosis, *J. Sound Vib.* 284 (2005) 567–582

[8] Liu P, Zhang H, Zhang K, Lin L and Zuo W, Multi-level wavelet-CNN for image restoration, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops* (2018) pp. 773–782

[9] Mallat S, *A Wavelet Tour of Signal Processing* (Elsevier, 1999)

[10] Ronneberger O, Fischer P and Brox T, U-Net: Convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (eds N Navab, J Hornegger, W M Wells and A F Frangi), *Lecture Notes in Computer Science*, Vol. 9351 (Springer, Cham, 2015) pp. 234–241

[11] Skodras A, Christopoulos C and Ebrahimi T, The JPEG 2000 still image compression standard, *IEEE Signal Process. Mag.* 18 (2001) 36–58

[12] Wu J, Li J, Yang J and Mei S, Wavelet-integrated deep neural networks: A systematic review of applications and synergistic architectures, *Neurocomputing* 657 (2025) 131648. 2

Computational Neuroscience and Machine Learning Analysis for Understanding Sex-Specific Differences in Microglial Morphology and Cognitive Function in Alzheimer's Disease

Avni Sriram and Padmanabhan Seshaiyer

CT

George Mason University

Alzheimer's disease (AD) disproportionately affects women, with females accounting for approximately two-thirds of all dementia cases (Rajan et al., 2021). Despite this striking disparity, the influence of biological sex on AD pathology and behavioral outcomes remains underexplored (Kolahchi et al., 2024). Microglia—the brain's resident immune cells—are critical for synaptic regulation, and their morphology reflects shifts in functional state. Therefore, detailed morphological analysis offers critical insight into how microglia influence AD pathology. In this study, we performed a computational investigation quantitatively comparing hippocampal microglial morphology and cognitive performance in female and male 5xFAD mice, a transgenic AD model, and their wild-type controls. Leveraging digitally reconstructed cells from NeuroMorpho.org, we analyzed >6,000 microglial cells (AD n=4,834; Control n=1,176) using ten morphometrics; we further integrated 5-Choice Serial Reaction Time Task (5CSRTT) data from MouseBytes, an open-access rodent touchscreen database (Female: AD n=394, Control n=1289; Male: AD n=598, Control n=1028) to examine behavioral and morphological changes in 5xFAD mice during gender stratification. Except for the number of processes, all morphometrics differed significantly between AD and Control groups (Welch's t-test, $p < 0.0001$). Principal component analysis (PCA) identified soma size and branching-based metrics as top contributors to PCA axes. Supervised models (random forest, gradient boosting; 5-fold CV) classified cells with 96% accuracy, while SHapley Additive ex Planations (SHAP) identified segments, soma radius, and segment length as key features in classification. Gender stratification revealed stronger AD-related effects in females (e.g. ~44% decrease in bifurcations and sections from control to AD), while males showed smaller changes across the same features. Analysis of 5CSRTT data revealed that males had larger differences in average threshold accuracy and average threshold condition, while females showed significantly larger differences in total perseverative correct responses (~34% decrease). ANOVA revealed that group (AD vs. Control) accounted for most variance in both morphology and behavioral metrics. Together, these results indicate that AD alters microglial morphology and cognitive performance in a sex-dependent manner, with females showing greater structural differences and both males and females exhibiting distinct behavioral deficits. These findings highlight the importance of considering sex in AD research and may inform the development of targeted, sex-specific interventions.

Multicriteria Approaches to Maximizing Static Multicommodity Network Flows

Bishwa Raj Adhikari^{1,}, Hari Nandan Nath², Tanka Nath Dhamala³*

CT

^{1,*}Department of Mathematics, Prithvi Narayan Multiple Campus, Tribhuvan University, Pokhara, Nepal.

²Department of Mathematics, Bhaktapur Multiple Campus, Tribhuvan University, Kathmandu, Nepal.

³Central Department of Mathematics, Tribhuvan University, Kathmandu, Nepal.

The maximum static multicommodity flow problem is a fundamental network optimization problem in which multiple commodities compete for limited shared network capacities. Although classical approaches maximize the overall network throughput, but may not adequately address the conflicting objectives associated with individual commodity flows. This paper develops a multicriteria optimization framework for the maximum static multicommodity flow problem to determine the set of nondominated (Pareto optimal) solutions for the maximum static multicommodity flow problem. The proposed approach uses maximum single commodity flow value at each sink to construct appropriate bounds and employs an ϵ -constraint based algorithm to generate nondominated solutions. Computational experiments on networks with multiple source sink pairs demonstrate the correctness and effectiveness of the proposed approach and show that it successively recovers the complete Pareto frontier.

An Optimal Control Framework for Adaptive Virtual Reality Interventions in Drug Rehabilitation

Charlotte Cheng and Padmanabhan Seshaiyer

CT

George Mason University

Substance use disorders remain a significant public health challenge, with relapse prevention continuing to be one of the central difficulties in rehabilitation. Traditional rehabilitation often faces barriers such as limited accessibility and difficulty recreating real world environments associated with cravings. As a result, researchers have begun to study virtual reality as a tool for drug rehabilitation. We introduce an Adaptive- Perturbed-Corrective-Relapse (APCR) compartmental model representing transitions between stable recovery, cue-induced perturbation, corrective intervention, and relapse states with an optimal control framework to model and optimize VR aided drug rehabilitation. We formulate the VR intervention intensity as a dynamic control variable $u(t)$, allowing the timing and magnitude of therapeutic exposure to be optimized. Using Pontryagin's Minimum Principle, we derived an optimal intervention strategy that balances reduction of cue induced perturbation against the cost of VR implementation across a 60-minute period. We developed a baseline model that incorporates behavioral biomarkers collected from VR simulations, including gaze duration toward drug-associated cues, to parametrize time-dependent relapse triggers. Numerical solutions obtained via a forward backward sweep method demonstrate the effectiveness of the optimal control strategy. This demonstrates the potential of mathematical optimization in the design of personalized VR rehabilitation while supporting the broader objective of improving health and wellbeing aligned with sustainable goal number three of the United Nations.

DHFLatTop For Stock Market Analysis

Ms. Deepa V and Dr. Babitha K V²*

CT

*Research Scholar, PG Dept & Research Centre of Mathematics, St. Joseph's College(Autonomous), Calicut University, Kozhikode, Kerala, India

²Assistant Professor, Department of Mathematics, Govt. College Madappally, Kozhikode, Kerala, India

Stock market investment involves uncertainty because different investors often have different opinions about the same stock. Some investors may believe that a stock will perform well, while others may think it will perform poorly. In many cases, investors are also unsure about their own opinions. Traditional fuzzy set models represent uncertainty using only one membership value and one non-membership value, which cannot fully describe such situations. Dual Hesitant Fuzzy Sets (DHFS), introduced by Zhu et al. [3], overcome this limitation by allowing multiple possible membership and non-membership values at the same time. This provides a more realistic way to represent hesitation and uncertainty in financial decision-making. Motivated by the need for a mathematically unified framework for uncertain financial decision-making, this paper introduces the category of Dual Hesitant Fuzzy Lattice Topological Spaces (DHFLatTop), which integrates dual hesitant fuzzy sets, lattice theory, topology, and category theory. In the proposed framework, each object is a dual hesitant fuzzy lattice-topological space, while morphisms are continuous lattice-preserving mappings representing information transformations such as aggregation, risk adjustment, ranking, and decision-making. We establish the categorical structure of DHFLatTop by defining its objects, morphisms, identity morphisms, and composition, thereby proving that DHFLatTop forms a well-defined category. The paper further develops the mathematical foundations of the framework by proving that the composition of continuous lattice morphisms is again a continuous lattice morphism. We also establish a characterization theorem showing that a mapping between DHFLatTop spaces is a continuous lattice morphism if and only if it preserves finite lattice joins and meets together with the underlying topological continuity condition. To model practical decision-making processes, we introduce matrix-induced morphisms and prove that positive matrix transformations satisfying suitable lattice-preserving assumptions define continuous lattice morphisms. These results provide a rigorous categorical foundation for sequential information processing and decision aggregation under uncertainty. A key contribution of this work is the introduction of a contravariant functor that connects dual hesitant fuzzy topological spaces with their corresponding lattice structures. This mathematical relationship allows continuous mappings between stock market structures to preserve important properties during analysis. In particular, the Alexandrov subcategory provides a complete logical structure where investment criteria can be combined naturally without requiring every possible combination to be defined in advance. This makes portfolio analysis and investment decision-making more flexible while maintaining mathematical consistency. The applicability of DHFLatTop is demonstrated through sev-

eral stock market problems, including stock classification, portfolio selection, volatility prediction, and trend clustering. In particular, expert opinions are represented as dual hesitant fuzzy objects, lattice operations aggregate hesitant information without loss of uncertainty, and matrix- induced continuous morphisms model expert aggregation, risk adjustment, and sector- based ranking as a single compositional decision pipeline. The framework also provides a common categorical setting in which several existing hesitant fuzzy decision-making approaches—including probabilistic hesitant fuzzy models [2], Grey Relational Analysis [1], cloud models, hesitant probabilistic linguistic methods, and neutrosophic soft set approaches—can be interpreted and systematically compared whenever they satisfy the required morphism properties. By combining dual hesitant fuzzy theory with lattice topology and category theory, DHFLatTop provides a mathematically consistent and extensible framework for uncertain information processing. The proposed framework not only preserves hesitant information throughout the decision-making process but also offers a common theoretical foundation for the analysis, comparison, and extension of fuzzy decision- making methodologies, with potential applications in financial modelling, engineering optimization, intelligent decision support, and uncertainty quantification.

References

- [1] Yang, S., & Ju, Y. A GRA method for investment alternative selection under dual hesitant fuzzy environment with incomplete weight information. *Journal of Intelligent & Fuzzy Systems*, 28(4) (2015), 1533-1543
- [2] Zhou, W., Liu, M., Xu, Z., & Skare, M. . Investment decision making based on the probabilistic hesitant financial data: model and empirical study. *Economic research-Ekonomska istrazivanja*, 34(1), (2021), 2812-2832
- [3] Zhu, B., Xu, Z., & Xia, M. Dual hesitant fuzzy sets. *Journal of Applied mathematics*, 2012(1) (2012), 879629

Maximum Fuzzy Flows in Lossy Network

Dipak Babu Amgain¹, Durga Prasad Khanal², and Tanka Nath Dhamala³

CT

^{1,3}Central Department of Mathematics, Tribhuvan University Name, Kathmandu, Nepal

²Saraswati Multiple Campus, Tribhuvan University Name, Kathmandu, Nepal

Fuzzy set theory, introduced by Zadeh [1], can help to model real-world network flow problems by handling uncertain and imprecise arc capacities with fuzzy numbers. We study the fuzzy maximum flow problems in lossy network by adopting a fuzzy arithmetic framework of Dubois and Prade [2] using LR representation. We extend the recent work of Dhamala et al. [3] from crisp to fuzzy arc capacities and propose models and algorithms by generalizing crisp idea to fuzzy. We also integrate priority-based storage of excess flow at intermediate nodes. The correctness theorems are proved and numerical validations are presented with small network examples.

References

[1] L. A. Zadeh, Fuzzy sets, Information and Control, 8(3) (1965), 338–353

[2] D. Dubois and H. Prade, Operations on fuzzy numbers, International Journal of Systems Science, 9(6) (1978), 613–626

[3] T. N. Dhamala, M. C. Adhikari, D. P. Khanal, and U. Pyakurel, Generalized maximum flow over time with intermediate storage, Annals of Operations Research, 335(1) (2024), 111–134

Budget constrained maximum flow with speed variation and intermediate storage

Durga Prasad Khanal, Tanka Nath Dhamala*

CT

Because of the severe traffic congestion in urban areas, organising facilities to those who are at risk is crucial challenges during evacuation. The aim of this presentation is to focus on the techniques to increase the number of evacuees by reducing traffic congestion via optimal use of limited budget and enhancing speed adjustment. From a set of bottleneck arcs, most appropriate arcs are first identified. The budget is then efficiently assigned to the selected arcs to boost their capabilities within specified constraints. The remaining arcs are then updated with speed adjustments to shorten the trip time. By placing evacuees in intermediate shelters, for people who might not possible to reach their ultimate destination because of network capacity or allowable time window restrictions, this concept increases the flow out from the risk zone. This innovative strategy helps to improve traffic management during crises, special events, and rush hour. The suggested models are used in a network of a part of the Kathmandu Valley to show their effectiveness.

Network Capacity Expansion Minimizing Total Cost

Giri Raj Paneru^{1,}, Hari Nandan Nath², Tanka Nath Dhamala³*

CT

¹Department of Mathematics, Trichandra Multiple Campus, Tribhuvan University, Kathmandu Nepal

² Department of Mathematics, Bhaktapur Multiple Campus, Tribhuvan University, Nepal

³Central Department of Mathematics, Tribhuvan University, Nepal

In a minimum-cost flow problem, the flow is sent through a network from supply nodes to demand nodes, minimizing the total cost of the flow without violating the capacities of the edges. There can be situations in which capacities of edges need to be increased to meet demand and supply requirements, or in which the total cost can be further minimized by increasing edge capacity. This work addresses the problem of minimizing the total cost of flow when both per-unit edge flow costs and capacity enhancement costs are taken into account. The study also identifies the set of edges in which capacity needs to be improved. We consider two cases: one with the linear capacity enhancement cost, which is formulated as a linear program, and the other with the non-linear capacity enhancement cost, which is formulated as a non-linear programming problem. Considering the non-linear enhancement cost as a piecewise linear cost, both problems are shown to be equivalent to a convex cost flow problem, based on which we propose polynomial time algorithms to solve the problems. Computational experiments on randomly generated network instances indicate that approaches with non-polynomial complexity perform efficiently on small problem sizes, while methods with stronger theoretical guarantees exhibit better performance for large network sizes with a large number of linearization pieces.

A Coupled Dynamical Systems Model of Climate Stress, Technology Adaptation, and Corn Yield

Grace Gao and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Climate change is intensifying heat and drought stress on U.S. corn production, and it remains unclear whether technological adaptation is keeping pace with that stress. This study develops a coupled system of three difference equations—modeling corn yield, accumulated climate stress, and technology adoption—to quantify the equilibrium relationship between climate stress and adaptive capacity across five major U.S. Corn Belt states. The model is calibrated using real historical climate records, corn yield statistics, and biotechnology-adoption data, with heat stress measured as degree-days above a fixed biological threshold (29°C) and drought stress measured using a standardized precipitation index (SPI). The technology-adaptation component models adoption as a saturating growth process, with adoption rate driven by both baseline growth and climate stress, approaching a fixed ceiling. Fit directly to real biotechnology-adoption data, this component fits well across all five states and shows that adoption has already nearly saturated in most of them. We derive closed-form steady states for all three variables and find a bifurcation in the yield equilibrium: once combined heat, drought, and accumulated-stress damage exceed the system's baseline growth rate, the nontrivial equilibrium doesn't shrink gradually but stops existing altogether, leaving zero yield as the only solution. Using real fitted parameters, we show that observed historical climate remains well below this threshold in most states. Sensitivity analysis shows that uncertainty in technology's effect on the yield ceiling often outweighs the full range of historical climate variability in its effect on equilibrium yield. We also distinguish which parts of the stress-accumulation mechanism are reliably identified from the data and which remain weakly constrained, using profile-likelihood scanning and out-of-sample cross-validation. These results indicate that technological adaptation, not climate variability alone, may be the dominant lever determining long-run corn yield stability in several Corn Belt states

Noise-Dependent Oversampling Trade-Offs in ESPRIT-Based Prony Recovery of Exponential Sums

Hari Rajagopal

CT

Faculty of Computer Science and Mathematics, University of Passau, 94032 Passau, Germany

Finite exponential sums model signals in spectroscopy, radar, and biomedical imaging. Prony-type methods recover their poles and amplitudes through structured Hankel and Vandermonde computations [1, 2]. With noisy samples, both the signal-subspace estimate and the Vandermonde amplitude-recovery stage can be sensitive to perturbations [3]. This work quantifies oversampling in an ESPRIT-based Prony pipeline. For an n -component signal, the factor ρ produces a $(\rho n + 1) \times (n + 1)$ Hankel matrix from $N = n(\rho + 1) + 1$ equally spaced samples. Singular value decomposition and ESPRIT estimate the poles [4], followed by a column-scaled Vandermonde least-squares solve for the amplitudes. A four-component damped signal is tested for $\rho \in 1, \dots, 10$ under real Gaussian noise with $\sigma \in [0, 0.5]$. Each noise level uses 1000 paired trials with common noise prefixes across ρ . Performance is measured using relative parameter errors, the condition number $\kappa(H)$ of the full noisy Hankel matrix, and fixed-window reconstruction RMSE against both clean and noisy data. Across the tested nonzero noise levels, the best factor reduces the mean exponent error by approximately 29 – 55% relative to $\rho = 1$. It shifts from $\rho = 8$ at $\sigma = 0.056$ to $\rho = 3$ at $\sigma = 0.5$, while $\rho = 3 - 6$ is best at intermediate noise levels. Mean amplitude errors decrease by approximately 17–29%, with minima at $\rho = 3$ or 4. For every nonzero noise level, mean $\kappa(H)$ decreases as ρ increases, even as the amplitude error rises beyond its minimum at moderate ρ . This empirical conditioning paradox separates the conditioning of the noisy Hankel matrix from the stability of the full estimation pipeline. The amplitude degradation is consistent with continuing sensitivity in the subsequent Vandermonde solve. On the common first $2n+1$ samples, clean-signal RMSE is minimized at $\rho = 3 - 5$, whereas the residual against noisy observations is smallest at $\rho = 1$. Thus a smaller data residual need not imply more accurate recovery of the underlying signal. Experiments for $n = 6$ and $n = 8$ show the same qualitative benefit for exponent recovery, while amplitude improvements plateau at larger ρ . Together, these findings support moderate, noise-dependent oversampling for the tested settings and identify a trade-off among parameter accuracy, data t , and Hankel conditioning that cannot be characterized by $\kappa(H)$ alone.

References

- [1] D. Potts and M. Tasche, "Parameter estimation for exponential sums by approximate Prony method," *Signal Processing*, vol. 90, no. 5, pp. 1631–1642, 2010
- [2] J. M. Peña and T. Sauer, "Stabilized recovery and model reduction for multivariate exponential polynomials," *Journal of Symbolic Computation*, vol. 125, article 102313, 2024

[3] W. Gautschi, "Norm estimates for inverses of Vandermonde matrices," *Numerische Mathematik*, vol. 23, pp. 337–347, 1974

[4] R. Roy and T. Kailath, "ESPRIT—estimation of signal parameters via rotational invariance techniques," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 7, pp. 984–995, 1989

Dynamical Systems Analysis of Diffusively Coupled FitzHugh–Nagumo Neural Networks: Synchronization, Stability, and Bifurcation on Weighted Graphs

Hrishikesh Shanmuganathan and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Understanding how individual neuron dynamics translate into collective network behavior remains a fundamental challenge in computational neuroscience. Analytical understanding of network dynamics is therefore often limited by assumptions regarding behavior or network topology, with many studies relying largely on numerical simulation. Abnormal neuron synchronization patterns are characteristic of several neurological disorders, such as epileptic seizures. Unlike existing analyses that rely on specific network architectures, we develop a topology-independent analytical framework for studying diffusively coupled FitzHugh–Nagumo networks on general weighted undirected graphs. Primarily using dynamical systems theory, we derive conditions for bifurcations—which characterize transitions between qualitatively distinct neuronal behaviors—of individual neurons with respect to model parameters and external current. We then study synchronous equilibria of the coupled n -neuron system, analytically deriving the parameter regions in which these equilibria are stable. We classify local single-parameter bifurcations of these equilibria in terms of eigenvalues of the graph Laplacian matrix, establishing a relationship between network structure and synchronous behavior. Using these Laplacian eigenvalue conditions, we derive upper bounds on maximum weighted degree for equilibrium stability and bifurcation regimes. Finally, we develop a generalized framework for extending stability and bifurcation analysis beyond fully synchronized states to more complex collective behaviors, including partial synchronization and non-synchronized states. These results provide a mathematically rigorous framework for understanding how connectivity patterns shape collective neural behavior and may inform future studies of abnormal synchronization associated with neurological disorders. This work aligns with the United Nations Sustainable Development Goal #3: Good Health and Well-Being

Wiener index and Matrix Methods in Optimal network design

Jicy N, Krishnaveni P and Shyama M P

CT

Distance matrices encode complex network connections into a single comprehensive grid to shortest paths. The k-open neighbourhood and k- closed neighbourhood matrices is a Boolean threshold variant of the classical distance matrix which will determine whether vertex pairs lie within k hops. Summing the entries yields k-Wiener index, $W_k(G)$, a flexible topological metric that ranges from total edge count to overall pair count at k. This matrix and Wiener index optimize network design by identifying hub locations and measuring overall connectivity without requiring full shortest path calculations

References

- [1] Andrey A Dobrynin, Ivan Gutman, Sandi Klavzar, Petra Zigert, Wiener index of hexagonal systems, *Acta Applicandae Mathematicae*. 72(2002), pp. 247-294.
- [2] Shyama M P, k- closed nbd and k-open nhd matrices of graphs and their energy, *International Journal management applied science*, Vol 11(2025), pp. 48-52
- [3] Sandi Klavzar, Ivan Gutman, Wiener number of vertex weighted graphs and a chemical application, *Discrete Applied mathematics*, 80 (1997), pp. 73- 81

On Topological Indices of Ideal Sumset Labelled Signed Graphs

Jincy P Mathai^{1,2,3,*} and Vinodkumar P B^{2,3}

CT

¹Vidya Academy of Science and Technology, Thrissur, Kerala, India

²Muthoot Institute of Technology and Science, Kochi, Kerala, India

³APJ Abdul Kalam Technological University, Thiruvananthapuram, Kerala, India

In this paper, we introduce degree-based topological indices for ideal sumset labelled signed (ISLS) graphs. Signed Zagreb, Forgotten, Sombor, and Harmonic indices are defined in terms of the signed degrees of the vertices. Some sharp bounds for these indices in terms of standard graph parameters are calculated. A characterization of the extremal ISLS graphs which attain these bounds are established. A comparison of various topological indices are also studied in the paper.

References

- [1] J. P. Mathai, N. Sudev and S. Sreedharan, (2021) On Ideal Sumset Labelled Graphs, *Proyecciones(Antofagasta, On line)*Vol.40, no.2, pp. 371-384
- [2] Juan Rada, Jos'e Manuel Rodr'guez, Jose M. Sigarreta, General properties on Sombor indices, *Discrete Applied Mathematics*,299(2021), 87-97
- [3] Tamas Reti, Tomislav Doslic, Akbar Ali, On the Sombor index of graphs, *Contributions to Mathematics*,3(2021), 11-18
- [4] J. B. Liu, M. M. Mateji'c, E. I. Milovanovi'c, I. Z. Milovanovi, Some New Inequalities for the Forgotten Topological Index and Coindex of Graphs, *MATCH Commun. Math. Comput. Chem.* 84 (2020) 719-738
- [5] Sudev Naduvath, Johan Kok, On certain topological indices of signed graphs, *MATH.GM*,(2020)
- [6] Lingping Zhong, The harmonic index for graphs, *Applied Mathematics Letters*,25(2012), 561-566

Targeted Containment vs. Structural Breakdown: Assessing Centrality Metrics for Epidemic Control in Real-World Networks

Junicca Anna George Matson¹, Sherin K P^{2,}, and Dhanyamol M V²*

CT

¹PG & Research Department of Mathematics, St. Peter's College Kolenchery, Ernakulam, India

²Department of Computational Science and Humanities, Indian Institute of Information Technology Kottayam, India

In a world where viruses and diseases spread rapidly, effective vaccination strategies are mandatory in the wake of an outbreak. The varying nature of connected networks demands tailored strategies for vaccination to be successful. This study evaluates how different centrality strategies like Degree, Betweenness, Closeness, and Eigenvector impact epidemic control when applied across three distinct compartment models (SIR, SIS, and SEIR) on real-world complex networks. Additionally, the paper utilizes Largest Connected Component (LCC) decomposition to assess how removing central nodes physically fragments the network to suppress disease propagation. The performance of these strategies is tested across three foundational epidemic frameworks. The Susceptible-Infected-Recovered (SIR) model, the Susceptible-Infected-Susceptible (SIS) model without permanent immunity, and the Susceptible-Exposed-Infected-Recovered (SEIR) model, which incorporates an incubation phase [3]. These dynamics were simulated across three distinct real world networks: the Dolphin Network (a dense, small scale social structure with 190 nodes, 1.1K edges, and a high clustering coefficient of 0.6588), the Crime Moreno Network (a sparse, low clustering network containing 829 nodes and 1.5K edges), and the Email Univ Network (a larger social dataset with 1.1K nodes, 5.5K edges, and moderate clustering) [1,2]. In the dense Dolphin Network, high overlap among centrality metrics is observed, which means vaccinating top degree nodes yields similar results to other strategies in SIR and SIS dynamics. However, Eigenvector Centrality achieves superior infection control in the SEIR model by keeping exposed and infectious populations lowest, while Betweenness Centrality proves most effective for physical network breakdown in LCC decomposition. In contrast, the sparse Crime Moreno Network sees Closeness Centrality dominate, keeping infection counts near zero and driving the steepest decline in LCC size, whereas Betweenness Centrality offers the strongest containment under SIS dynamics. For the larger Email Univ Network, Degree, Eigenvector, and Closeness centralities perform similarly in SIR models, but Betweenness and Eigenvector centralities stand out as most effective for controlling SEIR outbreaks and fragmenting the network physically [4]. The experimental results show that there is no single universal centrality strategy that outperforms across all network topologies and disease models. The optimal intervention depends heavily on specific network properties and epidemiological parameters such as transmission, recovery, and incubation rates. In dense networks, Eigenvector and Betweenness centralities excel at targeted containment and structural breakdown, while in sparse networks, Closeness Centrality is the dominant choice for SIR and SEIR dynamics. These findings

highlight that effective epidemic control cannot rely on simple degree-based interventions alone; public health strategies must tailor their centrality-based targeting to the specific structural properties of the contact network and the unique biological parameters of the spreading pathogen.

References

- [1] R.A.Rossi, N.K.Ahmed, The network data repository with interactive graph analytics and visualization in AAAI, 2015. URL<https://networkrepository.com>
- [2] R. Guimera, L. Danon, A. Diaz-Guilera, F. Giralt, A. Arenas, Self similar community structure in a network of human interactions, *Physical Review E* 68 (6) 065103, 2003
- [3] Eshtewy, N. A., Forootani, A., & Sisi, Z. A, Survey on mathematical modeling of infectious disease dynamics: insights and applications. *BMC Infectious Diseases*, 26, 672, 2026
- [4] Zuzek, L. G., & Stanley, H. E, A comparison of node vaccination strategies to halt SIR epidemic spreading in real-world complex networks. *Scientific Reports*, 12(1), 21323, 2022

Non Split Tree Domination in the Total Graphs of some Simple Graphs

Kavitha Mathew¹ and Sheeja T.K.²

CT

¹Department of Mathematics, Maharaja's College, Ernakulam, India

²Department of Mathematics, T.M. Jacob Memorial Govt. College, Manimalakunnu, India.

The concept of domination has been a prime area of research in graph theory. This paper focuses on tree dominating sets, which are dominating sets whose induced subgraphs are trees. Additionally, non-split tree dominating sets - tree dominating sets whose complements induce connected graphs - are investigated. The non-split tree dominating sets in the total graphs of certain simple graphs are discussed, and the corresponding non-split tree domination numbers are determined.

Automated DPDP Compliance Assessment Using Multi- Agent AI

Krithika S¹, Jia Elsa Kurian¹, Akshay B S¹ and Dr. Aswin V S¹.

CT

¹School of Digital Science, Kerala University of Digital Science, Innovation and Technology, Kerala.

The Digital Personal Data Protection (DPDP) Act, 2023 requires organizations to ensure that software implementations align with declared privacy policies. Existing compliance verification remains largely manual, making implementation gaps difficult to detect before deployment. This work proposes a multi- agent framework combining Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to automate DPDP compliance assessment for Python/Django web applications. The framework bridges the policy-to-code gap by extracting privacy commitments from policies, analysing implementation artifacts, aligning both through LLM-based reasoning, and validating findings against DPDP statutory provisions. It evaluates five compliance principles: transparency, consent management, purpose limitation, data minimization, and data retention. Five cooperating agents comprise the framework: a Privacy Policy Extraction Agent, an Implementation Extraction Agent, an Alignment Agent, a DPDP Legal Reasoning Agent grounded in RAG, and a Report Generation Agent that produces explainable compliance reports. Evaluation uses synthetic Django applications seeded with known DPDP violations alongside a compliant baseline, measuring precision, recall, alignment accuracy, false-positive rate, and report quality. The framework demonstrates a practical, explainable, and auditable approach to pre-deployment DPDP compliance assessment.

References

- [1] S. Das, N. Deb, "A Multi-Agent RAG Framework for Regulatory Compliance Checking of Software Requirements," ACM Transactions on Software Engineering and Methodology
- [2] Z. Zhang, Z. Li, et al., "PriAgent: A Collaborative Multi-Agent Framework for Auditing Android Privacy Compliance," Proceedings of the AAAI Conference on Artificial Intelligence

A numerical investigation of a novel mathematical shigellosis model using the wavelet method

Dr. Kumbinarasaiah S

CT

Bangalore University, Bengaluru, Karnataka, India
E-mail: kumbinarasaiah@gmail.com

Shigellosis is an intestinal infection caused by bacteria of the genus *Shigella* that causes diarrhea, abdominal pain, and fever; understanding its dynamics and transmission is necessary. In this study, we present a novel mathematical shigellosis model (SM) comprising susceptible, exposed, asymptomatic, symptomatic, treated, and recovered individuals; shigella concentration in the environment; the population average of memory B cells; and protective antibody index classes. These dependent variables are studied and solved using the wavelet technique to analyze the nature of the different classes. The existence and uniqueness of the presented model are also studied. The convergence analysis of the wavelet method is also presented as a theorem. The fractional derivative is also chosen to study the SM due to its inherent benefits in capturing the nature and effects at different fractional orders. The numerical results are presented with different techniques along with the wavelet method. The change in the nature of different classes is studied by varying different model parameters.

References

- [1] Stephen Edward, A fractional order model for the transmission dynamics of shigellosis, *Heliyon*, 10(10), 2024
- [2] Yeshwanth, R., Kumbinarasaiah, S. & Preetham, M.P. Wavelet-Based Mathematical Analysis of the Deadly Nipah Virus Transmission Model. *Proc. Natl. Acad. Sci., India, Sect. A Phys. Sci.* (2026)

Malayalam Handwritten Character Recognition Model using Kolmogorov Arnold Networks

M. R. Lekshmi Priya^{1,}, Angel George¹, Athulya Mannambath¹, Aswin V S¹* CT

¹School of Digital Science, Kerala University of Digital Sciences, Innovation and Technology, Thiruvananthapuram, India

Malayalam is a complex script characterized by its curved structures, visually similar characters and diverse handwritten variations making handwritten character recognition a challenging task. Traditional deep learning approaches may face difficulties in accurately distinguishing between visually similar Malayalam characters, particularly when variations in handwritten character recognition model based on Kolmogorov-Arnold Networks (KANs) to address these challenges and improve recognition performance. The proposed approach utilizes KANs as an alternative neural network architecture for learning complex patterns and nonlinear relationships within handwritten Malayalam character images. A dataset of Malayalam handwritten characters is preprocessed and used to train and evaluate the proposed model. The performance of the KAN-based model is assessed using standard classification metrics, with particular emphasis on recognition accuracy. The study investigates the ability of KANs to capture the complex structural characteristics of Malayalam handwritten characters and their potential to provide improved recognition compared with conventional neural networks approaches. The proposed model aims to contribute toward more accurate and reliable Malayalam handwritten character recognition, with potential applications in Malayalam optical character recognition (OCR), document digitization, and preservation of handwritten documents.

References

- [1] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, KAN: Kolmogorov-Arnold Networks, arXiv preprint arXiv:2404.19756, 2024
- [2] A. Alsayed, C. Li, M. Abdalsalam, et al., A hybrid model for Arabic character recognition using CNN and Kolmogorov Arnold Networks (KANs), Multimedia Tools and Applications, vol. 84, pp. 44973–44996, 2025
- [3] M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li, and F. Wei, TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models, arXiv preprint arXiv:2109.10282, 2021

Artificial Intelligence Integrated Numerical Study of Magneto-Casson Penta-Hybrid Nanofluid Flow with Activation Energy and Surface Catalytic Reactions

M. Shanmugapriya^{1,*}, R. Sundareswaran², A.S. Ashwinth Jeffrey²

CT

¹School of Science and Humanities (Mathematics), Shiv Nadar University, Chennai - 603 110, India.

²Department of Mathematics, Sri Sivasubramaniya Nadar College of Engineering, Chennai - 603 110, India.

This study presents a novel AI model to simulate magneto-Casson penta-hybrid nanofluid flow over a moving wedge. The Penta hybrid nanofluid (PHNF) comprises of five different types of nanoparticles such as Fe_3O_4 , Al_2O_3 , CuO , SWCNT, and MWCNT diluted in a base fluid H_2O . Using Shooting method, the velocity, temperature, concentration, homogeneous and heterogeneous profiles are interpreted against the influencing dimensionless variables. The influence of non-dimensional parameters on Cf_x , Nu_x , Sh_x , Sh_A and Sh_B for species A and B reactions is clearly illustrated in the graphs. It highlights that increase in magnetic field parameter (M), Casson parameter (β) increasing the velocity profile. Temperature profile is increased by volume fraction parameter ($\phi_{Fe_3O_4}$, $\phi_{Al_2O_3}$, ϕ_{CuO} , ϕ_{SWCNT} , ϕ_{MWCNT}), thermophoresis parameter (Nt) and Brownian motion parameter (Nb). Activation energy (E) contributes to enhancement in concentration profile. To improve predictive performance, AI models like ANN, and ANFIS were developed and trained using the numerically generated dataset. Their effectiveness was assessed through the regression correlation coefficient (R), which was found to be close to unity, signifying excellent agreement between the predicted outcomes and the computed results. Comparative analysis revealed that the ANN model achieved superior accuracy, yielding an average absolute error below 10^{-3} in estimating the fluid characteristics, thereby outperforming the ANFIS approach.

Perturbation Analysis and Numerical Stability of Quantized Rotary Positional Embeddings in Transformer Architectures

Malavika M^{1,*}, Neha Jose¹

CT

¹Department of Mathematics, Loyola College, Chennai, India

Modern Transformer-based Large Language Models (LLMs) rely on Rotary Position Embedding (RoPE) to preserve positional structure across long input sequences, encoding relative distance between tokens through orthogonal rotation of query and key vectors. To deploy these models efficiently by reducing their massive memory size and accelerating processing speeds for accessible hardware, engineers increasingly rely on post-training quantization (PTQ), compressing high-precision (FP16/BF16) representations into low-bit formats such as INT8 or INT4. Although this compression saves substantial memory, it introduces precision rounding errors into the underlying key-query matrix computations. Empirically, such quantization often leads to a sudden, severe drop in model's ability to handle long texts. Yet a rigorous mathematical analysis explaining why RoPE-based positional encoding's geometric space is so sensitive to this noise has been largely absent from the literature. This work addresses that gap by developing a formal backward error analysis framework, grounded in classical Numerical Linear Algebra, that models quantization as a bounded additive perturbation to the query and key matrices and studies its propagation through RoPE's orthogonal rotation structure. Because each rotation block is orthogonal and hence exactly norm-preserving, the ideal (unperturbed) attention mechanism is provably well-conditioned, with spectral condition number equal to one. We show that once quantization noise is introduced into the query and key matrices, the perturbed attention score decomposes into the true score plus cross-terms involving the noise and the orthogonal rotation. Applying the triangle and Cauchy-Schwarz inequalities, together with the norm-preservation property of orthogonal transformations, yields a closed-form upper bound on the resulting attention-score error, expressed purely in terms of the norms of the query and key matrices and the quantization noise matrices. This bound offers an analytically predictable criterion for the onset of positional-encoding instability under compression, rather than relying on empirical trial and error. To validate this theoretical bound, we propose a lightweight empirical validation pipeline using NumPy and PyTorch that does not require training or accessing a full pretrained language model. Synthetic query and key matrices are generated via Gaussian mixture distributions to accurately mimic real token embedding spaces. An algorithmic wrapper then simulates uniform precision formats (INT8, FP4, and INT4) under varying scaling factors. Finally, absolute backward error matrix norms and Singular Value Decomposition (SVD) profiles are computed across simulated token distances and measured directly against the derived analytical bound as a function of both bit-width and token distance. The expected significance of this study is threefold. First, it is expected to demonstrate that low-bit quantization artificially collapses the singular value spectrum of the attention matrix, revealing an immediate squishing of informational capacity under compression. Second, the framework is expected to map the

exact boundary where long-context instability concentrates. At large token distances, high spatial frequencies interact poorly with the coarse, rounded quantization grid, causing position data to degrade predictably into numerical noise. Third, by replacing purely empirical quantization heuristics with a provable, closed-form stability criterion, this work aims to give practitioners an analytical tool to predict, ahead of deployment, the bit-width and context-length thresholds at which a model's positional understanding is likely to degrade. More broadly, the study illustrates how classical backward error analysis, developed for numerical algorithms such as Gaussian elimination, extends naturally to the geometry of modern attention mechanisms, offering a principled bridge between numerical linear algebra and the stability analysis of large-scale deep learning systems.

References

- [1] Su, J., Ahmed, M., Lu, Y., Pan, S., Bo, W., & Liu, Y. (2024). RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 127063
- [2] Trefethen, L. N., & Bau III, D. (1997). *Numerical Linear Algebra*. SIAM
- [3] Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms*. SIAM
- [4] Qiu, H., & Yao, Q. (2025). Why Low-Precision Transformer Training Fails: An Analysis on Flash Attention. arXiv preprint arXiv:2510.04212

Vectorized Block Representation of Persistent Laplacian Diagram

Maneesha K K¹, Sasi Gopalan²

CT

^{1,2}Department of Mathematics, Cochin University of Science and Technology, Kochi, Kerala, India

Persistent Homology (PH) is a mathematical tool used in topological data analysis, which extracts the appearance and disappearance of topological features across the filtration. The output of PH is persistence diagrams. Persistent Laplacians provide spectral information that complements the birth-death information of persistence diagrams. Inke Jung, Wonwoo Kang, and Heehyun Park (2025) proposed the Persistent Laplacian Diagram (PLD) and Persistent Laplacian Image (PLI), incorporating spectral information through persistent Laplacians, and proved their stability under perturbations. Although PLDs provide richer spectral-topological information than persistence diagrams, there is currently no Vectorized Persistence Block representation for PLDs. Consequently, PLDs cannot directly benefit from the computational advantages of VPB- based feature extraction. We propose a Persistence Vectorized Block (PVB) representation for PLDs. The construction transforms the spectral information associated with persistent Laplacians into a structured vectorized representation by partitioning the relevant parameter domain into blocks and aggregating the corresponding persistence-laplacian information. The representation is designed to retain the essential topological and spectral characteristics while enabling efficient numerical computation. We establish a stability property of the proposed representation with respect to perturbations of the underlying PLD. In particular, the resulting vectorization varies continuously under an appropriate distance between persistence diagrams, providing a theoretical guarantee that small perturbations in the input produce controlled changes in the representation. The proposed PVB representation provides a bridge between Persistent Laplacian-based topological information and conventional vector-based computational methods. This makes PLD information more accessible for statistical analysis, machine learning, and other data- driven applications.

References

- [1] Chan, Kit C., et al. "A Computationally Efficient Framework for Vector Representation of Persistence Diagrams." (JMLR) (2022)
- [2] Jung, Inke, Wonwoo Kang, and Heehyun Park. "Persistent Laplacian Diagrams." arXiv preprint arXiv:2512.05463 (2025)
- [3] Adams., et al, 2017. "Persistence images: A stable vector representation of persistent homology". Journal of Machine Learning Research, 18(8), pp.1-35

A critical memory order in a delayed fractional rumour model

Manoj Kumar Singh*

CT

Department of Mathematics and Statistics, St. Xavier's College, Ranchi (Autonomous),
affiliated to
Ranchi University, Ranchi 834001, Jharkhand, India

We study the joint effect of fractional memory and time delay on the stability of the endemic state of a rumour propagation model, and report a criterion, read from the spectrum of a single matrix, that decides whether memory can immunise the endemic state against delay-induced oscillation. **Motivation.** Rumours spread by social contact, and compartmental models of the ignorant–spreader–stifler type describe this well. Two features of real communication are not captured by the classical rate laws. Waiting times before people act on information are broadly distributed rather than exponential, which the Caputo fractional derivative of order $\alpha \in (0, 1]$ models through its memory kernel. And responses are not instantaneous, which introduces a delay. We place a delay on the forgetting flux, the reversion of stiflers to the ignorant class, so that reversion responds to the stifler density a time τ earlier. **Problem.** Without delay the endemic state is stable for every memory order. Recent studies of delayed fractional rumour and epidemic models compute a critical delay and then vary the order to see the threshold move, usually finding that stronger memory postpones the loss of stability. This treats the order as a quantity that shifts a threshold. We ask a sharper question: for which memory orders can a delay destabilise the endemic state at all, and can strong memory remove the possibility entirely? **Approach.** The endemic state and a reproduction number are obtained in closed form, and boundedness follows from a conservation identity. Linearisation produces a system with an instantaneous coupling and a delayed coupling of rank one. Reducing the associated characteristic equation leads to a scalar crossing condition, whose analysis rests on an algebraic identity relating the constant term of a crossing polynomial to the determinants of the sum and the difference of the two coupling matrices. **Results.** The behaviour is governed by the eigenvalues of the difference of the two matrices. In one regime a delay destabilises the endemic state at every memory order, so no memory protects it, and stronger memory only shifts the critical delay. In the other regime a critical memory order exists, below which the endemic state stays asymptotically stable for every delay, however large. The crossing polynomial is found to depend on the order through a single trigonometric quantity, which makes the critical order exactly computable rather than merely estimated. A comparison of delay placements indicates that the mechanism responsible for this behaviour is tied to the mass-conserving delayed reversion. The findings are supported by numerical integration with a fine-step fractional scheme, and full proofs and simulations will be presented at the conference. **Significance.** The criterion is checkable from one matrix built at the endemic state, and it separates the case where memory can only delay oscillation from the case where memory can prevent it. The result is stated for a general class of delayed compartmental systems, so it applies beyond the

rumour setting.

References

- [1] D. J. Daley, D. G. Kendall, Epidemics and rumours, *Nature* 204 (1964) 1118
- [2] I. Podlubny, *Fractional Differential Equations*, Academic Press, San Diego, 1999
- [3] Y. Niu, A. Muhammadhaji, Dynamics of a fractional-order IDSR rumor propagation model with time delays, *Fractal and Fractional* 9 (2025) 242

Degree-dependent conditioning of Chebyshev Kolmogorov–Arnold networks

Manoj Kumar Singh^{1,*}, Lakshmi C², and Ankit Ruhi³

CT

¹Department of Mathematics and Statistics, St. Xavier's College, Ranchi (Autonomous), affiliated to

Ranchi University, Ranchi 834001, Jharkhand, India

³Department of Mathematics and Statistics, Dr. Harisingh Gour Vishwavidyalaya, Sagar, Madhya

We study how the collocation measure and the polynomial degree control the conditioning of a Chebyshev Kolmogorov–Arnold layer. The two Gram matrices involved are identified as compressions of reciprocal symbols. This gives exact orders of growth for both condition numbers. It also gives a choice of collocation nodes that removes one of them. A diagonal rescaling removes two powers of the degree from the other. **Motivation.** Kolmogorov–Arnold networks place trainable univariate maps on the edges of a network. Ordinary networks place fixed activations on the nodes instead. Liu et al. introduced these networks with B-splines on the edges [1]. Later variants replace splines by global bases. The Chebyshev variant is now the common choice in scientific computing. It is also the standard basis inside physics-informed Kolmogorov–Arnold networks [2]. The reason given for this choice is the same across the literature. The Chebyshev system is orthogonal. Its recurrence is cheap. Orthogonality is taken to guarantee stable training. We examine that last claim. **Problem.** The Chebyshev polynomials are orthogonal for the weight $w(x) = (1 - x^2)^{-1/2}$. A training loss is not measured in that weight. Collocation points are drawn uniformly or placed on a uniform grid. So the empirical loss approximates an integral against Lebesgue measure. A single layer is linear in its parameters. Its loss is therefore quadratic. Its Hessian is a Gram matrix. For the fitting term that matrix is $(G_k)_{mn} = \int_{-1}^1 \hat{T}_m \hat{T}_n dx$, where $\hat{T}_n$ is the Chebyshev polynomial normalised in $L^2((-1, 1), wdx)$. In the weighted inner product it would be the identity. It is not the identity here. For the spline architecture the matching question has a clean answer. There the bound is uniform in the grid size [3]. For the Chebyshev architecture the available work derives a neural tangent kernel. Its Gaussian expectations are left unevaluated. No eigenvalue estimate is extracted from it [4]. We ask whether the degree enters the conditioning at all. If it does, we ask at what order. **Approach.** The substitution $x = \cos \theta$ settles the matter. Under it $\hat{T}_n(\cos \theta)$ becomes the orthonormal cosine system on $L^2(0, \pi)$. The measure dx becomes $\sin \theta d\theta$. So G_k is exactly the compression of multiplication by $\sin \theta$ to the first $k + 1$ cosine modes. Its entries are rational. Two-sided estimates follow from a Nikolskii inequality for trigonometric polynomials. A Fejér kernel supplies the matching test function. The residual term is handled by the same route. Its Gram matrix is evaluated in closed form. The proof telescopes an integral through the identity $\cos((j - 2)\theta) - \cos(j\theta) = 2\sin((j - 1)\theta)\sin\theta$. **Results.** The symbol $\sin \theta$ is bounded above by one. It vanishes at both endpoints. So the largest eigenvalue of G_k rises to one.

The smallest decays like k^{-1} . The condition number is therefore of exact order k . This defect belongs to the sampling measure. It is not a defect of the architecture. Discretising at $k + 1$ Gauss–Chebyshev nodes makes the Gram matrix the identity. A rank argument shows that $k + 1$ points are the minimum budget under any placement. For a layer with d inputs the relevant eigenvalue ratio then equals d exactly. Under uniform collocation the same ratio is of order dk . The residual term behaves in the opposite way. Its Gram matrix has smallest eigenvalue trapped between 1 and $4/\pi$. The largest grows like k^3 . Changing the sampling measure makes matters worse here. The Chebyshev weight turns the angular weight $\sin^{-1} \theta$ into $\sin^{-2} \theta$. The growth is a Markov effect. It survives every positive weight. What can be removed is the algebraic factor carried by differentiation. Rescale the coefficient of the degree n mode by n^{-1} . This produces the compression of $\sin^{-1} \theta$ to the sine modes. Its condition number is of order k . The same reduction from order k^3 to order k holds for the full physics-informed Hessian with a boundary penalty. Numerical work assembles every matrix from the proved closed forms. So it tests the theorems directly. All four rates above are confirmed for k up to 1024. We also solve a first-order model problem by gradient descent to tolerance 10^{-8} at degree 64. The raw solver needs about 3.11×10^6 iterations. The rescaled solver needs about 1.14×10^3 . Second-order operators are worse. There the largest eigenvalue grows like k^7 . Full proofs and the complete numerical study will be presented at the conference. **Significance.** Three checkable prescriptions follow. Collocate at Gauss–Chebyshev nodes rather than uniformly. The gain is exact. It costs nothing. Rescale the degree n coefficient by n^{-1} whenever the loss carries a first-order residual. Prefer several low-degree subdomains to one high-degree domain. The condition number is set by the degree. It is not set by the total parameter count. The last point supplies a reason for the gains from domain splitting reported in applied work.

References

- [1] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljaci'c, T. Y. Hou, M. Tegmark, KAN: Kolmogorov–Arnold networks, International Conference on Learning Representations, 2025
- [2] F. Mostajeran, S. A. Faroughi, Scaled-cPIKANs: spatial variable and residual scaling in Chebyshev-based physics-informed Kolmogorov–Arnold networks, Journal of Computational Physics 537 (2025) 114116
- [3] Y. Wang, J. W. Siegel, Z. Liu, T. Y. Hou, On the expressiveness and spectral bias of KANs, International Conference on Learning Representations, 2025
- [4] S. A. Faroughi, F. Mostajeran, Neural tangent kernel analysis to probe convergence in physics-informed neural solvers: PIKANs vs. PINNs, arXiv:2506.07958 (2025)

A Study of Morphological Filtering Using Single-Valued Neutrosophic Hypergraphs

Miji Philip^{1,2}, Teena Liza John P¹

CT

¹Department of Mathematics,TKMCE,Kollam,
Affiliated to APJ Abdhul Kalam Technological University, Kerala

² Department of Applied Science and Engineering , TKMIT, Kollam,

Mathematical morphology (MM) provides a powerful set-theoretic framework for image analysis and shape description. While morphological operators have been successfully extended from binary images to graphs and hypergraphs, existing approaches operate on crisp structures, limiting their ability to handle uncertainty and imprecision inherent in real-world data. This paper introduces a novel framework for morphological filtering on Single-Valued Neutrosophic Hypergraphs (SVNHGs), which incorporate truth, indeterminacy, and falsity membership degrees. We extend the fundamental morphological operators—erosion, dilation, opening, closing, and Alternating Sequential Filters (ASF)—from crisp hypergraphs to the SVNHG domain. Experimental results on binary images demonstrate that the proposed SVNHG-based ASF achieves superior performance compared to traditional graph-based and crisp hypergraph-based filters, particularly at moderate noise levels. The framework shows promise for applications involving uncertain or imprecise data, such as medical imaging and remote sensing.

Dynamics of a Fractional-Order Climate-Driven Vector-Borne Zoonotic Disease Model with Environmental Contamination

Mitakshara Rawat^{1,*} and Dr. Bipin Kumar²

CT

¹Department of Mathematics and Statistics, Dr. Harisingh Gour University, Sagar, Madhya Pradesh, India

In this paper, we formulate and analyze a fractional-order mathematical model for a zoonotic vector-borne infectious disease under the influence of climate change. The proposed model incorporates fractional derivatives to capture the memory and hereditary characteristics associated with disease transmission dynamics. Climate-related factors, such as temperature variation and environmental fluctuations, are incorporated to investigate their effects on the spread and persistence of the disease. The qualitative behavior of the model is studied in detail. Initially, the positivity, existence, uniqueness, and boundedness of solutions are established to ensure the biological feasibility of the system. The equilibrium points of the model, including the disease-free equilibrium and endemic equilibrium, are obtained and analyzed. Local stability conditions for these equilibria are derived using fractional-order stability theory. Furthermore, the basic reproduction number is determined and utilized to study the disease transmission dynamics. The occurrence of bifurcation near the disease-free equilibrium is also investigated to understand the transition between disease eradication and disease persistence. Numerical simulations are carried out to support the theoretical findings and to examine the impacts of climate change parameters and fractional-order derivatives on disease transmission. The results reveal that both climate variability and memory effects play significant roles in influencing the spread, control, and long-term behavior of vector-borne zoonotic diseases. This study provides useful insights into the complex dynamics of infectious diseases and may contribute to the development of effective disease prevention and control strategies under changing environmental conditions.

References

- [1] Podlubny, I., Fractional Differential Equations, Academic Press, 1999
- [2] Diethelm, K., The Analysis of Fractional Differential Equations, Springer, 2010
- [3] Oldham, K. B. and Spanier, J., The Fractional Calculus: Theory and Applications of Differentiation and Integration to Arbitrary Order, Vol. 111, Academic Press (Elsevier), New York, 1974
- [4] van den Driessche, P. and Watmough, J., Reproduction Numbers and Sub-threshold Endemic Equilibria for Compartmental Models of Disease Transmission, Mathematical Biosciences, 180 (2002), 29–48
- [5] Hoan, Luu Vu Cam, et al., “A New Fractional-Order Compartmental Disease Model,” Alexandria Engineering Journal, Vol. 59, No. 5, 2020, pp. 3187–3196

[6] Almutairi, N. and Saber, S., "Existence of Chaos and the Approximate Solution of the Lorenz–Lu–Chen System with the Caputo Fractional Operator," AIP Advances, Vol. 14, No. 1, 2024

[7] Choi, S. K., Kang, B., and Koo, N., "Stability for Caputo Fractional Differential Systems," Abstract and Applied Analysis, Vol. 2014, Article ID 631419, 2014

[8] Althubayani, M., Adam, H. D. S., Alalyani, A., et al., "Understanding Zoonotic Disease Spread with a Fractional Order Epidemic Model," Scientific Reports, Vol. 15, Article No. 13921, 2025

[9] Eiman, et al., "On Mathematical Model of Infectious Disease by Using Fractals Fractional Analysis," Discrete and Continuous Dynamical Systems – Series A, 2024

PINNLab: An Interactive Framework for Physics-Informed Parameter Estimation in Differential Equations

Mohan Parthasarathy and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Undergraduate instruction in differential equations is typically organized around the forward problem: finding solution trajectories when the governing equation and its parameters are known. In scientific practice, however, inverse problems are often more relevant, requiring unknown parameters to be inferred from noisy observations while assessing whether a proposed model is consistent with the data. We introduce PINNLab, an open-source MATLAB dashboard designed to help undergraduate students explore inverse modeling through physics-informed neural networks (PINNs). PINNLab presents PINNs as a complementary data-driven framework that connects differential equations, optimization, empirical data, and scientific machine learning. The instructional sequence is organized around the 5E learning model and a T-shaped curriculum inspired by the Department of Labor's AI Literacy Framework. Students explore modules relating to ecology, fluid transport hemodynamics, rheology, and more. Throughout the curriculum, PINNLab makes the modeling process visible through real-time visualization of state reconstruction, parameter convergence, configurable initial guesses, and loss diagnostics. By emphasizing the distinction between data fit and physical consistency, the platform encourages students to evaluate, critique, and revise mathematical models in the presence of noisy empirical observations. Students thereby learn to treat differential equations not merely as equations to be solved, but as testable hypotheses to be confronted with finite, noisy, and sometimes inconsistent empirical data.

Analytical Investigation of Fracture in Porous Media Subjected to Moving Load

Muhammed Murshid AK^{1,*}, Srijita Das¹, and Sonali Mondal¹

CT

¹Department of Mathematics, SRM University AP, Amaravati, 522240, Andhra Pradesh, India

Fluid-saturated porous materials are widely encountered in geomechanics, petroleum engineering, biomechanics, and civil infrastructure, where the interaction between stress waves and cracks strongly influences structural integrity and failure. In many practical situations, such as hydraulic fracturing, drilling, moving machinery, and contact loading, cracks propagate under dynamic mechanical loads while interacting with pore fluids. The coupled effects of anisotropy, porosity, crack motion, and moving surface loading significantly complicate fracture analysis. The present work develops an analytical framework to investigate the dynamic fracture behaviour of porous media under Mode-I loading. The problem considers an orthotropic fluid-saturated porous strip of finite thickness containing two identical moving collinear Griffith with a moving punch load applied on the strip boundaries. The objective is to determine the Stress Intensity Factor (SIF) and Crack Opening Displacement (COD) while examining the influence of porosity, strip thickness, crack geometry, and external loading on fracture behaviour. The governing coupled equations describing the solid skeleton and pore-fluid motion are transformed into a moving coordinate system using the Galilean transformation, converting the transient problem into an equivalent steady-state formulation. Integral transform techniques are then employed to represent the displacement fields, and the resulting boundary-value problem is reduced to a system of singular integral equations through the finite Hilbert transform method. Closed-form analytical expressions are derived for the Mode-I Stress Intensity Factors at both crack tips as well as the Crack Opening Displacement. Numerical results demonstrate the effect of porosity, strip thickness, crack tip ratio, punch load on SIF and COD. Comparison with available limiting cases reported in the literature shows excellent agreement, confirming the accuracy and validity of the proposed analytical formulation. The closed-form solutions obtained for the SIF and COD may provide valuable insight into crack propagation in porous media and may assist in the design and integrity assessment of engineering systems encountered in hydraulic fracturing, enhanced oil recovery, rock mechanics, geotechnical engineering, and biomechanics.

Higher-Order Upwind Flux Decomposition MGD Equations Based on Kinetic Theory

N. Thakur¹, A. Ruhi¹, and M.K. Yadav¹

CT

¹Department of Mathematics and Statistics, Dr. Harisingh Gour Vishwavidyalaya, Sagar, India

Numerical schemes for magnetogasdynamics are generally formulated from macroscopic conservation laws using approximate Riemann solvers or characteristic-based decompositions. While effective in resolving ideal MGD wave structures, they do not explicitly capture the microscopic origin of acoustic and magnetic effects, as pressure and magnetic contributions are introduced mathematically rather than intrinsically. To address this limitation, we use a kinetic base formulation for MGD equations [1], a complex Maxwellian formulation where acoustic and Alfvén waves provide a unified and physically consistent description. In the complex Maxwellian kinetic formulation, the PDF is expressed as a function of $\xi_k - (u_k + i, v_\alpha)$, enabling a natural separation of kinetic (real) and magnetic (imaginary) effects. The macroscopic MHD equations are recovered from the Boltzmann equation by taking moments of the complex Maxwellian distribution. These moments yield the advective, acoustic-magnetic, and Alfvén fluxes, with the real part governing density, momentum, and energy, and the imaginary part governing the magnetic field. The Peculiar Velocity Upwind (PVU) scheme decomposes the flux into bulk velocity and peculiar velocity components. Which is developed by [2] The resulting PUV scheme is structurally analogous to the Euler equations. The PVU scheme derived from the complex Maxwellian kinetic framework is applied the 1D ideal MHD 7-equations. The results it has been validated 1D test cases. Overall, the scheme produces results comparable to classical MGD formulations while preserving stability and structural consistency.

References

- [1] B. Chetverushkin et al., "A kinetic model for magnetogasdynamics," *Mathematical Models and Computer Simulations*, vol. 9, no. 5, pp. 544–553, 2017
- [2] Deshpande, S. M., N. Balakrishnan, and S. V. Raghurama Rao. "PVU and wave-particle splitting schemes for Euler equations of gas dynamics," *Sadhana* 19.6 (1994): 1027-1054

A Preliminary Study of Reinforced Domination in Graphs

Najma Farsana A.¹, Manjusha O.T.²

CT

¹Department of Mathematics, St.Joseph's College (Autonomous) Devagiri, Kozhikode, India

²PG and Research Department of Mathematics, Govt. Arts and Science College Kondotty, Malappuram, India

Domination theory is an important area of graph theory with applications in communication networks, social networks, resource allocation, and influence systems. Classical domination models are mainly based on adjacency conditions, where every vertex outside a dominating set is required to be adjacent to at least one vertex in the set. However, in many real-world situations, the influence or support provided by a single neighboring vertex may not be sufficient to meet the requirements of a vertex. Motivated by weighted influence models and threshold-based network systems, we introduce a new domination concept called reinforced domination. In this model, each vertex of a graph is assigned a positive weight representing its capacity to provide support and a threshold representing the minimum level of support required by that vertex. A set $D \subseteq V(G)$ is called a reinforced dominating set if every vertex outside D receives a total weight from its neighbors in D that is at least its prescribed threshold. The minimum cardinality of a reinforced dominating set is called the reinforced domination number of G . In this work, we develop the basic framework of reinforced domination and investigate its fundamental properties. We first consider the uniform setting, in which all vertices are assigned the same weight and threshold, and then examine the reinforced domination number for several important classes of graphs. The influence of the relationship between the weight and threshold parameters, together with the structural properties of the underlying graph, is explored. The concept also provides a natural framework for potential applications in network support, resource allocation, influence propagation, and thresholdbased systems. Finally, some directions for further study, including reinforced domination under non-uniform weights and thresholds and its investigation in broader classes of graphs, are indicated.

References

- [1] R. T. Araújo, R. M. Sampaio, Domination and convexity problems in the Target Set Selection model, *Discrete Appl. Math.* 330 (2023) 14–23
- [2] C. Berge, *Theory of Graphs and Its Applications*, Methuen, London, 1962
- [3] G. Chartrand, P. Zhang, *Introduction to Graph Theory*, McGraw-Hill Higher Education, 2012
- [4] E. J. Cockayne, S. T. Hedetniemi, Towards a theory of domination in graphs, *Networks* 7 (1977) 247–261
- [5] D. Easley, J. Kleinberg, *Networks, Crowds, and Markets: Reasoning About a Highly Connected World*, Cambridge University Press, 2010

- [6] J. F. Fink, M. S. Jacobson, n -Domination in graphs, in: Y. Alavi, A. J. Schwenk (Eds.), *Graph Theory with Applications to Algorithms and Computer Science*, Wiley, New York, 1985, 283–300
- [7] T. W. Haynes, S. T. Hedetniemi, P. J. Slater, *Domination in Graphs: Advanced Topics*, Marcel Dekker, New York, 1998
- [8] D. Kempe, J. Kleinberg, E. Tardos, Maximizing the spread of influence through a social network, *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ACM, New York, 2003, 137–146
- [9] O. Ore, *Theory of Graphs*, American Mathematical Society Colloquium Publications, Vol. 38, Providence, RI, 1962
- [10] D. Peleg, Local majorities, coalitions and monopolies in graphs: A review, *Theoret. Comput. Sci.* 282 (2002) 231–257
- [11] N. J. Rad, J. Raczek, Vector domination in graphs, *Discuss. Math. Graph Theory* 31 (2011) 733–743
- [12] M. Richardson, P. Domingos, Mining knowledge-sharing sites for viral marketing, *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ACM, New York, 2002, 61–70
- [13] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001

Equitable Coloring of the Broom Corona Product $K_n \odot K_r$: Cyclic Latin-Rectangle Construction, Algorithmic Analysis, and Applications to Balanced Task Assignment

Naseema V A and J. Veninstine Vivik

CT

Department of Basic Science, KMEA Engineering College, Kuzhivelipady, Ernakulam-683 561, Kerala, India

The broom corona product $K_n \circ \bar{K}_r$ obtained from the complete graph K_n by attaching r independent pendant vertices to each of its vertices is a natural graph model for hub-and-cluster architectures in distributed computing and wireless resource assignment. In this paper we determine the equitable chromatic number and the equitable chromatic threshold of $K_n \circ \bar{K}_r$ in closed form, and we give an explicit cyclic Latin-rectangle construction that realises an equitable n -colouring in linear time. Specifically, we prove that $\xi = K_n \circ \bar{K}_r = n$ for all $n \geq 2$ and $1 \leq r \leq n - 1$, and we extend the construction to arbitrary $r \geq 1$ by an iterated cyclic shift. We further prove that $K_n \circ \bar{K}_r$ admits an equitable k -colouring for every integer k with $n \leq k \leq n(r + 1)$, establishing that the equitable chromatic threshold $\xi^* = K_n \circ \bar{K}_r$ coincides with the equitable chromatic number itself. Our algorithm runs in $O(nr)$ time and $O(n)$ auxiliary space, matching the trivial size of the output. We validate the construction by computational experiments up to $n = 1280$ and confirm the theoretical guarantees for equitability and properness on all instances tested. We conclude by discussing an application to balanced task assignment in synchronous distributed systems, where the cyclic construction produces a schedule whose worst-case round load is provably within one of optimum.

References

1. Balakrishnan, R., Ranganathan, K.: A Textbook of Graph Theory, 2nd edn. Springer, New York (2012)
2. Bollobás, B., Guy, R.K.: Equitable and proportional coloring of trees. J. Combin. Theory Ser. B 34(2), 177–186 (1983)
3. Chen, B.-L., Lih, K.-W., Wu, P.-L.: Equitable coloring and the maximum degree. European J. Combin. 15(5), 443–447 (1994)
4. Erdős, P.: Problem 9. In: Theory of Graphs and its Applications (Proc. Sympos. Smolenice), p. 159. Publ. House Czechoslovak Acad. Sci., Prague (1964)

A Graph-Restricted Ćirić-Type Fixed Point Theorem (GRFTFT) in Complete Fuzzy b-Metric Spaces

Neha

CT

Department of Computer Science and Engineering, SEST
Jamia Hamdard, New Delhi 110062, India
drnehavaid@jamiahamdard.ac.in

In this paper, a graph, based on a Ćirić type fixed point theorem, is presented in complete fuzzy b-metric spaces. Given a complete fuzzy b-metric space (W, ϑ, P_{min}) with a directed graph G with vertex set W and with all loops being in the edge set of G , under certain hypotheses (edge preservation, existence of an initial graph admissible point and compatibility of graph edges with convergence), we prove that, when there exists a graph-restricted contraction, then there exists a fixed point of the self-mapping. If any two fixed points are graph connected, then the uniqueness is obtained. The proof is based on the Cauchy-sequence criterion, but with the contraction constant $\lambda \in (0, 1)$, not quite so restrictive as $\lambda < 1/b$, as used in some of the previous proofs. When considered as the complete directed fuzzy b-metric space G , the proposed theorem reduces to a similar Ćirić-type theorem in an ordinary complete fuzzy b-metric space. Thus, the theorem unifies the graph-based and classical fuzzy b-metric fixed point theories, and it sets necessary condition for the contraction which is only for structurally related elements.

Depth-Adaptive LoRA for Transformer Architectures: A Comparative Study Against Uniform-Rank LoRA and PiSSA

Neha Jose^{1,*}, Malavika M¹

CT

¹Department of Mathematics, Loyola College, Chennai, India

Fine-tuning a large pretrained transformer for a downstream task by updating every weight is often impractical, since it requires storing and updating gradients and optimizer states for the full parameter count, at substantial memory and compute cost. Low-rank adaptation methods such as LoRA address this by updating each weight matrix through a low-rank correction W instead of the full matrix, relying on the assumption that the change required during fine-tuning lives in a low-dimensional subspace. In most implementations, this correction is assigned the same rank at every layer, regardless of depth. Contrary to this uniform practice, studies reveal a rank collapse, where token representations in deep attention-only transformer stacks lose rank as depth increases. If this collapse is present in the query, key, and value projections in a normal transformer that LoRA typically adapts, then giving every layer the same rank budget may be wasteful at some depths and insufficient at others. There exist adaptive-rank methods such as AdaLoRA which adjust each layer's rank dynamically during fine-tuning, using gradient-based importance scores computed as training proceeds. This work takes a different approach to the same question: rather than adapting rank during training, we ask whether the rank each layer needs can be estimated directly from the pretrained model itself, before any fine-tuning begins, by measuring how much effective rank its attention projections actually use at each depth, giving our proposed method, Depth-Adaptive LoRA. The central question addressed is a comparative one: under a fixed total parameter budget, does allocating LoRA rank per layer according to a pretraining-time effective-rank diagnostic, rather than fixing the same rank everywhere, outperform, match, or underperform standard uniform-rank LoRA and PiSSA, a related low-rank initialization method, on downstream fine-tuning tasks? We study this on GPT-2 Small, with plans to expand the evaluation to other architectures and baseline against adaptive methods like AdaLoRA as time permits. Measured on GPT-2 Small, effective rank varies substantially and non-uniformly with depth: representations collapse towards their lowest effective rank in the early-to-middle layers before partially recovering towards the output, and this pattern differs sharply between the value projection and the query/key projections. To estimate the rank each layer needs, we use tools from numerical linear algebra, specifically the stable rank $\text{sr}(M) = \|M\|_F^2 / \|M\|_2^2 = \sum_i \sigma_i^2 / \sigma_1^2$, a standard proxy for effective rank obtained from the singular values σ_i of a matrix M , rather than the algebraic rank of the weight matrices, which is essentially full rank by construction and therefore uninformative here. For a pretrained transformer and a randomly initialized model of the same architecture (which serves as a baseline), we compute this quantity at every layer for the residual-stream input to the attention block and for the query, key, and value projection outputs, using a batch of natural-language probe sentences, and plot how it varies with depth. This depth profile is then used to fix a rank for each layer and each

projection matrix in advance. We evaluate this fixed schedule against standard uniform-rank LoRA and PiSSA at matched total parameter budgets. A rank budget informed by this pretraining-time structure should, in principle, allocate parameters more efficiently than either a uniform schedule or the training-time, gradient-driven schedule used by adaptive methods such as AdaLoRA, since it targets rank at the specific layers and projections where the pretrained model already shows it is used. Because the schedule is fixed before fine-tuning begins, Depth-Adaptive LoRA also avoids the additional computational overhead of tracking importance scores throughout training, making it a computationally lighter alternative to training-time rank adaptation, while remaining grounded in a measurable property of the pretrained model rather than an arbitrary uniform choice.

References

- [1] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-Rank Adaptation of Large Language Models, International Conference on Learning Representations (ICLR), 2022
- [2] F. Meng, Z. Wang, M. Zhang, PiSSA: Principal Singular Values and Singular Vectors Adaptation of Large Language Models, Advances in Neural Information Processing Systems (NeurIPS), 2024
- [3] Q. Zhang, M. Chen, A. Bukharin, P. He, Y. Cheng, W. Chen, T. Zhao, AdaLoRA: Adaptive Low-Rank Adaptation for Parameter-Efficient Fine-Tuning, International Conference on Learning Representations (ICLR), 2023
- [4] Y. Dong, J.-B. Cordonnier, A. Loukas, Attention is Not All You Need: Pure Attention Loses Rank Doubly Exponentially with Depth, International Conference on Machine Learning (ICML), 2021
- [5] T. Nait Saada, A. Naderi, J. Tanner, Mind the Gap: A Spectral Analysis of Rank Collapse and Signal Propagation in Attention Layers, arXiv:2410.07799, 2024
- [6] L. N. Trefethen, D. Bau III, Numerical Linear Algebra, SIAM, 1997

Is it Real? Optimal Control of an SIR Model of Misinformation through Educational Campaigns

Nehal Joshi and Padmanabhan Seshaiyer

CT

[George Mason University](#)

While misinformation has long been a societal challenge, the modern accessibility of the internet has accelerated its spread, slowing consensus on critical global issues and driving societal polarization. To quantitatively evaluate and mitigate the viral spread of false narratives, this study introduces an epidemiologically inspired Susceptible-Critically Literate-Misinformed (SCM) compartmental model. The system utilizes a set of ordinary differential equations that incorporate targeted educational campaigns as an active intervention variable. To accurately reflect digital behavior, the model also introduces a distinct natural exit rate from the misinformed compartment, isolating individuals who leave the system due to platform churn or extreme polarization. Mathematical analysis guarantees the system's validity through proofs of positivity and boundedness. Additionally, the Misinformation-Free Equilibrium is derived using the Next Generation Matrix method to compute the basic reproduction number (R_0). To ground the theoretical framework in real-world dynamics, Physics-Informed Neural Networks (PINNs) were applied to the PHEME dataset—comprising over 105,000 tweets across multiple crisis events—to extract empirical transmission rates. Subsequent sensitivity analysis revealed that the viral transmission rate is the overwhelmingly dominant factor driving the system. To combat this spread efficiently, the study employs optimal control theory, utilizing Pontryagin's Maximum Principle and a quadratic cost function to determine the most effective educational intervention strategy. The results demonstrate that implementing an optimal quadratic educational control, which never exceeds an effort intensity of 64%, successfully reduces the peak misinformed population by 99.1% and decreases the total spread of misinformation across the population by 98.0%.

Optimization of School Bus Transport using Application of Graph Theory and Python

Nihar Venkat Ramtej, Bhavya Sri Santhoshi and Kanakadurga Devi

CT

¹Hyderabad Public School, 3-8-152, A-K, Ramanthapur, Hyderabad, Telangana-500013

²Delhi Public School Beeramguda, Survey No:11, Near Indiramma Colony, Patelguda, Sangareddy, Hyderabad-502319 ³Principal, Delhi Public School Beeramguda, Survey No:11, Near Indiramma Colony, Patelguda, Sangareddy, Hyderabad-502319

Student transport has saturated due to on one side the student population and the other the few available buses. This paper investigates the application of graph theory using Dijkstra's algorithm and Python to optimize school transportation by determining efficient bus transport routes for each bus based on the vehicle route to transport the maximum number of students to their destination by reducing the maximum flow of children with minimum cost. The school transportation network is modeled as a weighted graph, where bus stops are represented as vertices and road distances as weighted edges. Shortest-path algorithms are used to identify routes that minimize travel distance and travel time while ensuring that students arrive at school safely and punctually. The proposed approach demonstrates how graph-theoretic techniques can improve route planning, reduce transportation costs, and enhance the overall efficiency of school bus operations. This study highlights the practical value of graph theory using Dijkstra's algorithm and Python in solving real-world transportation problems and provides a foundation for future research on intelligent school transport systems. All calculations will be carried out using Python. The result of the algorithm will be extracted and scientifically analyzed

References

- [1] Afdhaluzzikri, M., Santriawan, L. D., Sapni, E., Nugroho, M. S., Romdhini, M.U. (2024). Application of Dijkstra Algorithm in Determining Transportation Costs for Tourist Attractions in Lombok Island Based on the Shortest Path. *Jurnal Pariwisata Nusantara (JUWITA)*, 3(2), 67-73. <https://doi.org/10.20414/juwita.v3i2.11017>
- [2] N., Hussain, A., & Bokhari, S. W. A. (2025). Optimizing maritime routes. *Croatian Operational Research Reviw*, 16(1), 73-83. <https://doi.org/10.17535/corr.2025.0007>
- [3] Bao, Y. (2024), Proposed Outlook based on Different Obstacle Avoidance and Pathfinding Algorithms. *Applied and Computational Engineering*, 97(1), 83-88, <https://doi.org/10.54254/2755-2721/97/20241268>
- [4] Dhanashri Kor Nisha Satpute, Nayana Joshi, Snehal Kulkarni, Komal Walguda, & Neha Dhadiwal. (2024). Numerical Data Processing by The Implementation of Trees and Graphs. *International Research Journal on Advanced Engineering and Management (IRJAEM)*, 2(11), 3256-3260. <https://doi.org/10.47392/IRJAEM.2024.0479>

Targeting Operational Efficiency: Mathematical Modeling of Heavy Metals in Water – Treatment Optimization Through Difference Equations

Penelope Grewell and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Heavy metal contamination in drinking water remains a significant global challenge, with pollutants such as arsenic, lead, and mercury threatening the health of millions of people worldwide. While treatment and filtration technologies provide effective solutions, economic inequality and maintenance limitations often prevent sustainable implementation. This research addresses the challenge of identifying water treatment strategies that minimize long-term contamination while maximizing financial efficiency. In alignment with the United Nations Sustainable Development Goal 6.3, we developed a mathematical model using difference equations to evaluate the effectiveness of four treatment strategies under varying contamination conditions. The strategies included CT1, a constant treatment rate; CT2, a constant treatment rate with a weekly treatment boost; CT3, an adaptive treatment strategy based on heavy metal concentration levels; and CT4, a degrading treatment system representing insufficient maintenance. Across four contamination scenarios, CT2 was identified as the most cost-efficient strategy under typical contamination conditions, while CT3 demonstrated the greatest resilience across long-term and variable contamination scenarios. CT1 and CT4 failed to sustain effective treatment, highlighting the importance of adaptive and well-maintained water management systems. This model provides a framework for evaluating sustainable treatment strategies and supporting equitable access to safe drinking water.

Radical Construction and Characterization of r-Neutrosophic Primary Submodules

Preema T S¹ and Binu R²

CT

¹Research Scholar, Rajagiri School of Engineering and Technology, APJ Abdul Kalam Technological

University(KTU), Kerala, India

²Rajagiri School of Engineering and Technology, APJ Abdul Kalam Technological University(KTU),

Radical theory is an important area of algebra that studies rings, modules, and their substructures through special objects called radicals. These radicals help reveal important information about the structure of an algebraic system. Several radicals, including the Jacobson radical, prime radical, and nilradical, have been widely studied in classical algebra. They have contributed significantly to the understanding of prime ideals, prime submodules, primary decomposition, and quotient structures. Because of its ability to describe structural properties, radical theory has become a useful tool in many branches of algebra. As mathematical models have evolved to represent uncertainty and incomplete information, many classical algebraic concepts have been extended to more general settings. In particular, intuitionistic fuzzy algebra has provided a framework for studying algebraic structures with uncertainty. Within this framework, radical concepts have been introduced for intuitionistic fuzzy ideals and submodules, leading to results that resemble those found in classical radical theory. These developments have shown that radicals remain valuable in studying generalized algebraic systems. **Objective** This paper aims to develop a radical-theoretic approach for the study of r-neutrosophic submodules. The notion of the radical of an r-neutrosophic submodule is introduced as a natural extension of classical radical concepts. The work seeks to establish the fundamental properties associated with this new structure and to examine its algebraic behavior. Special emphasis is placed on understanding the connection between radicals and r-neutrosophic ideals. The associated r-neutrosophic characteristic function of the radical is also investigated. In addition, the study explores the role of radicals in the characterization of r-neutrosophic primary submodules. Several relationships between radicals and primary structures are derived and analyzed. The paper further aims to provide a unified framework for studying these concepts within neutrosophic module theory. Through these investigations, new insights into the structural properties of r-neutrosophic modules are obtained. The overall objective is to broaden the scope of radical theory in the neutrosophic setting and stimulate further research in this direction. **Methodology** The results presented in this paper are obtained through a theoretical and proof-oriented approach. The study begins by reviewing the basic concepts required for the development of the theory, including neutrosophic sets, r-neutrosophic sets, ideals, and submodules. A suitable definition of the radical of an r-neutrosophic submodule is then formulated. Fundamental properties of the newly defined radical are established through rigorous mathematical proofs.

The associated characteristic function is examined to provide additional structural information. The developed theory is subsequently applied to investigate the properties of r -neutrosophic primary submodules and their behavior under module homomorphisms. **Scope and Contribution** This study contributes to the advancement of neutrosophic algebra by extending radical-theoretic concepts to the setting of r -neutrosophic submodules. It provides a detailed study of radicals and establishes several of their essential properties. A characterization of the radical through its corresponding r -neutrosophic ideal and characteristic function is obtained. The paper further examines the interaction between radicals and r -neutrosophic primary submodules. Various results describing these relationships are proved and discussed. The preservation of primary properties under homomorphic images and inverse images is also established. These findings strengthen the theoretical foundation of r -neutrosophic module theory. Furthermore, the concepts and techniques developed in this work may serve as useful tools for future investigations involving radicals, primary structures, and related neutrosophic algebraic systems.

A Difference-Equation Framework for Modeling Organ Transplantation Dynamics

Radha Srinivasula and Padmanabha Seshaiyer

CT

George Mason University

Organ transplantation is a life-saving process in which patients with failing or damaged vital organs receive donor grafts to improve their condition. However, in the United States, the demand for organs consistently exceeds the available supply, leading to long waiting times and limited access to transplantation. To better understand these dynamics, this project develops a difference equation model to examine the waitlist and donor populations, median waiting time, and 3-year survival rates. Data from the Organ Procurement and Transplantation Network (OPTN) on living, donors, total donors, and transplant procedures was used to estimate key model parameters, while additional parameters were determined through biologically and clinically motivated assumptions. The results demonstrate how differences between donor availability and transplant demand influence waitlist growth, while changes in median waiting time and survival rates illustrate additional factors affecting transplant system dynamics. Equilibrium analysis identified conditions required for system stability, while donor-scarcity analysis characterized waitlist behavior when donor supply acts as the primary limiting constraint. Overall, this model provides a mathematical framework for understanding the intricacies of the transplantation system, aligning with the UN Sustainable Development Goal 3: Good Health and Well-Being.

MDS Spectral Codes Beyond Generalized Reed-Solomon Codes

Ramsujithra R S¹ (), Suresh Babu N² and Preenu C S³*

CT

¹ Department of Mathematics, Christian College, Kattakada, Thiruvananthapuram, India

² Department of Mathematics, University of Kerala, Kariavattom, Thiruvananthapuram, India

³ Department of Mathematics, University College, Thiruvananthapuram, India.

Error-correcting codes are essential tools for ensuring reliability in data storage and communication systems. Among them, Maximum Distance Separable (MDS) codes are considered optimal. A large and important class of MDS codes is given by Generalized Reed–Solomon (GRS) codes. In this paper, we investigate and analyze MDS spectral codes. We establish conditions under which the resulting spectral codes achieve the MDS bound and investigate whether they belong to the GRS family or not. In particular, the Schur product and Schur-square dimensions of the codes are employed as invariants under monomial equivalence. The results demonstrate that spectral constructions provide a systematic approach for obtaining MDS codes beyond the classical GRS framework and contribute to the broader study of the structure and classification of non-GRS MDS codes.

References

[1] W Cary Huffman, Jon-Lark Kim, and Patrick Sol'e. Concise encyclopedia of coding theory. Chapman and Hall/CRC, 2021

[2] Florence Jessie MacWilliams and Neil James Alexander Sloane. The theory of error- correcting codes, volume 16. Elsevier, 1977

[3] N Suresh Babu, B Ravivarma, EM Elsayed, and KG Sreekumar. Spectral codes of real symmetric operators for error correction. Discrete Mathematics, Algorithms and Applications, 17(04):2450055, 2025.

K-Preorder, C-Preorder and Intuitionistic Fuzzy Filter

Resmi G Nair*

CT

Research center of Mathematics, St.Thomas College Palai, Kottayam, Kerala, India.

This paper introduces the concept of K-preorder and C-preorder in the framework of intuitionistic fuzzy sets. We investigate their fundamental properties and examine their relationship with intuitionistic fuzzy topological space. In particular, we study the structures of intuitionistic fuzzy topologies induced by these preorders and analyse the connections between preorder relations and the corresponding topological properties. Furthermore, we derive the relationship between K-preorder and intuitionistic fuzzy filters, highlighting their role in the characterisation of intuitionistic fuzzy topological structures.

References

- [1] Atanassov K. , Intuitionistic fuzzy sets, Fuzzy Sets and Systems, vol20, no.1, (1986), 87-96
- [2] Atanassov K. , Intuitionistic fuzzy sets theory, Springer, Berlin, (1999)
- [3] Chang C. L. , Fuzzy topological space, Journal of mathematical analysis and applications, 24, (1968), 182-190
- [4] Coker D. , An introduction to intuitionistic fuzzy topological space, Fuzzy Sets and Systems, 88, (1977), 81-89
- [5] Madhavan Namboothiri N. M. , Order induced L-fuzzy topological properties, Advances in Fuzzy Sets and Systems, Vol. 9, no.2, (2011), 167-197
- [6] Sadik Bayhan and Coker D. , On separation axioms in intuitionistic topological space, IJMMS, 27(10), (2001), 621-630
- [7] Selvanayagi S. , Gnanambal Ilango, Homeomorphisms in intuitionistic topological spaces, Annals of fuzzy mathematics and informatics , Vol. 11, No.6, (June 2016), pp. 957-966
- [8] Thirumalaiswami M. , Ramesh K. , On some new continuous mappings in intuitionistic topological space , vol 2, (2013), 99-108. Zadeh L. A. , Fuzzy sets, Inform. Contra. 8, (1965), 338-353

Convex Skeletons and Reliable Centrality: A Case Study on the 2025 Papal Conclave Network

Rintu Jose*, Divya Sindhu Lekha, Rubell Marion Lincy G

CT

Department of Computer Science and Engineering, Indian Institute of Information Technology Kottayam, Palai-686635, Kerala, India

Complex real-world networks are often dense, with many overlapping and redundant connections. Computing standard measures such as shortest paths, connectivity, or centrality on such networks can be computationally expensive. The cost grows quickly as network size grows. A natural question is whether such a network can be simplified into a smaller, sparser structure that preserves its essential connectivity and clustering properties, making these computations considerably cheaper. The mathematical notion of network convexity offers a promising route to this problem. Marc and ubelj [1] define an induced subgraph as convex if every shortest path between its own nodes lies entirely within it. They show that many real networks split into a non-convex core and an already convex periphery, detectable through a simple convex hull expansion procedure. Building on this, ubelj [2] defines a convex skeleton as the largest part of a network that becomes convex after removing the fewest possible redundant edges. Edges are pruned based on how little damage they cause to local clustering. This turns the dense core into a tree of cliques. Tree-like and clique-like structures make many otherwise expensive graph problems solvable in polynomial time [4],[5]. So, this skeleton offers a mathematically grounded way to reduce computational cost. Real-world social networks provide a natural setting to test this idea. One recent example is the College of Cardinals network, studied by Soda, Iorio, and Rizzo [3]. The Cardinals network was assembled before the 2025 papal conclave, combining shared Vatican committee memberships and episcopal consecration ties into a single graph. Using eigenvector and betweenness centralities, the authors identified likely papal candidates. Notably, they found that Cardinal Robert Prevost, largely overlooked by public predictions, held one of the strongest positions in the network. Surprisingly, he was elected Pope Leo XIV shortly after. We identified the convex skeleton of this network of 186 nodes and 2884 edges by applying the convex hull expansion [1] and edge pruning [2]. The convex hull expansion procedure detected a densely connected core of 104 cardinals within an already convex periphery of 82 cardinals. We then extracted a convex skeleton by removing redundant edges from the core. This removed 1032 of 2149 possible core edges, which is about 48 percent. Comparing centrality before and after pruning, Cardinal Prevost remains ranked first in eigenvector centrality throughout, and his betweenness rank improved from fourth to first. We also tested the rank correlation between the two networks. Across all cardinals, the correlations between the original and pruned networks remain high for both centralities (Spearman's $\rho = 0.93$ for eigenvector centrality and 0.80 for betweenness centrality). These results are promising for two reasons. First, a convex skeleton contains only about half of the original edges and thus can be analysed at a fraction of the original computational cost. It still preserves essentially

the same centrality ranking. Second, and more strikingly, some genuinely influential nodes rise in rank once redundant, overlapping connections are pruned away. This suggests that convex skeletons do not merely approximate a network cheaply, but can actively sharpen its true structural signal. This positions convex skeleton extraction as a promising tool for the efficient and reliable analysis of large, dense real-world networks.

References

- [1] T. Marc, L. Šubelj, Convexity in complex networks, *Network Science*, 6(2) (2018), 176-203.
- [2] L. Šubelj, Convex skeletons of complex networks, *Journal of the Royal Society Interface*, 15(145) (2018), 20180422.
- [3] G. Soda, A. Iorio, L. Rizzo, In the network of the conclave: Social connections and the making of a pope, *Social Networks*, 83 (2025), 215-232.
- [4] F. Gavril, The intersection graphs of subtrees in trees are exactly the chordal graphs, *Journal of Combinatorial Theory, Series B*, 16(1) (1974), 47-56.
- [5] N. Robertson, P.D. Seymour, Graph minors. II. Algorithmic aspects of tree-width, *Journal of Algorithms*, 7(3) (1986), 309-322.

Solving for Optimal Staffing: An Inverse Queuing Approach to Equity-Weighted Physician Allocation Across New Jersey Eds

Rithvik Sharma and Padmanabhan Seshaiyer

CT

[George Mason University](#)

New Jersey emergency departments face persistent problems with patient wait times, jeopardizing both patient health and the operational efficiency of medical centers. Many existing allocation models aim to solve this problem by reducing wait times; however, most assume that all hospitals are equally deserving, failing to account for the fact that understaffing burdens vulnerable populations disproportionately. This leads to algorithms allocating more physicians to medical centers with greater resources, since those resources yield higher returns on allocation when compared to their understaffed counterparts, while hospitals with fewer resources receive less physician allocation and face increased understaffing and overcrowding. We model each ED as an M/M/c queue and use root-finding (brentq) to back-calculate each hospital's implied physician capacity from observed CMS wait times. Arrival rates are estimated via a bed-share allocation of county-level ED visit volumes (NJSHAD) to individual facilities (HCRIS). We then formulate an equity weighted greedy allocation algorithm, provably within $(1-1/e)$ of optimal by submodularity, that distributes additional physician capacity across hospitals in a network, critically accounting for counties with higher CDC Social Vulnerability Index scores within our distribution algorithm. Applying this methodology, we achieved an average wait time reduction of 41.30 minutes across 22 NJ hospitals in our study upon distributing 40 doctors across the network, while only sacrificing 0.1 minutes of mean wait time reduction for 5 points of redirected allocation towards more vulnerable counties. This approach enables a more equitable and effective distribution of physicians across NJ emergency departments, leading to improved wait times and better patient care.

An ε -Uniform L1-Finite Element Method for Singularly Perturbed Time-Fractional Delay Parabolic Problems

Roselin Priscilla Stephen and Franklin Victor

CT

^{1*2}Department of Mathematics, Bishop Heber College,
Affiliated to Bharathidasan University, Tiruchirappalli, 620017, Tamil Nadu, India .

1. Motivation and Problem Addressed Singularly perturbed parabolic partial differential equations involving non-local memory kernels and delay mechanisms model multi-scale physical processes such as subdiffusive microscale thermal transport in biological tissue, viscoelastic fluid dynamics, and neural signal conduction with transmission lags [1]. The simultaneous presence of a small perturbation parameter $\varepsilon \in (0, 1]$, a Caputo time-fractional derivative of order $\alpha \in (0, 1)$, and constant delays in time $\tau_1 > 0$ and space $\tau_2 \in (0, 1)$ introduces three severe numerical difficulties: A weak initial temporal singularity at $t = 0$ where derivative blow-up $\partial_t^k u \sim t^{\alpha-k}$ degrades standard time-stepping schemes, boundary layers of width $O(\sqrt{\varepsilon} \ln N)$ near the spatial endpoints $x = 0$ and $x = 1$ that induce spurious spatial oscillations on uniform grids, and an interior interface layer across the spatial delay transition line $x = \tau_2$ caused by derivative jump discontinuities. This work considers the initial-boundary value problem on $\Omega \times (0, T] = (0, 1) \times (0, T]$:

$${}_0^C D_t^\alpha u(x, t) - \varepsilon \frac{\partial^2 u}{\partial x^2}(x, t) + a(x)u(x, t) = b(x)u(x - \tau_2, t) + c(x)u(x, t - \tau_1) + f(x, t), \quad (1)$$

subject to Dirichlet boundary conditions and historical initial functions on $[-\tau_2, 0] \times [-\tau_1, T] \cup [0, 1] \times [-\tau_1, 0]$. **Methodology** To construct a parameter-uniform numerical scheme that resolves all localized layers without requiring ε -dependent step sizes, we formulate a coupled discretization: The Caputo derivative is approximated via the classical L1 scheme over a temporally graded mesh

$$t_k = T \left(\frac{k}{M} \right)^r, \quad k = 0, 1, \dots, M,$$

where

$$r \geq \frac{2 - \alpha}{\alpha}$$

compensates for the initial singularity at $t = 0$. The spatial domain is discretized using a continuous piecewise linear (P_1) Galerkin Finite Element Method over a piecewise-uniform condensed Shishkin mesh with transition parameters

$$\lambda_1, \lambda_2 = O(\sqrt{\varepsilon} \ln N)$$

strategically placed around the boundaries and the internal delay point $x = \tau_2$. **Key Results** We establish an asymptotic decomposition of the continuous solution into regular, boundary layer, and interior layer components. By deriving a discrete fractional

coercivity property and employing a discrete fractional Grönwall inequality tailored for delay terms, unconditional discrete stability in the energy norm is proven. Furthermore, we prove the parameter-uniform error estimate: $\max_{1 \leq k \leq M} \|u(\cdot, t_k) - u_h^k\|_\varepsilon \leq C (M^{-\min\{r\alpha, 2-\alpha\}} + N^{-1} \ln N)$. where $\|\cdot\|_\varepsilon$ is the standard energy norm and C is independent of ε , N , and M . Numerical experiments confirm the sharpness of the $\mathcal{O}(M^{-(2-\alpha)} + N^{-1} \ln N)$ rate.

References

- [1] M. Stynes, E. O’Riordan, and J. L. Gracia, Error analysis of a finite difference method on graded meshes for a time-fractional diffusion equation, *SIAM J. Numer. Anal.*, 55(2):1057–1079, 2017.

Study of a Depleting Resource and Consumers Model Using Physics-Informed Neural Networks with Parameter Estimation

Rovan Vaz¹, P. Danumjaya²

CT

^{1,2}Department of Mathematics, BITS-Pilani KK Birla Goa Campus, Goa, India

This study develops a system of non-linear coupled differential equations to model the dynamic interactions between a depleting natural resource such as food, minerals, etc., and its consumers, drawing structural inspiration from the cooperative network frameworks. We first establish the existence, uniqueness, and asymptotic stability of the system solutions and derive the conditions on the parameters required for global asymptotic stability. To overcome the limitations of classical numerical solvers, we employ Physics-Informed Neural Networks (PINNs) to simulate and predict system state trajectories. We estimate the key parameters involved in the model using the inverse problem. The results demonstrate that PINNs provide trajectories that consistently converged asymptotically to the exact steady-states predicted by our analytical stability analysis. Several examples are provided to illustrate the results established.

References

- [1] M. Raissi, P. Perdikaris, and G. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics*, pp 686-707 (2019)
- [2] P. Raja Sekhara Rao, K.V. Ratnam, and G. Shirisha, Mathematical Model on Influence of Past Experiences on Present Activities of the Human Brain, *Computational and Mathematical Biophysics*. pp 1-23 (2025)
- [3] V. S. H Rao, and P. Raja Sekhara Rao, Cooperative and Supportive Neural Networks, *Physics Letters A* 371, pp 101-110 (2007). 1

Bi-Criteria Minimum-Charge FlowLoc Model with Excess Storage Using the ε -Constraint Method

Sachin Wagle and Tanka Nath Dhamala

CT

¹Institute of Engineering, Purwanchal Campus, Tribhuvan University, Dharan, Nepal ²Central Department of Mathematics, Tribhuvan University, Kathmandu, Nepal.

In a network, locating a facility on an arc reduces its capacity, which may consequently decrease the network's maximum flow value. However, this reduction can be minimized by locating the facility at an optimal location. On the other hand, if there is a limited budget for installation costs, the facility cannot be located arbitrarily. Moreover, due to the bottle-neck capacity, the entire flow leaving the source may not reach the sink. If the intermediate nodes have storage capacities, the excess flow can be stored there. In this presentation, by integrating the above concepts, we investigate a mathematical formulation of a bi-criteria optimization problem in which one objective minimizes the facility locating charge, while the other minimizes the reduction in the maximum flow value by placing the facility at an optimal location under the given budget. We present algorithms based on the ε -constraint method to solve the proposed problem

References

- [1] Ehrgott M. Multicriteria optimization. Springer Science & Business Media, 2005.
- [2] Hamacher H.W., Heller S. and Rupp B. Flow location (FlowLoc) problems: dynamic network flows and location models for evacuation planning. Annals of Operations Research 207, 2013, 161–180.
- [3] Pyakurel U. and Dempe S. Network flow with intermediate storage: models and algorithms, SN Operations Research Forum, 1, 2020, 1–23

Comparative Study of Arterial and Venous Characteristics in the Riemann Problem for Blood Flow with Transport

Sahadeb Kuila

CT

Department of Mathematics, SRM Institute of Science and Technology,
Kattankulathur, India

This study investigates the Riemann problem for one-dimensional arterial and venous blood flow models with passive scalar transport and vessel-specific tube laws. Using the method of characteristics, we analyze shock waves, rarefaction waves, and contact discontinuities, establish the existence and uniqueness of Riemann solutions, and derive conditions for vacuum formation. We show that vacuum states may occur in arteries but not in veins, compare their wave structures in the phase plane, visualize all elementary wave combinations, and numerically quantify wave intensities for both vessel types.

References

- [1] A. Spilimbergo et. al., Flux vector splitting schemes applied to a conservative 1D blood flow model with transport for arteries and veins, *Comput. Fluids*, 271 (2024) 106165
- [2] B. Brook et. al., Numerical solutions for unsteady gravity-driven flows in collapsible tubes: evolution and roll-wave instability of a steady state, *J. Fluid Mech.* 396 (1999) 223-256
- [3] J. A. Smoller, *Riemann Problems and the Theory of Shock Waves*, Springer, New York, (1994)

From Susceptibility to Survival: A Compartmental Approach to Maternal Health Under Climate Stress

Sahasra Rao and Padmanabhan Seshaiyer

CT

[George Mason University](#)

Maternal mortality remains unreasonably high, and reflects a major shortcoming of public health. This is especially the case for lower and lower-middle income countries, with cases of maternal mortality much higher in these areas. While about one woman dies every two minutes from childbirth or other maternal complications, more than 80% of these cases are deemed preventable. Motivated by UN SDG 3.1, this model utilizes a compartmental system of ordinary differential equations to formalize the flow of women in maternal care within each state, all the way from pregnancy, to a safe exit, or death. We rigorously prove the existence of exact solutions through the Cauchy-Lipschitz theorem, establish exact solutions through the integration factor method, and evaluate the model's stability and equilibrium points to determine points of intervention. This model further utilizes nonlinear least squares regression to accurately estimate parameters through Python. Furthermore, we accurately derive the probability of death from this model using data-driven parameters from Sub-Saharan Africa and compare it to empirical data. We establish the most sensitive factors through a sensitivity analysis, thus establishing points of intervention and care. Building on this, we add an aid compartment that is a function of foreign aid and the parameter it aims to improve, enabling a shift from a broad point of intervention to a numerical and actionable form of aid. By merging aid intensity and parameter sensitivity, we can determine which interventions lead to the most reduction for a given unit of aid, turning the model into a cost-efficient model rather than just predictive

Uncertainty-Aware Diagnosis Support using Monte Carlo Dropout for Confidence-Calibrated Risk Prediction

Sandra K S¹

CT

¹Department of Mathematics, Amrita Vishwa Vidyapeetham, Coimbatore, India

The classical approach of utilizing neural network classifiers gives a single point estimate without providing any information about the level of confidence of the provided estimate. This is a very serious drawback that may lead to negative consequences in the area of clinical diagnostics, since in some cases, a confident wrong prediction may result in greater problems than an uncertain one. The application of MC Dropout, an efficient variational approximation to Bayesian inference, to create a diabetes risk predictor trained on the Pima Indians Diabetes dataset and thus gain uncertainty estimates is considered. Dropping neurons at the inference stage and 100 stochastic forward passes per sample allow obtaining not a point but a probability distribution of the prediction that leads one calculate such measures of uncertainty as predictive mean, predictive standard deviation, and predictive entropy at the cost of minor computation. Test accuracy reached with the model is 72.7% with the Area Under the Curve on Receiver Operating Characteristics being equal to 0.817. It is revealed that the higher the predictive uncertainty is, the less likely the patient is to be predicted correctly—misclassified patients have the predictive standard deviation and predictive entropy of 0.117 and 0.535, respectively, while correctly classified ones have 0.087 and 0.312, correspondingly. A reliability diagram enables us to measure calibration of the model; its expected calibration error equals 0.101.

References

- [1] Asgharnezhad, H., et al., Enhancing Monte Carlo Dropout Performance for Uncertainty Quantification, *Neural Computing and Applications*, 38 (2025)
- [2] Kurz, A., et al., Uncertainty Estimation in Medical Image Classification: Systematic Review, *JMIR Medical Informatics*, 10 (2022)
- [3] Loftus, T., et al., Uncertainty-Aware Deep Learning in Healthcare: A Scoping Review, *PLOS Digital Health*, 1 (2022)
- [4] Hassan, M.M., Ismail, H.R., Bayesian Deep Learning Applied to Diabetic Retinopathy with Uncertainty Quantification, *Heliyon*, 11 (2025)

A Mathematical Framework for Weighted Neutrosophic Principal Components in Machine Learning

Shereeta D'cotha¹ and Binu R²

CT

¹ Research Scholar, Rajagiri School of Engineering and Technology, APJ Abdul Kalam Technological University (KTU), Kerala, India

²Rajagiri School of Engineering and Technology, APJ Abdul Kalam Technological University

The increasing complexity and uncertainty associated with modern machine-learning datasets suggest the need for effective mathematical approaches for data representation, dimensionality reduction, and preservation of essential information. Conventional dimensionality reduction techniques such as Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) are widely used for feature extraction, data compression, and representation learning. However, these techniques are primarily formulated for precise numerical data and may not suitably represent indeterminacy and vague information present in many real-world datasets. Neutrosophic sets provide a mathematical framework for representing truth, indeterminacy, and falsity components simultaneously. This study proposes a weighted neutrosophic PCA–SVD framework for dimensionality reduction and machine-learning applications. The proposed approach represents the dataset through truth, indeterminacy, and falsity components and applies SVD and PCA to identify important features. The selected neutrosophic principal components are then combined using optimized weights corresponding to their truth, indeterminacy, and falsity components. The weights are determined in a data-driven manner so that the relative contribution of each component is incorporated into the reduced feature representation. The resulting weighted neutrosophic principal components are used as inputs to a machine-learning model. The proposed framework establishes a systematic connection between neutrosophic representation, dimensionality reduction, and machine learning, with the aim of preserving uncertain and indeterminate information while obtaining a relatively compact feature representation. **Objective** The primary objective of this study is to develop a mathematical framework for dimensionality reduction of neutrosophic data by integrating SVD and PCA with a weighted combination of truth, indeterminacy, and falsity components. The study aims to identify important neutrosophic principal components based on their information contribution, develop a data-driven weighted combination of the selected components, and investigate the effectiveness of the resulting reduced representation when used as input features for machine-learning models. A further objective is to compare the proposed approach with conventional PCA-based dimensionality reduction in terms of predictive performance and the number of retained components. **Methodology** The proposed study follows a mathematical and computational approach. A numerical dataset is first represented in neutrosophic form using three components corresponding to truth, indeterminacy, and falsity information. SVD is applied to the three component matrices to identify their underlying structure, while

PCA is employed to determine the dominant principal directions and their contribution to the data variation. The relationship between SVD and PCA is used to establish a unified dimensionality-reduction procedure for the neutrosophic representation. The resulting neutrosophic principal components are ranked according to their contribution to the information content, and the most important components are retained to obtain a lower-dimensional representation. For each selected principal component, the corresponding truth, indeterminacy, and falsity projections are combined using a weighted aggregation. The weights are determined through a constrained, data-driven optimization procedure, allowing the relative contribution of the three neutrosophic components to be learned from the data. The resulting weighted neutrosophic principal components are then supplied to a machine-learning model for prediction. The proposed framework will be evaluated using suitable performance measures and the number of retained components. Comparative experiments with conventional PCA-based dimensionality reduction will be conducted to assess the effectiveness of the proposed approach.

Conclusion The scope of this study is focused on the mathematical development and computational investigation of a neutrosophic dimensionality-reduction framework based on SVD and PCA for machine-learning applications. The initial study will consider numerical datasets and a regression-based machine-learning model to demonstrate how truth, indeterminacy, and falsity information can be incorporated into feature extraction and prediction. The main contribution is the development of a weighted neutrosophic PCA-SVD framework in which the three components of selected neutrosophic principal components are combined using data-driven weights determined through constrained optimization. This provides a systematic connection between neutrosophic matrices, SVD, PCA, dimensionality reduction, and machine learning while retaining information associated with indeterminacy and conflicting data. The framework can provide a foundation for future extensions to classification, nonlinear machine-learning models, larger datasets, and real-world applications involving uncertain, incomplete, or inconsistent information.

Modeling Tuberculosis Transmission, Policy, and Optimal Control in Resource-Limited Communities

Siddharth Vishwanath and Padmanabhan Seshaiyer

CT

George Mason University

Tuberculosis (TB) remains one of the deadliest infectious diseases, especially in resource-limited communities found within Asia and Africa. Standard compartmental models like the SIR model focus on disease dynamics but rarely include or focus on the effects of a public-health policy that governs disease dynamics. We introduce SLIDR-P, a policy-coupled compartmental model that extends the SIR framework with a non-infectious latent stage (L), a disease-specific mortality compartment (D), and a public-health policy variable (ρ) that simultaneously affects transmission, progression, recovery, and mortality. Through mathematical analysis, we show that the model is well-posed with non-negativity and boundedness. We also consider ρ , not as a fixed input or reactive differential equation, but as a control, applying Pontryagin's Maximum Principle to derive an optimal policy trajectory minimizing active infection and mortality, considering the non-linear cost of improved policy effectiveness. We note that against reactive policy curves, optimal policy curves increasingly reduce deaths and peak active infections as we increasingly prioritize minimizing active infection and death relative to policy cost. To address the lack of specific data that categorizes TB data as we do in our model, we implement a physics-informed neural network (PINN) that recovers key transition rates through synthetic data that is transformed under Gaussian noise. SLIDR-P combines compartmental modeling, optimal control, and PINNs to describe and inform TB.

A Parameter Robust Nonstandard Finite Difference Scheme for the One-dimensional Burgers' Equation

Sona Santhosh¹, Ansamol M. A.¹, and Ashish Awasthi¹

CT

¹Department of Mathematics, National Institute of Technology Calicut, Kerala, India

This paper presents a parameter-robust numerical method for solving the one-dimensional singularly perturbed Burgers' equation, which represents a special class of quasilinear parabolic problems with a small viscosity coefficient. The governing equation is first discretized in time using the implicit Backward Euler method on a uniform temporal mesh, resulting in a nonlinear system at each time level. To avoid solving the nonlinear system directly, the nonlinear convective term is linearized using a time-lag (lagged coefficient) approach, in which one of the solution components in the nonlinear convection term is evaluated at the previous time level, thereby transforming the nonlinear problem into a sequence of linear systems. The resulting linearized problem is then discretized in space using an appropriate Nonstandard Finite Difference (NSFD) scheme. Richardson extrapolation is further employed to improve the accuracy and convergence behavior of the proposed method. A rigorous theoretical analysis is carried out to establish the reliability of the proposed numerical scheme. Several numerical experiments are presented to demonstrate the accuracy, efficiency, and robustness of the method.

References

- [1] M.K. Kadalbajoo, K.K. Sharma and A. Awasthi, A parameter-uniform implicit difference scheme for solving time- dependent Burgers' equations, *Applied Mathematics and Computation*, 170 (2005), 1365–1393
- [2] M.K. Kadalbajoo and A. Awasthi, A numerical method for singularly perturbed Burgers' equation, *Applied Mathematics and Computation*, 182 (2006), 1430–1442
- [3] R.E. Mickens, *Nonstandard Finite Difference Models of Differential Equations*, World Scientific Publishing, Singapore, 1994
- [4] J.J.H. Miller, E. O'Riordan and G.I. Shishkin, *Fitted Numerical Methods for Singular Perturbation Problems*, Revised Edition, World Scientific Publishing, Singapore, 1996

Profile-Conditioned Adaptive-Risk TD3 for Personalized Multi-Asset Portfolio Management

Sourudra Nag^{1,} and Aswin V. S.¹*

CT

¹School of Digital Science,
Kerala University of Digital Sciences, Innovation and Technology,
Thiruvananthapuram, Kerala, India

Portfolio management is defined as a technique for allocating capital between several financial assets in terms of the investment return and risks. Deep Reinforcement Learning (DRL) appears to be a promising method to use for portfolio management due to its capability to make sequential decisions regarding asset allocation through interaction with the market; TD3 has already been used for multi-asset trading with continuous actions and transaction costs [1]. However, the risk aware and personalized reinforcement learning methods used by investors usually express their preferences through individual strategies, predefined risks, or objective risks, thus providing some room for a strategy that can keep changing its behavior depending on investor risk preferences. In order to solve the problem stated above, we consider an approach of Profile-Conditioned Adaptive-Risk Twin Delayed Deep Deterministic Policy Gradient (PCAR-TD3) which can be used for personal multi-asset portfolio management. Particularly, our framework is built upon: (1) profile-conditioned policy learning where risk tolerance of the investor is modeled using a continuous variable and included as part of the state space along with market information and current portfolio state, allowing one TD3 policy to be able to adopt a portfolio allocation strategy under conservative and aggressive investor profile without adding any portfolio constraint tailored for the two different types of investors; and (2) adaptive risk-aware reward learning where portfolio return is taken into account along with downside risk, transaction cost, portfolio turnover, and market stress, and the risk penalty is determined based on the investor's risk tolerance and market conditions. Consequently, conservative investors can be guided towards strong downside-risk mitigation while investors with high risk tolerance can have relatively high portfolio risk. Initial findings reveal that the policy suggested in this paper is sensitive to shifts in risk appetite among investors, and portfolios with more risk-tolerant profiles tend to be associated with more risk exposure and high portfolio risks compared to those with conservative profiles, despite sharing the same constraints on portfolio construction. Moreover, the results show that such differences arise due to the ability of the profile-conditioning method to learn from data rather than setting constraints on portfolios based on different profiles manually.

References

- [1] T. Kabbani and E. Duman, "Deep Reinforcement Learning Approach for Trading Automation in The Stock Market," IEEE Access, 2022. doi: 10.1109/ACCESS.2022.3203697
- [2] Y. Jiang, J. Olmo, and M. Atwi, "Deep Reinforcement Learning for Portfolio Selection," Global Finance

An Algorithm for γ_P -sets in Mycielskians of n-spiders

Sreethu K¹, Seema Varghese¹ and Seethu Varghese²

CT

¹Department of Mathematics, Government Engineering College, Thrissur, A P Abdul Kalam Technological University, Kerala, India,

²Department of Mathematics, Bharata Mata College, Thrikkakara, Kerala, India

Power domination arose from placing Phasor Measurement Units in electric power networks. Haynes et al. [1] introduced its graph model using domination and propagation rules. A power dominating set is a vertex set that monitors the whole graph. A minimum such set is called a γ_P -set. Its size is the power domination number $\gamma_P(G)$. The power domination problem is NP-complete for general graph classes [1]. Exact results for structured graph families are therefore important. They help identify the graph features that control monitoring and propagation. Mycielski [2] introduced the Mycielskian while studying triangle-free graphs with large chromatic number. For a graph G , the Mycielskian (G) contains the original vertices, their twins, and a root. An n -spider SP_n is formed from n spiders whose heads lie along a path. Each attached spider is called a kid. Kids are grouped as e -kids, s -kids, or m -kids based on their odd legs. Their order along the path gives the head sequence of SP_n . Varghese, Varghese, and Vijayakumar [3] studied power domination in Mycielskians of spiders. They classified the spiders whose Mycielskians have power domination number one or two. Sreethu, Varghese, and Varghese [4] extended this study to Mycielskians of n -spiders. They established $1 \leq \gamma_P(SP_n) \leq n$. They also identified the cases where either bound is attained. The work showed that every value from 1 to n can occur. However, a general method for finding a minimum set was still needed. The present paper fills this gap. It gives a linear algorithm for finding a γ_P -set of (SP_n) . This study gives a complete method for finding a γ_P -set of any Mycielskian of an n -spider. Earlier work gave bounds and solved selected cases. The present paper covers every possible head sequence. The study also identifies the structural features that control the minimum set. The main factors are m -bundles, odd groups of e -bundles, and nearby s -bundles. Their order determines where extra selected vertices are required. This paper identifies a graph family where the power domination problem has an exact and efficient solution. The same structural approach may also guide the study of related Mycielskian graph classes.

Limiting Behaviour of the Sample Autocorrelation and Partial Autocorrelation Functions in the Presence of a Doublet Outlier and Its Impact on Parameter Estimation

Suresh R

CT

Department of Statistics, Bangalore University, Bengaluru-560056, India

The occurrence of outliers in time series is common in many practical applications, and their detection and appropriate treatment are essential because they can adversely affect model identification, parameter estimation, and statistical inference. This paper investigates the effect of a doublet outlier on the Sample Autocorrelation Function (SACF) and the Sample Partial Autocorrelation Function (SPACF). The limiting forms of the SACF and SPACF are analytically derived as the magnitude of the doublet outlier, denoted by δ , tends to infinity, and are shown to be independent of the underlying time-series model and its parameter values. The implications of this model-independent limiting behaviour for the estimation of the parameters of the underlying AR, MA, and ARMA models are subsequently investigated. Since the limiting results are obtained under the theoretical condition $\delta \rightarrow \infty$, a comprehensive simulation study is carried out to determine the minimum magnitude of δ required for the finite-sample SACF and SPACF to closely approximate their limiting forms for different sample sizes, model classes, and parameter values. The simulation study validates the theoretical results and identifies practical thresholds beyond which the limiting behaviour provides an accurate approximation. The analytical and simulation results provide a comprehensive understanding of the influence of a large doublet outlier on autocorrelation-based diagnostics and parameter estimation, thereby offering useful insights into the robustness of time-series model identification and estimation procedures in the presence of extreme outlying observations

References

- [1] Box, G. E. P., Jenkins, G. M., Reinsel, G. C., and Ljung, G. M. (2015), Time Series Analysis: Forecasting and Control, 5th ed., Wiley.
- [2] Fox, A. J. (1972), Outliers in Time Series, Journal of the Royal Statistical Society: Series B, 34, 350–363.
- [3] Chan, W. S. (1995), Understanding the Effect of Time Series Outliers on Sample Auto correlations, TEST, 4, 179–186.
- [4] Maronna, R. A., Martin, R. D., and Yohai, V. J. (2006), Robust Statistics: Theory and Methods, Wiley

A Spatially Coupled q-Rung Orthopair Fuzzy TOPSIS Framework for Flood-Risk Prioritization in Idukki District

Dr. Suriyapriya K¹ and Mary Pauly K²

CT

¹Assistant Professor, Department of Computational Science and Humanities, IIIT Kottayam, Kerala, India

²Research Scholar, Department of Computational Science and Humanities, IIIT Kottayam

Floods are one of the major natural disasters that cause serious damage to human life, infrastructure and the environment. In a geographically diverse region like Idukki district, flood risk is influenced by several environmental, topographical and socio-economic factors. Most of the existing flood assessment studies consider the regions independently and do not effectively include the influence of neighbouring areas and uncertainty present in the available data. Hence, this study proposes a spatially coupled q-rung orthopair fuzzy TOPSIS framework for prioritising flood-risk areas in the Idukki district. The study area is divided into 54 alternatives consisting of 52 Grama Panchayats and 2 municipalities. Different flood-related criteria such as extreme rainfall, elevation, slope, topographic wetness index, drainage density, distance from rivers, built-up area, population density, vulnerable population and accessibility are considered under hazard, exposure, vulnerability and adaptive capacity. The required spatial and socio-economic data are collected from suitable GIS and open-data sources and processed at the administrative-unit level. To represent the uncertainty in the collected data, the normalized criterion values are transformed into q-rung orthopair fuzzy information by considering the reliability of each data source. Further, a spatial adjacency matrix is introduced to include the influence of neighbouring administrative regions in the decision-making process. The spatially adjusted q-rung orthopair fuzzy decision matrix is then evaluated using TOPSIS to obtain the flood-risk priority ranking of all 54 regions. The robustness of the proposed approach is examined through sensitivity analysis with respect to the rung parameter and spatial influence parameter. The obtained results are also compared with conventional and non-spatial fuzzy TOPSIS approaches and validated using historical flood information. The proposed framework is expected to provide a more realistic and uncertainty-aware decision-support approach for identifying high-priority flood-risk areas and supporting local-level disaster preparedness and mitigation planning.

References

- [1]. Fan, Jianping and Han, Yanlong and Wu, Meiqin, CRITIC-TOPSIS method based on pairwise preference graph neural similarity and its application in vehicle selection, *International Journal of Fuzzy Systems*, 28(2026), 397-411
- [2]. Khan, Zohaib Ahmed and Jhamnani, Bharat, Development of flood susceptibility map using a GIS-based AHP approach: a novel case study on Idukki district, India, *Journal of Spatial Science*, 69(2024), 409-442
- [3]. Saravanan, Subbarayan and Abijith, Devanantham and Reddy, Nagireddy Masthan and Kss, Parthasarathy

and Janardhanam, Niraimathi and Sathiyamurthi, Subbarayan and Sivakumar, Vivek, Flood susceptibility mapping using machine learning boosting algorithms techniques in Idukki district of Kerala India, Urban Climate, 49 (2023), 101503

[4]. Pathan, Azazkhan Ibrahimkhan and Girish Agnihotri, Prasit and Said, Saif and Patel, Dhruvesh, AHP and TOPSIS based flood risk assessment-a case study of the Navsari City, Gujarat, India, Environmental Monitoring and Assessment, 194(2022), 509

[5]. Selvaraj, A and Dash, Saroj Kumar and Punithavelan, N and Felix, A, Fuzzy TOPSIS approach to identify the flood vulnerability region in South Chennai, International Journal of Pure and Applied Mathematics, 118 (2018), 667-674

Some Upper Bound of Level Energy of a Rooted Tree

Unni M S

CT

DB Pampa College, Parumala

In this paper, we study the concept of level energy of a rooted tree and establish certain upper bounds related to it. The level matrix of a rooted tree is introduced as a symmetric matrix whose entries represent the level differences between vertices. Using spectral properties and inequalities such as Cauchy-Schwarz, we derive relations between the level energy, level matrix, and distance energy of a rooted tree. Furthermore, we present several bounds involving the eigenvalues, level sums, and spectral radius of the level matrix. These results provide insights into the spectral characteristics of rooted trees and their structural properties.

A Regularized Fractional Dynamical Embedding of the Midilli–Küçük Drying Model ($n = 1$): Well-Posedness and Global Existence

Vaisakh Venu^{1,2}, Vijitha Mukundan¹

CT

¹ Department of Mathematics, Sacred Heart College (Autonomous),
University of Calicut, Chalakudy, Kerala, India

² Kelappaji College of Agricultural Engineering & Food Technology,

The Midilli–Küçük model [1] is extensively used in the mathematical modeling of thin-layer drying due to its strong agreement with experimental data. Being purely empirical, however, it lacks an underlying dynamical equation, offering no framework for studying existence, uniqueness, well-posedness, or memory effects that arise in heterogeneous porous media. A Caputo fractional differential equation [2] with a non-homogeneous forcing term is considered whose Laplace-domain representation recovers the classical Midilli–Küçük law (the case $n = 1$) in the integer-order limit. The forcing term is weakly singular at $t = 0$, producing an unbounded initial rate of change. A small time-shift regularization yields the regularized fractional dynamical Midilli–Küçük (RFDKM) model, retaining the qualitative behaviour of the original equation and eliminating the singularity. The RFDKM problem is expressed as a Volterra integral equation of the second kind on $C([0, T])$. Banach's fixed-point theorem enabled us to prove local existence, uniqueness, and continuous dependence on the initial data (Hadamard well-posedness). Employment of the equivalent Bielecki norm leads to removal of the restriction on the length of the time interval and establishes global existence and uniqueness on any finite horizon [3]. An explicit solution is obtained by Laplace transformation, which is then expressed by means of Mittag–Leffler functions and a Volterra convolution integral. This representation makes the influence of fractional memory transparent, recovering the classical model when the fractional order tends to one. The resulting framework supplies a rigorous mathematical foundation for the fractional Midilli–Küçük equation and facilitates advanced numerical approximation, parameter identification, inverse problems, and extensions to arbitrary exponents n .

References

- [1] A. Midilli, H. Kucuk, Mathematical modeling of thin layer drying of pistachio, *Journal of Food Engineering* (2002)
- [2] I. Podlubny, *Fractional Differential Equations*, Academic Press, 1999
- [3] E. Kreyszig, *Introductory Functional Analysis with Applications*, Wiley Classics Library, John Wiley & Sons, New York, 1989, reprint of the 1978 edition

Modeling the Dynamics of Behavioral Episodes and Applied Behavioral Analysis in Autism Spectrum Disorder

Vanisha Waikul and Padmanabhan Seshaiyer

CT

George Mason University

Applied Behavior Analysis (ABA) is a primary intervention for addressing behavioral episodes associated with Autism Spectrum Disorder (ASD); however, current clinical evaluations often rely on retrospective data tracking rather than dynamic models capable of forecasting long-term treatment trajectories. Although ABA provides significant therapeutic benefits, high financial costs and unpredictable treatment timelines create accessibility challenges for families seeking to evaluate progress and optimize care. To address this gap, this study develops an epidemiologically inspired compartmental framework to quantitatively model the temporal dynamics of behavioral episodes and therapeutic outcomes in ASD. Using a system of coupled ordinary differential equations, we construct a compartmental model that represents patient progression across six distinct clinical and behavioral states. Mathematical analysis establishes solution non-negativity, system boundedness, existence, and uniqueness, while identifying the theoretical Disease-Free Equilibrium of the system. Forward numerical simulations and sensitivity analyses are performed to identify key parameters influencing long-term patient stabilization and graduation. To overcome the challenge of limited clinical datasets and address the inverse problem of parameter estimation, Physics-Informed Neural Networks (PINNs) are trained on noise-injected synthetic data to recover unknown compartmental transition rates. This framework provides a foundation for predicting long-term behavioral trajectories, identifying critical treatment factors, and supporting future data-driven personalization of ABA interventions for individuals with ASD.

Precursor-Informed DTW Analog Ensembles for Early Forecasting of Quasi-Periodic Time Series

Varun Soman and Amaranathan Sabarinath

CT

¹Department of Mathematics, Maharaja's College, Ernakulam, Kerala, India ²Research Scholar, University of Kerala, Thiruvananthapuram, Kerala, India ³Vikram Sarabhai Space Centre, Thiruvananthapuram, Kerala, India

Forecasting the amplitude of a new cycle before the cycle has actually begun is one of the hardest problems in cyclic time-series prediction. Shape-based analog methods are useful when part of the target cycle has already been observed, because they can compare the developing cycle with similar historical cycles. However, before the cycle begins, no observed shape is available. In that situation, the method must rely on a generic reference shape, and the amplitude forecast becomes less reliable. In our recently published Dynamic Time Warping (DTW) analog framework for solar-cycle forecasting [4], the pre cycle peak-amplitude error was substantially larger than the error obtained after five years of observations were available. This limitation is important because many practical decisions in space-weather planning, satellite operation, and communication infrastructure must be made before a new solar cycle is well developed. This work addresses the pre-cycle forecasting problem by adding a causal physical pre cursor to the DTW analog framework. The main question is whether information available before cycle onset can improve the amplitude estimate of a shape-based analog forecast. The study combines two sources of information: a geomagnetic precursor derived from the aa index and a DTW-based analog estimate obtained from historical solar-cycle shapes. The first part of the method constructs a causal precursor from the aa geomagnetic index. Geomagnetic activity near solar minimum is a widely studied indicator of the strength of the following solar cycle, because it is related to the Sun's polar magnetic field, which acts as the seed of the following cycle's amplification. To avoid information leakage, only backward-looking smoothing is used, and the precursor window ends at the onset of each cycle. For every validation target, the precursor regression is fitted using only earlier cycles. During development, two possible sources of leakage were identified: non-causal smoothing of the geomagnetic series and indirect use of the realised cycle length. Both were removed from the final pipeline. The second part combines the precursor estimate with the DTW analog estimate. Each estimate is treated as an uncertain measurement of the unknown cycle amplitude. The uncertainty of each source is estimated from its own past walk-forward errors. The final combined estimate gives more weight to the source with smaller historical error variance and less weight to the less reliable source. Thus, the combination does not require a manually chosen mixing weight. It also remains interpretable, because the contribution of the precursor and the analog forecast can be examined separately. The third part retains the uncertainty-quantification structure of the analog method. Prefix-to-prefix DTW matching is used when part of the target cycle has been observed, so that the early portion of the target is compared with corresponding early portions of historical cycles.

Prediction intervals are obtained using a phase-aware residual bootstrap, where past forecast errors are resampled within comparable phases of the cycle. This allows the interval width to reflect the fact that forecasting difficulty changes across the cycle. The framework is evaluated using the monthly smoothed international sunspot number series under a strict walk-forward protocol. For each target cycle, only data and model fits that would have been available at the forecast date are used. On the validation cycles, the precursor and the DTW analog estimate show complementary behaviour. Before cycle onset, when no target-cycle shape is available, the combined method reduces the peak amplitude mean absolute error from 44.1 smoothed sunspots for the DTW analog alone to 22.1 smoothed sunspots. This is a reduction of about one half. When a 60-month observed prefix is available, the combination still improves the analog estimate from 9.1 to 7.6 smoothed sunspots, while the precursor alone remains near 24. This shows that neither source is uniformly best across all forecast stages. Instead, the combined method follows the stronger source as information accumulates. The method is also applied retrospectively to Solar Cycle 25. The pre-cycle combined estimate is 159.8 smoothed sunspots, compared with a provisionally observed smoothed maximum of 160.9. The corresponding DTW-only pre-cycle baseline is 188.7. In addition, the forecast issued from the published framework in December 2024 has been checked against twelve subsequent monthly observations from 2025, all of which fall within its 80% prediction interval. The evaluation is reported together with its limitations. The walk-forward validation set is small, because only a limited number of complete cycles allow strict causal precursor construction. One validation cycle behaves as an outlier for the geomagnetic precursor, and its effect is measured rather than removed. The precursor and analog errors are also not completely independent, so sensitivity to this assumption is examined. Leave-one-analog out tests are used to study the dependence of the analog forecast on individual historical cycles. Although the application here is solar-cycle prediction, the method suggests a broader approach for cyclic scientific time series. When a system has both a causal scalar precursor available before cycle onset and a library of historical cycle shapes, the two can be combined to produce early forecasts that are typically more accurate than either source alone. The resulting framework remains transparent, updateable, and suitable for uncertainty aware forecasting of quasi-periodic physical processes

References

- [1] M. Lockwood, A. Chambodut, L. A. Barnard, M. J. Owens, E. Clarke, and V. Mendel, A homogeneous aa index: 1. Secular variation, *Journal of Space Weather and Space Climate*, 8 (2018), A53.
- [2] K. H. Schatten, P. H. Scherrer, L. Svalgaard, and J. M. Wilcox, Using dynamo theory to predict the sunspot number during Solar Cycle 21, *Geophysical Research Letters*, 5 (1978), 411–414.
- [3] H. Sakoe and S. Chiba, Dynamic programming algorithm optimization for spoken word recognition, *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1) (1978), 43–49.
- [4] V. Soman and A. Sabarinath, Analog-Based Forecasting of Solar Cycles Using Dynamic Time Warping and Residual Bootstrapping: Real-Time Application to Cycle 25, *Solar Physics*, 301 (2026), Article 86

Interplay of Gyrotactic Motility and Solid Particles in Bio-Thermal Convective Instability

Vashavi A K^{1,*} and Virendra Kumar¹

CT

¹Department of Mathematics, Central University of Tamil Nadu, Thiruvarur, India

This study investigates the linear stability of bio-thermal convection, considering the effects of gyrotactic motile cells and solid particles within a dilute fluid confined to a finite-depth horizontal layer of infinite extent. Rigid-free boundary conditions are imposed at the layers to examine stationary convection. Particle settling, cell movement, and a thermal gradient are used to assess how dimensionless factors and critical conditions affect the system. Linear stability analysis is utilized for evaluating the onset of instability. The Galerkin method is employed to address the ensuing eigenvalue issue after normal-mode expansion, thus deriving the critical Rayleigh number and the wave number. Parameters such as the Péclet and Gyrotaxis numbers play a crucial role in examining the stability of this non-oscillatory system. The current findings indicate that the transport of settled solid particles, influenced by temperature gradients, stabilizes the system, while the orientation parameter fosters the formation of bioconvection.

References

- [1] K. Srikanth, Virendra Kumar and Vashavi A. K., Coupled thermal-vibrational effects on the instability of gravitactic swimmers, *Nonlinear Science* 6 (2026), (100112)
- [2] A. K. Kushwaha, Y. D. Sharma and Shivani Saini, Instability analysis of vibrated thermo-convection in Casson nanofluid suspension, *Modern Physics Letters B* 38 (2024), 2350219(16 pages)
- [3] B. C. Prasannakumara, J. K. Madhukesh and G. K. Ramesh, Bioconvective nanofluid flow over an exponential stretched sheet with thermophoretic particle deposition, *Propulsion and Power Research* 12 (2023), (284 -296)
- [4] Sanjalee and Rajesh Sharma, Analysis of Magneto-Thermo-Bioconvection of Nanofluid Containing Gyrotactic Microorganisms Through Porous Media, *Journal of Nanofluids* 11 (2022), (979–987)
- [5] Sohail Ahmada, Muhammad Ashrafa and Kashif Alib, Bioconvection due to gyrotactic microbes in a nanofluid flow through a porous medium, *Heliyon* 6 (2020)
- [6] A. V. Kuznetsov and A. A. Avramenko, Effect of Small Particles on the Stability of Bioconvection in a Suspension of Gyrotactic Microorganisms in a Layer of Finite Depth, *International Communications in Heat and Mass Transfer* 31 (2004), (1 -10). 1

A Cubic Trigonometric B Spline Collocation Method for Time-Fractional Convection–Diffusion Equations with Variable Coefficients

Zainab Kammappa¹ and Ashish Awasthi²

CT

¹Department of Mathematics, National Institute of Technology Calicut, Kerala, India

²Department of Mathematics, National Institute of Technology Calicut, Kerala, India

Time-Fractional Convection–Diffusion Equations (TFCDEs) model memory-dependent transport in applications like porous-media flow and heat conduction, but a cubic trigonometric B-spline (CuTBS) collocation scheme — despite its known advantages — has not yet been developed for the variable-coefficient case, motivating the present work. This paper presents a numerical approach based on CuTBS interpolation for solving TFCDEs involving the Caputo fractional time derivative. The CuTBS-based scheme effectively combines accurate spatial interpolations with a robust finite-difference discretization for the fractional derivative, ensuring high precision in both temporal and spatial domains. The method is unconditionally stable and demonstrates second-order convergence in space and $2 - \alpha$ order convergence in time. Numerical experiments are conducted to validate the applicability and feasibility of the technique. Applicability to a practically motivated problem is also demonstrated.

References

- [1] Kammappa, Z., Awasthi, A., Trigonometric cubic B-spline collocation method for time fractional diffusion equation, Computational Methods for Differential Equations (2025)
- [2] Okeke, A.A., Hamid, N.N.A., Ong, W.E., Tumba, P., Abbas, M., An efficient extended cubic B-spline collocation method for time-fractional convection-diffusion equations with applications in porous media transport, Indian Journal of Pure and Applied Mathematics, 1–27 (2026)
- [3] Yaseen, Muhammad, Muhammad Abbas, and Muhammad Bilal Riaz, A collocation method based on cubic trigonometric B-splines for the numerical simulation of the time-fractional diffusion equation, Advances in Difference Equations 2021, no. 1 (2021), 210

INTERNATIONAL CONFERENCE ON COMPUTATIONAL MATHEMATICS, MODELING & ARTIFICIAL INTELLIGENCE (CMMAI-2026)

SPONSORED BY



International Council for
Industrial & Applied Mathematics



Council of Scientific and
Industrial Research

CO-SPONSORED BY



Marino Design and Engineering Solutions

IN ASSOCIATION WITH



Kerala Mathematical
Association



National Institute of Technology
Calicut